EXACT CYCLICITY TWO OF THE DEGENERATE QUADRATIC GRAPHIC DF_{1b}

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Abstract. We prove that, in the full local space of planar quadratic vector fields, the degenerate graphic DF_{1b} has whole-graphic cyclicity exactly two for every fixed $0 < b < 2$. In the normalized unfolding, the upper bound counts actual minimal-return multiplicity for periodic orbits close to the entire graphic in two-sided Hausdorff distance, is uniform on compact modulus intervals, and imposes no relative-rate restrictions on the parameters. The proof controls a weighted derivative crossing through slow-root collision and a flat-source double-saddle composition with signed splitting at simultaneous endpoint degeneration. A complement on a full signed source rectangle realizes two cycles of the same field, one repelling and one attracting; their hyperbolic continuations yield nonempty open coefficient sets approaching the base field, with both cycles converging separately to the entire graphic.

1. The graphic and the exact cyclicity statement

The quadratic graphic DF_{1b} consists of a fast arc and a segment of equilibria ending at a nilpotent contact. Periodic orbits approaching this entire graphic can have unbounded passage times as several unfolding scales tend to zero. We prove that its whole-graphic cyclicity is exactly two and obtain a common upper bound without restricting the relative rates of the unfolding parameters. Huzak explicitly excludes this contact endpoint from the results of [15, Remark 2, p. 1307] and identifies it as requiring different techniques. We estimate its complete return through root collision and simultaneous endpoint degeneration.

Consider the real quadratic family

$$\dot z = y(1 + bz - y) + \mu_1 + \mu_2 z + \mu_3 z^2, \qquad \dot y = zy + \mu_4, \qquad 0 < b < 2, \tag{1.1}$$

where $\mu = (\mu_1, \mu_2, \mu_3, \mu_4)$ is the raw unfolding parameter and b is a modulus. At $\mu = 0$, division by y in the positive half-plane gives the fast system

$$z' = 1 + bz - y, \qquad y' = z. \tag{1.2}$$

Let γ_b be its positive-time orbit from $(0, 0)$ to its first subsequent intersection $(f_b, 0)$ with the axis. Then $f_b < 0$ and

$$\Gamma_b = \Gamma_0^b := \gamma_b \cup ([f_b, 0] \times \{0\}) \tag{1.3}$$

is a Jordan curve. The fast solution and the sections in Figure 1 are constructed in Appendix A. This is DF_{1b} in the quadratic graphic classification.

Write d_{H} for two-sided Hausdorff distance. If L is an isolated periodic orbit, its multiplicity $m(L)$ is the order of the zero of $P(s) - s$ for the actual finite minimal Poincaré return on a physical analytic transverse section. Analytic changes between physical sections preserve this order. The multiplicity will be one or two in the proof.

Throughout this paper cyclicity refers to convergence to the entire graphic. More explicitly, for fixed b set

$$C_{\mathrm{H}}(b) = \inf_{\eta > 0} \sup_{\|\mu\| < \eta} \#\{L : \ L \text{ is an isolated periodic orbit of } X_{b,\mu},\ d_{\mathrm{H}}(L, \Gamma_b) < \eta\}.$$

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The whole-Hausdorff condition also places L in any prescribed phase neighborhood when η is sufficiently small. The stronger upper assertion below counts actual multiplicity.

Theorem 1.1. *For the family* (1.1) *the following assertions hold.*

(1) *For every compact interval* $J \Subset (0,2)$ *there are a common compact phase neighborhood* K *of* $\bigcup_{b\in J}\Gamma_b$*, an open raw parameter neighborhood* $V \subset \mathbb{R}^4$ *of zero, and* $\rho_{\mathrm{H}} > 0$ *such that, for every* $(b,\mu) \in J \times V$*,*

$$\sum_{\substack{L \text{ isolated} \\ L\subset K,\ d_{\mathrm{H}}(L,\Gamma_b)<\rho_{\mathrm{H}}}} m(L) \le 2. \tag{1.4}$$

(2) *For each fixed* $b \in (0,2)$ *there are positive sequences* $c_n, \varepsilon_n, \nu_n$ *such that*

$$\mu_n = (\varepsilon_n^2 c_n, -\varepsilon_n^2, \varepsilon_n^2, -\nu_n) \longrightarrow 0, \tag{1.5}$$

and the same field X_{b,μ_n} *has two distinct isolated positive periodic orbits* L_n^-, L_n^+ *satisfying*

$$d_{\mathrm{H}}(L_n^-,\Gamma_b) \longrightarrow 0, \qquad d_{\mathrm{H}}(L_n^+,\Gamma_b) \longrightarrow 0. \tag{1.6}$$

After deletion of finitely many terms, both orbits have multiplicity one; L_n^- *is hyperbolic repelling and* L_n^+ *is hyperbolic attracting. Thus* Γ_b *has cyclicity exactly two for each fixed* $b \in (0,2)$*.*

The upper assertion permits arbitrary relative rates of the raw parameters and is uniform on compact modulus intervals. The realizing sequences are chosen for each fixed b, and their periods may increase. The whole-graphic count differs from the count of all cycles contained in a tube: a cycle shrinking to a point of Γ_b is excluded by the two-sided Hausdorff condition.

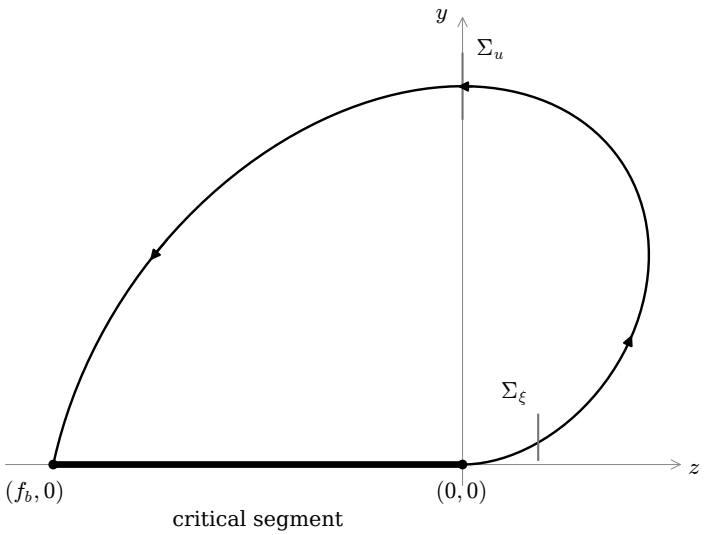


FIGURE 1. The graphic Γ_0^b for $b = 1/2$, plotted from (A.1) with equal coordinate scales. The solid curve is the fast arc; arrows indicate its direction. Every point of the thick horizontal segment is an equilibrium of the unperturbed quadratic field. The short vertical lines indicate transverse sections. The graphic is not a periodic orbit.

1.1. **Relation with earlier graphic problems.** The Dumortier–Roussarie–Rousseau program reduces the quadratic limit-cycle problem to local cyclicity problems for a finite classification of graphics [12]. Theorem 1.1 gives the sharp count for cycles converging to this entire graphic. This is the Hausdorff notion used in the actual counting statements of [13, Theorem 3.1] and [15, Theorems 1.1–1.2]. The analytic affine and time normalization of [13, Proposition 2.1] identifies

a neighborhood in the full twelve-dimensional quadratic coefficient space with this family and the coordinate and time changes. Using that known normalization, Corollary 10.1 transfers the sharp whole-graphic count to the full coefficient space: the total actual multiplicity is at most two, and two simultaneous whole-graphic orbits realize the bound. Corollary 10.2 further gives nonempty open sets of coefficients approaching each fixed base field, with one attracting and one repelling cycle converging to the graphic. [1]

For normal slow–fast double-saddle cycles, [2, Theorem 2.25(ii)] gives an upper bound two. That result assumes two persistent hyperbolic corners, a common slow scale with distinct scaled characteristic ratios, and strict passage conditions between the corners; Lemma 4.11 there retains the signed splitting in the connection. Theorem 2.23 of the same paper also treats certain interior slow saddle-node passages. In the present graphic the right corner is the turning point itself. As a positive saddle approaches that point, its characteristic ratio, the physical source coordinate, and the connection scales degenerate together. Applying a normal double-saddle comparison in this limit requires uniform bounds for the complete composed return, including its source derivatives and signed splitting.

For generic balanced multilayer canards with every side comparison side-preserving, [10, Theorem 4.10, pp. 1346–1347] models the actual rescaled return by translated-power compositions with a controlled remainder on compact subsets of their positive domain; Corollary 4.11 gives finite cyclicity bounds in that setting. The all-side-preserving restriction is clarified in [6]. The separate corrigendum [9, p. 1995] gives the smooth functions in equation (8) full layer-variable dependence; their single-variable dependence holds only for their infinite jets at the layer origin.

For ordinary non-transitory canards crossing two open fast layers, De Maesschalck, Dumortier and Roussarie treat all five connection diagrams, including side-reversing connections [7]. Their Theorem 2 (p. 25), with Definitions 9–10 (p. 24), gives the bound $\mathrm{Cod}(X) + 1$, counting multiplicity, for any unfolding of a special finite codimensional system X. On its fixed compact layer rectangle, the bound holds for all parameters in a common neighborhood and all $0 < \varepsilon \leq \varepsilon_0$, with no restriction on their relative rates [7, proof of Theorem 2, Claim, p. 27]. Obtaining an upper bound two from this theorem requires a corresponding model with $\mathrm{Cod}(X) \leq 1$. The original Γ_b has only an attracting slow segment and a fast arc issuing from the contact, so it is not itself an ordinary two-layer canard. The estimates below follow the signed physical connector through the degenerating parameter windows, retaining its negative offset, actual returning domain, and weighted derivatives.

The local contact results of [3, 4] provide finite-order transition and scaled-divergence estimates for finite-order singular slow dynamics. The global count in [4, Theorem 2.7] uses normal entry and exit corners, while Theorems 3.1–3.2 exclude simultaneous zeros of the slow dynamics at both corners. For the quadratic graphics Γ_x^b whose fast orbit starts at $(x, 0)$, the applications in [3, Theorem 7] and [15, Theorems 1.1–1.2] take x in a compact subset of $(0, \infty)$. Their source parametrization is regular in that domain. In the $b = 0$ case of [15], for example, the upper-section height is $1 + \sqrt{1 + x^2}$, whose derivative vanishes at $x = 0$. We use a regular physical section on the fast arc of Γ_b and estimate the return at this contact endpoint directly.

Exact smooth normal linearization for specific skew products whose center dynamics is independent of the normal variables is proved in [11, Theorems 1.1 and 1.3, pp. 41–42]; the one-center result preserves a multiple zero at every parameter value. Exact finite-smooth normal linearization is also available in [1, Theorem 3] for a skew product with a fixed normal spectral gap and sufficiently small center spectrum.[2] Our phase and amplitude constructions retain the selected invariant graph, a common tube with finite phase-derivative bounds, the small coupling factor in the coordinate shear, and the derivatives after passage to physical sections (Lemma 3.1). These quantitative outputs are used in the subsequent return comparisons.

[1] The numbering for [13] refers to the November 30, 2007 author version, `https://www.dms.umontreal.ca/~rousseac/Dumortier_Rousseau.pdf`.

[2] For [1], this is Theorem 3, p. 4, of the October 1, 2010 author version; the published article is cited in the bibliography.

1.2. **The counting mechanisms and the proof.** Two coupled degenerations determine the upper proof. At a collision of slow roots, a through-passage can magnify the incoming height by an unbounded factor. We retain that factor and cancel it in a weighted derivative identity. The resulting size and logarithmic-curvature bounds force one derivative ratio to cross its critical value in one direction on the actual returning interval (Lemma 4.1). For negative forcing, the preceding slow segment places every counted return in a common small source interval before the collision. The through-passage and fiber-exit descriptions there satisfy the same strict bounds across their finite junction (Lemma 4.2). This gives one bound for the field across the collision.

When both endpoints degenerate, the intermediate map is $H(a) = \Psi(\eta_0 u(a)^R)$, where $\Psi(0) = -B_{\mathrm{conn}} < 0$. We retain the complete nonlinear connection. Its negative offset and small positive derivative give $0 < H < u^R$ on the returning interval; a weighted second-derivative bound controls its contribution to the return curvature. Source flatness makes the double-saddle comparison uniform both for a small weak characteristic ratio and as that ratio approaches one (Lemma 4.3). The actual source interval and the required weighted second-derivative bounds are verified for this quadratic family. Once each parameter window controls all counted cycles of one field, a finite cover of the raw parameter directions gives the common upper bound.

For sharpness, the weak saddle passage is an exact power map with exponent below one. Its complement is exponentially contracting on a full signed source rectangle, including the unstable outlet (Lemma 8.1). A small negative splitting produces a negative–positive–negative displacement on one actual return interval. Its two zeros give two orbits of the same field; the image bounds on the rectangle show that each orbit follows the fast arc and the complete slow segment.

Sections 2–4 fix the physical return, normal-product coordinates and scalar comparisons. Sections 5 and 6 verify the complete collision and connector estimates, and Section 7 gives their common parameter cover. The appendices supply the geometric and local passage estimates. Section 8 realizes two whole-graphic orbits using the full complement proved in Appendix G; Section 9 obtains hyperbolicity from the multiplicity bound. Section 10 transfers the result to the full quadratic coefficient space. The finite differentiability order in each coordinate argument is fixed before the parameter windows are shrunk.

2. The actual return and its multiplicity

A positive function is called a positive unit when it and its reciprocal are bounded. Phase and section derivatives keep the field parameters fixed unless a parameter derivative is specified. Fix $J \Subset (0, 2)$ and write $u = u_{\mathrm{up}}$ for the reserved physical coordinate on the upper section. On the fast arc choose

$$\Sigma_u^b = \{(0, y_b + u) : |u| < u_0\}, \qquad y_b = 1 + \exp(\pi b/\sqrt{4 - b^2}),$$

and a small outgoing section Σ_ξ at the lower intersection with $z = \xi > 0$. The source and target cores are fixed before the parameters are shrunk. Let T_- be the backward regular fast transition and T_+ the forward transition through descent, the slow segment, and contact. The displacement

$$\Delta(u) = T_+(u) - T_-(u) \tag{2.1}$$

is defined only where both passages exist in finite time. The target uses the fast first integral E_b of Lemma A.1. In these orientations $T_-' = -j$, with j and its logarithmic derivatives bounded on the fixed regular core.

Lemma A.2 proves that every whole-graphic cycle lies in $y > 0$ and meets each prescribed regular section exactly once. One Hausdorff inclusion places the orbit near every graphic marker, forcing a crossing inside the prescribed core; the other confines all crossings to that core. Jordan separation and the fixed transverse sign then give uniqueness. The regular holonomies are defined for all initial values in a smaller core, independently of periodicity.

2.1. **Pullback to actual minimal-return multiplicity.** For two physical sections of the same finite orbit, regular holonomy h gives $P_2 = h \circ P_1 \circ h^{-1}$. Near a return zero,

$$h(P_1(s)) - h(s) = (P_1(s) - s) \int_0^1 h'(s + t(P_1(s) - s))\, dt.$$

The last factor is nonzero at the zero. A signed energy target or a scaled section change has the same property on its actual finite-flow domain. Uniform invertibility at a singular parameter limit is unnecessary; fixed-field nonzero derivatives suffice. The regional estimates detect simple zeros or stationary zeros with nonzero second derivative. Their finite-order section coordinates therefore determine the order of the physical analytic return zero.

Equilibria and infinite saddle passages bound the actual domain. At a finite event, two passage formulas describe the same flow at one source value. Each simple displacement zero has actual multiplicity one, and each stationary zero with nonzero second derivative has multiplicity two.

2.2. **Normalized coefficient directions.** For each $\mu \neq 0$ define $\varepsilon > 0$ by

$$\frac{\mu_1^2 + \mu_2^2 + \mu_3^2}{\varepsilon^4} + \frac{\mu_4^2}{\varepsilon^6} = 1. \tag{2.2}$$

The left side strictly decreases from infinity to zero, so the radius is unique. Put $E_i = \mu_{i+1}/\varepsilon^2$ and $D = \mu_4/\varepsilon^3$. This is a re-expression of the same field; it does not alter phase coordinates or time. Its direction belongs to S^3, and $\varepsilon(\mu) \to 0$ as $\mu \to 0$.

Write, exactly,

$$F = y(1 + bz - y) + \varepsilon^2 P(z), \qquad G = zy + \varepsilon^3 D, \qquad P(z) = E_0 + E_1 z + E_2 z^2. \tag{2.3}$$

All regional estimates below hold on open coefficient windows with fixed finite phase order, fixed section cores, and one whole-graphic phase neighborhood.

3. A common normal-product coordinate

At a simple saddle, a slow-root collision and the lower endpoint, we start from a selected invariant graph with a fixed normal spectral gap. The construction below separates the base and normal equations while preserving that graph. It gives a common tube, uniform finite phase derivatives, and the small coupling factor needed when passing to physical sections. Each application first constructs its selected graph.

The persistence result of [14, Theorem 3, p. 215] assumes a C^r field, a compact connected properly embedded C^r invariant manifold, a hyperbolic splitting, and the source rate inequalities $\sigma^{\pm}(m) < 1/r$ for every point of that manifold. We apply it to compact boundaryless extensions after establishing a fixed normal gap and finite-order rate domination. Its direct conclusion is invariant-manifold existence. Selection by the tail or incoming separatrix, the sharp forcing factor, uniform phase jets, and physical passage are adaptations proved in Sections D, 5, 6, and G.

Lemma 3.1. *Fix an integer $j \geq 4$ and nested compact intervals $I_0 \Subset I_1 \Subset I_2$. Suppose a chosen invariant graph has been translated to $u = 0$ and the equations are*

$$\dot{x} = \tau v(x) + \rho c(x,u)u, \qquad \dot{u} = B(x,u)u, \qquad 0 < \tau \leq \tau_0, \quad 0 \leq \rho \leq 1. \tag{3.1}$$

On a fixed tube over I_2, suppose v, c, B have common C^{2j+6} bounds in (x,u) and B has one sign with $|B| \geq a_* > 0$. Then, after decreasing τ_0 and the tube radius, there are exact C^j coordinates on a common tube over I_0 in which

$$\dot{\theta} = \tau v(\theta), \qquad \dot{\eta} = b(\theta)\eta, \qquad b(\theta) = B(\theta, 0). \tag{3.2}$$

The coordinate maps and their inverses have common bounds through order j. They preserve the graph and satisfy

$$x = X(\theta,\eta) = \theta + \rho\eta L_1(\theta,\eta), \qquad u = \eta + \eta^2 L_2(\theta,\eta),$$

where L_1, L_2 have common C^{j-1} bounds. In particular $X_\eta = O_{C^{j-2}}(\rho)$ and the normal coordinate is uniformly transverse to the graph. All bounds are for phase derivatives with the field parameters fixed. Zeros of v, including multiple zeros, are allowed.

Proof. Put $\bar\tau := |\tau|$. If $B \geq a_*$, first apply the argument to the reversed vector field, replacing v by $-v$ so that its base scale is the positive $\bar\tau$. All estimates below use this $\bar\tau$; reversing the orientation at the end restores the stated equation with the original positive τ. We may therefore assume $B \leq -a_*$. Extend the already translated equations outside I_2 with fixed smooth cutoffs, equal to the original coefficients on I_2, keeping $u = 0$ invariant, $B \leq -a_*$, and the factor τ in its base flow. Choose the extension of B by a smooth interpolation to a negative constant, and taper v, c smoothly only outside I_2; no extension of a prescribed boundary condition is being asserted. This gives bounded coefficients on the whole base line. A smaller normal tube is forward invariant. Write φ_t for the complete flow of $\dot x = \tau v(x)$ and (x_t, u_t) for the full flow. The asymptotic phase is the convergent limit

$$\theta(x,u) = \lim_{t\to\infty} \varphi_{-t}(x_t) = x + \rho \int_0^\infty D\varphi_{-t}(x_t) c(x_t,u_t) u_t \, dt. \tag{3.3}$$

Indeed $|u_t| \leq |u| e^{-a_* t}$ and $\|D^k \varphi_{\pm t}\| \leq K_k (1+t)^{N_k} e^{K_k \bar\tau t}$. The required differentiated estimates follow directly from the flow variations. At first order they have coefficient matrix

$$\begin{pmatrix} \tau v_x + \rho c_x u & \rho(c + u c_u) \\ B_x u & B + u B_u \end{pmatrix}.$$

Variation of constants and $\int_0^\infty |u_t| dt \leq |u|/a_*$ bound the longitudinal variations by a polynomial times $e^{K\bar\tau t}$. Every forcing term in a normal variation contains either u_t or a lower normal variation. Induction through order $j+3$ consequently bounds the differentiated integrand in (3.3) by

$$K_j (1+t)^{N_j} e^{-(a_* - K_j d - K_j \bar\tau)t},$$

where d is the normal tube radius. Choose $K_j d < a_*/4$ and then $K_j \bar\tau_0 < a_*/4$. The integral and these derivatives converge uniformly. In particular $\theta = x$ on $u = 0$, $\theta - x = \rho u L(x,u)$ with bounded finite derivatives, and $\theta_x = 1 + O(d)$. Thus (θ, u) are uniformly invertible coordinates. The flow identity in the limit gives $\dot\theta = \tau v(\theta)$ exactly.

In these coordinates write the second equation as $\dot u = b(\theta) u + q(\theta,u) u^2$ and put

$$\eta = \lim_{t\to\infty} u_t \exp\left(-\int_0^t b(\varphi_s \theta) ds \right) = u \exp\left(\int_0^\infty q(\varphi_t\theta, u_t) u_t \, dt \right). \tag{3.4}$$

The same induction, now with the base equation independent of u, gives an integrable majorant for the required derivatives. The last expression is $u + O(u^2)$ and has positive normal derivative on the smaller tube. Shifting the limit in time proves $\dot\eta = b(\theta)\eta$. The inverse function theorem gives the asserted common coordinate bounds. The input order $2j+6$ leaves more than the derivatives lost in division by u, these two coordinate changes, and the section changes below. Reversing time back proves the assertion for either normal sign. □

A product of order J requires C^{2J+6} coefficients and, when a coefficient contains one derivative of the selected graph, a C^{2J+7} graph. Fix these orders before shrinking the parameter window. The simple-contact and positive-collision applications take $J = j+2$, with graph order at least $2j+11$ and C^{2j+10} coefficients; this retains the C^j small shear needed for physical sections and the full collision output N. The lower-endpoint product and the lower B output take $J = j+1$, with graph order $2j+9$ and C^{2j+8} coefficients. Only finite phase derivatives with the field parameters fixed are used.

To use the product for physical transitions, we need two section changes. If $\rho = \tau$ and $|v(\theta_b)| \geq v_* > 0$ at an exterior base, the change from $\theta = \theta_b$ to $x = \theta_b$ has uniformly bounded finite derivatives. In fact its hitting equation, divided by τ, is

$$\int_0^T v(\varphi_t\theta_b) dt + \eta_T L_1(\varphi_T\theta_b, \eta_T) = 0, \qquad \eta_T = \eta \exp \int_0^T b(\varphi_t\theta_b) dt.$$

Its derivative in T is bounded away from zero for small η; therefore $T = O(\eta)$ with common derivative bounds. The required lower bound on $|v|$ is at the exterior base; v may vanish inside the collision strip.

For $\rho = 1$, or after a further uniformly smooth change to physical coordinates, write $z = X(\theta, \eta)$ and $y = H(\theta, \eta)$. The scaled variable $\eta = \tau V$ gives uniform section bounds in V. Assume again $|v(\theta_b)| \geq v_* > 0$. Suppose X_θ is a positive unit and the normal direction is transverse to the graph, with $a_l := H_\eta - h_z X_\eta$ a positive unit at $(\theta_b, 0)$, where $y = h(z)$ is the physical graph. Divide $X(\varphi_T\theta_b, \eta_T) - X(\theta_b, 0) = 0$ by τ after this substitution. The implicit function theorem gives bounded derivatives in V for $|V| \leq d_1$, where $d_1 > 0$ is fixed and small. To retain the normal factor, write $H(\theta, \eta) - h(X(\theta, \eta)) = \eta a(\theta, \eta)$ and set $E_T(V) = \exp \int_0^{T(V)} b(\varphi_t\theta_b)dt$. Exactly,

$$\frac{\Phi(\tau V) - h(z_b)}{\tau} = VE_T(V)a\big(\varphi_{T(V)}\theta_b, \tau V E_T(V)\big).$$

The right side has a common C^{j-1} bound and derivative a_l at zero. Taylor's formula therefore gives

$$\Phi(\eta) = h(z_b) + a_l\eta + \frac{\eta^2}{\tau}E(\eta/\tau), \qquad \|E\|_{C^{j-3}} \leq K, \quad \frac{|\Phi^{(k)}|}{\Phi'} \leq K_k\tau^{1-k} \quad (2 \leq k \leq j-1). \tag{3.5}$$

Here $z_b = X(\theta_b, 0)$ and Φ' is a positive unit after decreasing d_1. Thus unscaled section derivatives need not remain bounded as $\tau \to 0$.

Finally, if v has a simple zero θ_s and $|v'(\theta_s)|$ is a positive unit, division of time by $|b(\theta)|$ gives $\dot\theta = \tau g(\theta)$ and $\dot\eta = \operatorname{sgn}(b)\eta$. With $\beta = g'(\theta_s)$, the coordinate

$$\xi = \pm(\theta - \theta_s) \exp \int_{\theta_s}^{\theta} \left(\frac{\beta}{g(t)} - \frac{1}{t - \theta_s} \right) dt$$

with the sign chosen to make the incoming coordinate positive, has common finite derivatives and satisfies $\dot\xi = \tau\beta\xi$. The integrand has a removable singularity; the small factor τ never enters its denominator. When the normal direction is expanding and $\beta < 0$, the passage from $\xi = \xi_0$ to $\eta = \eta_0$ is exactly $D(s) = \xi_0(s/\eta_0)^r$, with $r = -\tau\beta$. This last scalar linearization is used only at simple roots.

4. Comparisons for the complete return

Three scalar comparisons give the upper bound on the complete actual returning interval of one fixed field. They compare a simple-contact passage with its complement, control a derivative ratio through slow-root collision, and compose two endpoint passages with their signed connector. The physical domains and estimates needed to apply them are proved in Sections 5 and 6, and Appendix E.

4.1. **A simple-contact passage and its complement.** For a simple contact with positive preceding slow speed, the bound two comes from a derivative comparison of the local passage with its complement. Near the central saddle-node, let W be the scalar speed in the orbital product coordinates. The passage from the fixed incoming section $w = w_0 > 0$ to $s = s_+$ is $D(s) = w_0 \exp(-\int_s^{s_+} dt/W(t))$; write its increasing complement as $H(a)$. On a subinterval containing every return zero, the finite-passage condition and complement estimates give $W(H) > 0$ and

$$\frac{d}{da}\log\frac{D(H(a))}{a} = \frac{Z(a)}{aW(H(a))}, \qquad Z = aH' - W(H), \qquad Z' = H'\left(1 - W_s(H) + a\frac{H''}{H'}\right) > 0.$$

The logarithmic displacement therefore has at most one stationary point, a strict minimum. In the saddle ranges, the corresponding comparison makes every zero of $(D \circ H)' - 1$ a strict

downward crossing. For the weak saddle this follows directly from the exact power $D(s) = \xi_0(s/\omega_0)^r$, $0 < r \le r_0 < 1$: at $(D \circ H)' = 1$,

$$(D \circ H)'' = \frac{r-1}{rD(H)} + \frac{H''}{H'} < 0,$$

because the complete complement satisfies $r|H''/H'| = o(1)$. The finite-ratio Dulac estimates give the same conclusion, including ratios approaching one. In each case a tangent zero is exactly quadratic and exhausts the local count two.

In the weak-saddle range, early departures form the upper interval of the physical source, and their displacement is strictly decreasing. The late interval has the single-maximum property just proved. After two simple late zeros, or one quadratic late zero, its displacement is negative up to the finite threshold; continuity and strict decrease then exclude a threshold or early zero. Otherwise each interval contributes at most one simple zero. This gives total multiplicity two on the whole return domain. The local conjugacy is used only where its outgoing continuation and inverse are both defined; all late outputs lie in that common image. Appendix E proves these domain statements and the derivative estimates above.

4.2. **A weighted derivative ratio through slow-root collision.** Let q_0 be the scaled incoming height and β the fixed-field separator height on that cut. On the no-root side, β is the trace of the physical outgoing tail; on the root side it is the trace of the selected incoming stable or parabolic separator. In both cases the actual returning domain has source distance $s = q_0 - \beta > 0$. The incoming reference $q_c(0)$ controls distortion through $w_\beta = \beta - q_c(0)$; β determines the returning boundary. Section 5 constructs these objects and proves the incoming and outgoing bounds used in the following comparison.

Lemma 4.1. *Suppose that on an actual returning interval $s = q_0 - \beta > 0$, where β is independent of u_{up}, $w_\beta = \beta - q_c(0) > 0$, and $w_0 = s + w_\beta \le C_0$. Write $M(s) = \partial_s T_+ > 0$. Assume that, for some $\delta > 0$,*

$$sM(s) \le \varepsilon^2\delta \quad (s \in I), \qquad -s(\log M)_s \ge c_M > 0 \quad (s \in I \cap (0, s_0]), \tag{4.1}$$

where I is the actual returning interval and $s_0, c_M > 0$ are fixed. Assume also that the incoming transition has

$$\begin{aligned} s_{u_{\mathrm{up}}} &= -\varepsilon^{-2}\lambda(s + w_\beta), \qquad 0 < c_\lambda \le \lambda \le C_\lambda, \quad 0 < c_j \le j \le C_j, \\ |(\log\lambda)_{u_{\mathrm{up}}}| &\le L_\lambda\varepsilon^{-2}, \qquad |(\log j)_{u_{\mathrm{up}}}| \le L_j. \end{aligned} \tag{4.2}$$

on I, where all bounds are fixed. These are the incoming and short-transition estimates from Lemma 5.2 and (A.4)*. Fix $0 < \varepsilon \le \bar\varepsilon$ and put $K = L_\lambda + C_\lambda + \bar\varepsilon^2 L_j$. Require*

$$4C_\lambda\delta \le c_j, \qquad \delta \le \frac{c_M c_j}{4K}. \tag{4.3}$$

If the curvature bound is only available for $s \le s_0$, require also

$$\frac{4C_\lambda\delta}{c_j} w_\beta < s_0. \tag{4.4}$$

This last condition may be replaced by $2C_\lambda\delta C_0/c_j < s_0$; it is unnecessary when the curvature bound holds throughout I. Then the isolated return zeros of $T_+ - T_-$ have total actual multiplicity at most two.

Proof. Write $u = u_{\mathrm{up}}$ and let $R = M|s_u|/j$. The size and incoming bounds give

$$R \le \frac{C_\lambda\delta}{c_j}\frac{s + w_\beta}{s}.$$

Where $R \ge 1/2$, the first condition in (4.3) therefore gives

$$s \le \frac{2C_\lambda\delta}{c_j - 2C_\lambda\delta} w_\beta \le \frac{4C_\lambda\delta}{c_j} w_\beta. \tag{4.5}$$

Before absorption the same inequality gives $s \le 2C_\lambda\delta(s + w_\beta)/c_j \le 2C_\lambda\delta C_0/c_j$. Thus either cutoff above places these points in the curvature range. Its logarithmic derivative is

$$(\log R)_u = (\log M)_s s_u + (\log \lambda)_u - \varepsilon^{-2}\lambda - (\log j)_u.$$

At these points $M|s_u| \ge j/2$, so $|s_u|/s \ge c_j/(2\varepsilon^2\delta)$. The second condition in (4.3) now yields

$$(\log R)_u \ge \frac{1}{\varepsilon^2}\left(\frac{c_M c_j}{2\delta} - K\right) \ge \frac{c_M c_j}{4\varepsilon^2\delta} > 0.$$

Thus $R_{u_{\mathrm{up}}} > 0$ wherever $R \ge 1/2$. It crosses one at most once, strictly upward. Since $(T_+ - T_-)_{u_{\mathrm{up}}} = j(1 - R)$, there is at most one stationary point, a strict maximum. At a stationary return zero, $(T_+ - T_-)_{u_{\mathrm{up}}u_{\mathrm{up}}} = -jR_{u_{\mathrm{up}}} < 0$. Such a zero is quadratic and cannot coexist with another return zero. Otherwise there are at most two simple zeros. Section 2 transfers these orders to actual minimal-return multiplicity. □

In the collision product coordinate, $\eta_0 > 0$ is the fixed normal exit height. Write A and B for the normal amplitudes at the left and right bases. Lemma 5.6 gives an open physical returning interval I_{ret} on which $0 < A < \eta_0$; every counted cycle has $A < \eta_0/2$. On the no-root side $B = m_b A$, where $m_b > 1$ is the normal expansion through the strip, without a uniform upper bound. The through and fibre-exit pieces are the intersections of the actual source with $0 < A < \eta_0/m_b$ and $\eta_0/m_b < A < \eta_0$, respectively. They are ordered by increasing source distance s, opposite to u_{up}, and have at most one finite common boundary. On the root side only the fibre-exit piece occurs. Figure 2 depicts this partition. Lemma 5.9 includes empty pieces and the finite boundary.

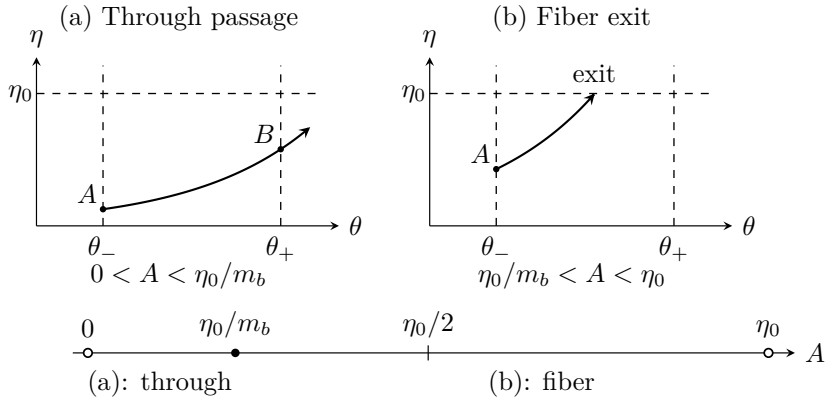


FIGURE 2. Schematic no-root passages in the product coordinates (5.33). Dashed lines are the fixed bases and exit level. Trajectory arrows give positive-time direction; the lower line records increasing A. It shows one possible position of the sole seam $A = \eta_0/m_b$ in $0 < A < \eta_0$; intersect the displayed branch ranges with the actual source image. Here $m_b > 1$ has no uniform upper bound, and the seam or a branch may be absent from that image. The half-amplitude tick marks localization of counted return zeros, not a passage event. When a root is present, only fiber exit remains; $B = m_b A$ is the no-root through relation.

The through and fibre-exit formulas describe one physical passage, so the derivative comparison must hold across their finite boundary. The next lemma gives the common constants and cutoff condition needed for this step. Section 5 verifies its flow and domain hypotheses.

Lemma 4.2 (Finite-event gluing for the collision return map)**.** *Fix one negative-forcing field in the collision coefficient box, and let I_{ret} be the restricted physical source from Lemma 5.6, after the common target-core restriction. On the no-root side, the through and fibre-exit formulas*

in (5.40) *are restrictions of one flow-defined* C^2 *map* $\widehat{T}_+$ *on* I_{ret}*, with* $M = \partial_s \widehat{T}_+ > 0$*. Thus* M *and* M_s *are continuous at the possible finite event* $A = \eta_0/m_b$*. Assume the incoming and short-transition bounds of Lemma 4.1 on* I_{ret}*. Suppose the estimates* (4.1) *hold on the nonempty open pieces, with constants* (δ_ℓ, c_ℓ) *for* $\ell \in \{\mathrm{through}, \mathrm{fibre}\}$*. Set* $\delta = \max_\ell \delta_\ell$ *and* $c_M = \min_\ell c_\ell$*. Require* (4.3) *with the common incoming bounds. On a row whose curvature bound is restricted to* $s \le s_{0,\ell}$*, require in addition* $4C_\lambda \delta_\ell w_\beta / c_j < s_{0,\ell}$*, or* $2C_\lambda \delta_\ell C_0 / c_j < s_{0,\ell}$*. No such restriction is needed when a row's curvature bound holds throughout its returning interval. Then the estimates extend to the finite event, and the complete displacement* $\widehat{\Delta} = \widehat{T}_+ - T_-$ *has total actual isolated zero multiplicity at most two on* I_{ret}*. On the root side the same conclusion holds under these estimates for the single fibre-exit piece, with no internal finite event.*

Proof. At the finite event the returning flow has finite flight time, so the two formulas have the same value and first two derivatives. Since $s > 0$ and $M > 0$, continuity of M and M_s extends sM and $-sM_s/M$ to that value. Where the derivative ratio in Lemma 4.1 is at least $1/2$, (4.5) with δ_ℓ, or its C_0 alternative, puts the source in the range of the row's curvature bound. The proof of that lemma gives the crossing inequality with a common positive margin on both pieces and, by continuity, at the event. Its single-maximum argument consequently applies on the whole source interval. On the root side it applies directly to the single fibre-exit interval. □

4.3. **A two-saddle comparison with a flat source.** For simultaneous endpoint degeneration, we compare the positive Dulac passage with the complete nonlinear connector $H = \Psi(\eta_0 u^R)$. The scalar comparison permits a nonpositive baseline $\Psi(0) \le 0$. For the quadratic family, Proposition 6.4 constructs the connector on its full signed source interval with a strictly negative baseline and the required weighted derivative bounds. Source flatness and second derivatives suffice, including when the positive characteristic ratio approaches one.

Lemma 4.3 (Two-saddle comparison with a flat source)**.** *Fix the vector field parameters and let* $I \subset (0, \infty)$ *be an open actual returning interval. Let* $u \in C^2(I)$ *satisfy*

$$0 < u < 1, \qquad u' > 0, \qquad |u''/u'| \le M.$$

For fixed $R > 1$ *and* $\eta_0 > 0$*, put* $t = \eta_0 u^R$ *and* $H = \Psi(t) > 0$*. Assume that* $\Psi \in C^2([0, t_*])$*,* $t(I) \subset (0, t_*)$*, and*

$$\Psi(0) = -B_{\mathrm{conn}} \le 0, \qquad \Psi' > 0, \qquad \eta_0 \sup_{[0,t_*]} \Psi' \le 1, \qquad \left| t \frac{\Psi''(t)}{\Psi'(t)} \right| \le L_\Psi.$$

Let $\mathcal{D} \in C^2$ *with* $\mathcal{D}' > 0$ *on a positive interval containing* $H(I)$*. All derivatives are taken with the field parameters fixed. The equation* $\mathcal{D}(H(a)) = a$ *has total isolated zero multiplicity at most two on* I *in either of the following situations:*

1. $\mathcal{D}(s) = c_D s^r$*,* $0 < c_D \le C_D$*,* $0 < r \le r_0 < 1$*,* $Rr \ge 1$*,* $0 < \mathcal{D}(H) \le a_0$*, and* $Rr \sup_I u' \le \eta_*$*. The bounds* C_D, r_0, M, L_Ψ *are fixed, and* $a_0, \eta_* > 0$ *are sufficiently small in terms of them;*
2. $r \in [r_*, r^*] \Subset (0, \infty)$*,* $Rr \ge 1 + \kappa$ *for a fixed* $\kappa > 0$*,* $0 < H' \le m$*,* $R \sup_I u' \le M_f$*, and*

$$0 < \mathcal{D}' \le M_D s^{r-1}, \qquad \frac{s\mathcal{D}''}{\mathcal{D}'} = r - 1 + E(s), \qquad |E| \le M_D s^\alpha, \tag{4.6}$$

on the positive domain of $\mathcal{D}$*. Here* $\alpha, M_D, M_f > 0$ *and* r_*, r^*, M, L_Ψ *are fixed,* E *is continuous, and* $m > 0$ *is sufficiently small in terms of these bounds and* κ*.*

Every stationary zero has nonzero negative second derivative. In the second case the conclusion is uniform as r *approaches one, and if* $r \ge 1$ *there is at most one zero, which is simple.*

Proof. The signed-connector hypotheses give

$$0 < H \le \Psi(t) + B_{\mathrm{conn}} = \int_0^t \Psi'(v)\, dv \le u^R < 1.$$

Put $v = (\mathcal{D} \circ H)'$, $U = u''/u'$, $W(s) = s\mathcal{D}'(s)$, $\sigma(s) = s\mathcal{D}''(s)/\mathcal{D}'(s)$, and $\Lambda(t) = t\Psi''(t)/\Psi'(t)$. Direct differentiation gives

$$\begin{aligned} \frac{H''}{H'} &= [R - 1 + R\Lambda(t)]\frac{u'}{u} + U, \\ v' &= \frac{H'}{H} J \quad \text{at } v = 1, \end{aligned} \tag{4.7}$$

where

$$J = \sigma(H) + W(H)U + W(H)[R - 1 + R\Lambda(t)]\frac{u'}{u}.$$

We prove $J < 0$ at every such point. Set $C_\Psi = 1 + L_\Psi$. In the first case $W(H) = r\mathcal{D}(H)$ and $\sigma = r - 1$. Since $u^{Rr-1} \le 1$, including when $Rr = 1$,

$$RW(H)\frac{u'}{u} \le C_D Rr u' u^{Rr-1} \le C_D \eta_*.$$

Take a_0 first, then η_*, so that $r_0 M a_0 + C_\Psi C_D \eta_* \le (1 - r_0)/2$. Consequently

$$J \le -(1 - r_0) + r_0 M a_0 + C_\Psi C_D \eta_* \le -(1 - r_0)/2 < 0.$$

In the second case $r \ge 1$ gives $v \le M_D m H^{r-1} \le M_D m < 1$ once $M_D m < 1$, so only $r < 1$ needs consideration. Write $\beta = 1 - r$ and $T = \log(1/H)$. At $v = 1$, the bound on $\mathcal{D}'$ implies

$$\beta T \ge L_m := \log \frac{1}{M_D m}.$$

Since $Rr \ge 1 + \kappa$, put $\gamma_\kappa = r_*\kappa/(1 + \kappa) > 0$. Then

$$r - \frac{1}{R} \ge \frac{r\kappa}{1 + \kappa} \ge \gamma_\kappa.$$

Using $u \ge H^{1/R}$ gives

$$RW(H)\frac{u'}{u} \le M_D M_f H^{r-1/R} \le M_D M_f H^{\gamma_\kappa}.$$

It follows that

$$J \le -\beta + M_D H^\alpha + M_D M H^{r_*} + C_\Psi M_D M_f H^{\gamma_\kappa}.$$

Put $\zeta_\kappa = \min\{\alpha, \gamma_\kappa\} > 0$ and $C = M_D(1 + M + C_\Psi M_f)$. Choose

$$0 < m \le \frac{1}{M_D} \exp\left(-\frac{\max\{1, \log(4C)\}}{\zeta_\kappa}\right).$$

Then $M_D m < 1$, $L_m \ge 1/\zeta_\kappa$ and $Ce^{-\zeta_\kappa L_m} \le 1/4$. At the critical points $T \ge L_m$, since $0 < \beta < 1$, and hence

$$\frac{H^{\zeta_\kappa}}{\beta} \le \frac{Te^{-\zeta_\kappa T}}{L_m} \le e^{-\zeta_\kappa L_m}.$$

Thus $J \le -3\beta/4 < 0$ with one choice of m depending on κ, uniformly as $r \to 1$.

Every zero of $v - 1$ is a strict downward crossing by (4.7). There is at most one: two downward crossings would require an intervening upward or nontransverse crossing. The displacement is therefore strictly monotone, or increases before its unique critical point and decreases afterward. It has at most two simple zeros or a single quadratic zero with negative second derivative; a zero interval is impossible. Section 2 transfers these orders to actual minimal-return multiplicity. □

5. Positive contact constant and slow-root collision

The remaining contact scaling has two independent small parameters:

$$F = y(1 + bz - y) + \varepsilon^2 P(z), \qquad G = zy + \kappa^3 d, \qquad P(z) = z^2 + arz + r^2, \qquad \kappa = r\varepsilon. \tag{5.1}$$

Throughout this section, a, d range in fixed compact intervals and $r, \varepsilon > 0$ tend to zero independently. In particular, estimates must allow r to be smaller than every power of ε.

Proposition 5.1. *On every compact (a, d, b)-set with $0 < b < 2$, all sufficiently small fields* (5.1) *have total actual minimal-return multiplicity at most two for isolated cycles in one Hausdorff neighborhood of the whole limiting graphic. The parameter and Hausdorff neighborhoods are uniform on that compact set. For $d \geq 0$ the bound is one and every counted zero is simple.*

Near a positive double root, the slow passage can amplify the incoming height by an arbitrarily large factor. We retain this multiplier in product coordinates. Weighting the complete outgoing derivative by the incoming distance from the separating trajectory cancels the multiplier and gives size and logarithmic-curvature bounds. Combined with the incoming distortion estimate, these bounds force the derivative ratio with the short map to cross one only upwards (Lemma 4.1).

Lemma 5.2 controls the incoming trace, and Lemma 5.3 estimates the logarithmic derivative from departure to the target. Before applying the comparison at a collision, Lemma 5.6 places every counted cycle in one small source interval. Lemma 5.9 divides it into through passages and fiber exits; on the root side only the fiber exit occurs (Figure 2). At their possible finite junction, the two formulas represent the same physical return. Lemma 4.2 therefore gives the bound two on the whole interval.

5.1. The incoming map in a regular coordinate.

We first transport the upper-section source to the inner cut $z = 0$. The signed variation of its height will control the derivative of the complete contact map.

Use the signed coordinate u_{up} on the upper section from Lemma A.2. Every counted cycle meets this section exactly once. The same lemma supplies the contact passage in $y > 0$, $F > 0$, from the decreasing-height section $y = K\varepsilon^3$ to a fixed positive section, on the interval of sources for which that passage is defined. No sign or scaled-size condition is imposed on u_{up}. Set

$$z = \kappa\zeta, \quad y = \kappa^2 q, \quad q_0(u_{\mathrm{up}}) = q(0; u_{\mathrm{up}}), \qquad q_\zeta = \frac{\zeta q + d}{V(\zeta, q)}, \quad V = \mathsf{A} + \mathsf{B}q - \mathsf{C}q^2, \tag{5.2}$$

where

$$\mathsf{A} = 1 + a\varepsilon\zeta + \varepsilon^2\zeta^2, \qquad \mathsf{B} = 1 + b\kappa\zeta, \qquad \mathsf{C} = \kappa^2.$$

All derivatives below keep the vector field fixed.

Lemma 5.2. *The actual incoming values satisfy $0 < q_0 \leq Q_0$, with Q_0 uniform. There is a particular solution q_c, independent of u_{up}, such that $q_0 = q_c(0) + w_0$, $w_0 > 0$. For $d \geq 0$,*

$$q_c(0) \geq 0, \qquad \frac{|(q_0)_{u_{\mathrm{up}}}|}{q_0 + d} \leq C\varepsilon^{-2}. \tag{5.3}$$

For $d = -m < 0$, a cycle is possible only if

$$a < 0, \qquad 0 < m \leq C\exp(-c/\varepsilon^2). \tag{5.4}$$

On this set $-4m \leq q_c(0) < 0$ and

$$(\log w_0)_{u_{\mathrm{up}}} = -\varepsilon^{-2}\lambda, \qquad c \leq \lambda \leq C, \qquad |(\log\lambda)_{u_{\mathrm{up}}}| \leq C\varepsilon^{-2}. \tag{5.5}$$

The passing and target restrictions have interval preimages in u_{up}.

Proof. We first justify the full incoming passage. On a fixed negative interval, $P(z) \geq z^2/2$ for small r. Put $x = -z$ and subtract the forcing graph $\kappa^3 d/x$. For $x \geq Mr$, with M fixed and large, the two sides of

$$\left| y - \frac{\kappa^3 d}{x} \right| \leq k\varepsilon^2 x^2$$

point inward; this follows by substituting the two equalities in $dy/dz = G/F$, using $P \asymp x^2$ and then choosing M large. It carries the fixed negative entrance to $Z = z/r = -M$, with bounded small normal coordinate in

$$Y = \frac{y}{r^2\varepsilon^2}, \qquad w = Y + \frac{\varepsilon d}{Z}.$$

On a fixed negative Z-interval the normal eigenvalue is bounded away from zero. The invariant graph and strong foliation give an exact scalar base equation $\dot{\theta} = \varepsilon^2 U Q_\varepsilon(\theta)$, where $U > 0$ and, before the near-identity base change,

$$Q_\varepsilon(Z) = Z^2 + aZ + 1 - \frac{\varepsilon d(1 + brZ)}{Z} - \frac{r^2\varepsilon^4 d^2}{Z^2}.$$

Its second derivative is positive on this fixed strip for sufficiently small parameters, so it has at most two roots. The normal coordinate contracts at a fixed rate. The base at entry lies in $(-M - 1/2, -M + 1/2)$; choosing M larger puts this entire interval to the left of every possible negative root. A first root cannot be crossed by the scalar base. Its approach takes infinite time and normal contraction makes this alternative terminal. If there is no root, the exit at a fixed $Z = -\zeta_0 < 0$ satisfies

$$|w| \leq C\varepsilon^3 + Ce^{-c/\varepsilon^2}.$$

The two sides of $|q + d/\zeta| \leq K/\zeta^2$ point inward on $-\zeta_0/\varepsilon \leq \zeta \leq -L$, and V lies between two positive constants there. This gives a bounded entrance at $\zeta = -L$. These assertions use the strong spectral gap, not a lower bound for a weak root derivative. The fixed-gap graph transform, as in the proof of Lemma D.5, first supplies the invariant graph and its finite phase bounds on this negative strip. Lemma 3.1 then applies with $\tau = \rho = \varepsilon^2$ and negative normal sign. No scalar base linearization is made, so its conclusion includes a negative double root as well.

Here is a direct bound on the last inner ordinate. In $X = z/r$ and $Y = y/(r^2\varepsilon^2)$ the equations are

$$X' = \varepsilon^2(YQ + p_a(X)), \quad Y' = XY + \varepsilon d, \quad Q = 1 + brX - r^2\varepsilon^2 Y, \quad p_a = X^2 + aX + 1.$$

The preceding upper barrier gives $Y(-L_1) \leq K_1$ for a fixed large L_1. Choose $T > K_1 + 4\max_{[-L_1,0]} |p_a| + 1$. Above $Y = T$, $YQ + p_a \geq Y/4$. If $d \leq 0$, Y decreases for $X < 0$. If $d > 0$, an upcrossing of $Y = T$ requires $X \geq -\varepsilon d/T$; thereafter

$$\frac{d(Y^2)}{dX} \leq \frac{8d}{\varepsilon}.$$

The remaining interval has length at most $\varepsilon d/T$. Thus $Y(0)^2 \leq T^2 + 8d^2/T$. This proves the uniform bound on q_0.

Select the particular solution y_c by zero strong amplitude on the incoming invariant graph and continue the same solution through the taper and the inner funnel. The strong-graph equation gives, on every fixed negative interval,

$$y_c = -\kappa^3 d/z + O(\varepsilon^2\kappa^3|d|), \qquad \partial_z^j y_c = O(\kappa^3|d|), \quad j \leq 2.$$

The construction is made on the nested negative intervals with the same graph, so these are restrictions of one solution. For $d = 0$ it is exactly zero. For $d \neq 0$ its sign on the incoming half is the sign of d: at $y = 0$ the scalar velocity has that sign, and the selected forced trace cannot cross to the other side. Its continuation to $z = 0$ has the same sign. At the decreasing-height entrance $z = \Lambda(u_{\mathrm{up}})$,

$$W_i := K\varepsilon^3 - y_c(\Lambda) \asymp \varepsilon^3 > 0.$$

Scalar uniqueness preserves $W = y - y_c > 0$ along every passing interval.

For the needed variational estimates write, exactly,

$$\begin{aligned} W_z &= A(z,W)W, \quad A(z,W) = \int_0^1 H_y(z, y_c + tW)\,dt, \\ L_z &= WA_W L, \qquad\qquad L = (\log W)_{u_{\mathrm{up}}}, \qquad H = G/F. \end{aligned} \tag{5.6}$$

Because the entrance moves, its fixed-phase initial value is

$$L_i = \frac{(W_i)_{u_{\mathrm{up}}}}{W_i} - A(\Lambda, W_i)\Lambda_{u_{\mathrm{up}}}. \tag{5.7}$$

The moving entry satisfies the common C^2 estimate (A.6), proved by integrating (A.8); its error from the fast-foot map is $O(\varepsilon^2(1 + |\log\varepsilon|))$. At this section

$$A(\Lambda, W_i) = \frac{\Lambda}{\varepsilon^2 P(\Lambda)}(1 + O(\varepsilon)).$$

It follows that $|L_i| \le C\varepsilon^{-2}$ and, for negative d, $L_i = -\varepsilon^{-2}\lambda_i$, where λ_i is a positive unit with bounded logarithmic u_{up}-derivative.

To propagate the initial variation through the shrinking scales, we bound the feedback integrals. On the fixed negative interval,

$$|H_{yy}| \le C\varepsilon^{-4}, \quad |H_{yyy}| \le C\varepsilon^{-6}, \quad W \le C\varepsilon^3 e^{-c(z-\Lambda)/\varepsilon^2}.$$

On the taper from x_g to Mr,

$$\begin{aligned} |H_{yy}| &\le C\varepsilon^{-4}x^{-3}, \qquad |H_{yyy}| \le C\varepsilon^{-6}x^{-5}, \\ W(-x) &\le W_g(x/x_g)^N, \qquad N = c/\varepsilon^2, \qquad W_g \le C\varepsilon^3 e^{-c/\varepsilon^2}. \end{aligned}$$

Integrating these powers down to zero bounds the first feedback by $CW_g/(\varepsilon^2 x_g^2)$ and the second by $CW_g^2/(\varepsilon^4 x_g^4)$. On a passing negative root interval, the exact strong equation $\dot\eta = -b_n(\theta)\eta$, $b_n \ge c > 0$, bounds the first feedback by $C\int |\eta(0)|e^{-ct}\,dt$. The endpoint changes $Z = \theta + O(\varepsilon^2\eta)$ have the same bound. To control a fixed physical section without dividing by a small base speed, write $\dot\theta = \varepsilon^2 v(\theta)$ on a passing branch, where $v > 0$. The entrance speed is a positive unit, v' is bounded, and

$$\frac{d}{dt}\log\frac{|\eta|}{v} = -b_n - \varepsilon^2 v' \le -c/2.$$

Consequently $|\eta|/v \le C|\eta(0)|e^{-ct/2}$, which controls the ratio occurring in the physical-section correction. Indeed, the product inverse has $Z = \theta + \varepsilon^2\eta L_1$ and $u = \eta + \eta^2 L_2$, with bounded finite jets. Thus $v(Z)$ is comparable with $v(\theta)$ and the physical speed $v(Z) + c(Z,u)u$ is a positive unit times $v(Z)$, provided the entrance amplitude is fixed sufficiently small. For the divided physical coefficient $a = B/[\varepsilon^2(v + cu)]$, direct differentiation gives

$$|ua_u|\,dZ \le C\left(|u| + \frac{|u|}{v(Z)}\right)dt, \quad |u^2 a_{uu}|\,dZ \le C\left(u^2 + \frac{u^2}{v(Z)} + \frac{u^2}{v(Z)^2}\right)dt.$$

The preceding decay bounds make both integrals finite with a common bound. The physical feedback equations are differentiated at fixed Z; the product coordinates are only used to estimate their coefficients. Weak flight time need not be bounded; a base root reached only at infinite time is terminal and is excluded from this passing branch. For negative d, the restriction $a < 0$ proved below gives $p_a(Z) \ge 1 + Z^2$ for $Z < 0$. On the fixed negative scaled interval the scalar equation then has $A \le -c/\varepsilon^2$ and its first two normal derivatives are $O(\varepsilon^{-2})$, which also bounds the second feedback.

For $d = -m < 0$, a bound proportional to m will localize the later ratio estimate. At a fixed small negative contact point $z = -x_*$, the selected incoming graph has $q_c(-x_*/\kappa) = -\kappa m/x_* + O(\varepsilon^2\kappa m)$. On the interval from that point to zero, $a < 0$ gives $\mathsf{A} \ge 1$. The graphs $-4m/(1-\zeta)$ and zero are lower and upper barriers. Indeed, writing $t = -\zeta$, at the lower graph the numerator $\zeta q - m$ is nonnegative for $t \ge 1/3$. For $0 \le t \le 1/3$, $V \ge 1/2$ bounds the scalar

right-hand side below by $-2m$, while the lower graph has derivative at most $-9m/4$. The initial value is between the two barriers; at the upper graph that right-hand side is $-m/\mathsf{A} < 0$. Thus $-4m \le q_c(0) < 0$, with no reciprocal small scale or root gap.

In the funnel set $q = q_c + w$. Its exact divided coefficient is

$$a_f = \frac{\mathsf{N}}{V(\zeta, q_c)V(\zeta, q_c + w)}, \quad \mathsf{N} = \zeta\mathsf{A} - d\mathsf{B} + \mathsf{C}\{\zeta q_c^2 + 2dq_c + w(\zeta q_c + d)\}.$$

On the funnel ending at $\zeta = -L$, with L fixed and large, $\mathsf{N} \le -c(|\zeta|+d)$ for $d \ge 0$; in the negative exponential range it is at most $-c|\zeta|$. The speeds are positive units, and $B_f = (a_f)_w/a_f$ and $(B_f)_w$ are bounded. Since $w_\zeta = a_f w < 0$ and $(a_f)_{ww} = a_f(B_f^2 + (B_f)_w)$, the feedback integrals are bounded by $C\int(-w_\zeta)d\zeta$. The remaining bounded inner interval $[-L, 0]$ has bounded third height derivatives; the sign of a_f need not stay negative there. Consequently

$$\int |WA_W|\, dz \le C, \qquad \int (|WA_W| + |W^2 A_{WW}|)\, dz \le C \quad (d < 0). \tag{5.8}$$

Linear changes of phase leave these integrals unchanged. The strong coordinate calculation above includes the endpoint terms when that coordinate is used in the negative root interval.

Equations (5.6)–(5.8) preserve $|L| \le C\varepsilon^{-2}$. For $d \ge 0$, $q_0 + d \ge w_0$ proves (5.3). For $d < 0$ write $L(0) = L_i e^{J(u_{\mathrm{up}})}$, $J = \int_{\Lambda(u_{\mathrm{up}})}^{0} WA_W\, dz$. Then

$$J_{u_{\mathrm{up}}} = -[WA_W]_\Lambda \Lambda_{u_{\mathrm{up}}} + \int L(WA_W + W^2 A_{WW})\, dz = O(\varepsilon^{-2}).$$

The first term is $O(\varepsilon^{-1})$; the second is controlled by (5.8). This proves (5.5).

It remains to justify the restriction used on negative d. Lemma B.2 excludes $a \ge 0$. For $d = -m < 0$ and $x \ge L_1 r$, the upper barrier and $F \le C\varepsilon^2 x^2$ give

$$y(-L_1 r) \le K\varepsilon^3 (L_1 r/x_g)^{c/\varepsilon^2}.$$

The height decreases to $z = -\kappa$. Division by κ^2 loses only r^{-2}: for $N > 2$, $r^{-2}(L_1 r/x_g)^N$ is increasing in r. Choosing $r_0 < x_g/(2L_1)$ gives $q(-1) \le Ce^{-c/\varepsilon^2}$, uniformly even for arbitrarily small r. On $-1 \le \zeta \le 0$, $q_\zeta \le -m/2$ as long as $q > 0$. Positivity at zero proves (5.4). Finally, scalar order, $F_y > 0$, and the fixed target interval show that every intervening solution between two retained ones is retained. Thus the restrictions are intervals. □

Let $T_-(u_{\mathrm{up}})$ be the short map in the energy coordinate E_b, and $T_+(u_{\mathrm{up}})$ the map obtained by the contact passage and outgoing arc. Lemma A.2 gives

$$(T_-)_{u_{\mathrm{up}}} = -j(u_{\mathrm{up}}), \qquad c \le j \le C, \qquad |(\log j)_{u_{\mathrm{up}}}| \le C. \tag{5.9}$$

Their intersections correspond to captured periodic orbits; their isolated intersections are precisely the limit cycles being counted.

Actual-domain scope on the nonnegative side. For $a > 0$ the negative scaled strip may have a base root. The incoming product in Lemma 5.2 then cannot pass its first root: its base approaches only in infinite time while the normal deviation contracts. Such a field has no retained passage to $z = 0$, also at $d = 0$. On a no-root branch that lemma controls feedback on the whole returning interval. The outgoing and short-map constants can be fixed on fixed enlarged physical cores before shrinking ξ; regular energy holonomy has a uniform unit bound there. After these constants are fixed choose ξ, then r, and finally ε.

5.2. **Nonnegative forcing and a comparison criterion.** For nonnegative forcing, a weighted variational identity makes the complete contact derivative smaller than the short-map derivative.

For $d \ge 0$, the required outgoing estimate holds even when P has positive roots. Let $J = \partial_{q_0} q$ in (5.2) and $h = q_0 + d > 0$. On $0 \le \zeta \le 1$, direct integration gives $q(1) \ge ch$ and $0 < J(1) \le C$. For $\zeta \ge 1$ put

$$\mathcal{R} = hJ\frac{V(\zeta, q)}{q}.$$

The exact cancellation

$$(\log \mathcal{R})' = \frac{(V_\zeta)|_q - d/q}{V} \tag{5.10}$$

is obtained by differentiating the scalar and variational equations. The function $\Phi = \mathsf{B}q - \mathsf{C}q^2$ increases, since $q' \geq 0$ and $\mathsf{B} - 2\mathsf{C}q = F_y > 0$. The integral of $\mathsf{B}'q/V$ is bounded: outside a negative well of A, or when q is large, $q/V \leq C$; on the remaining bounded-height part use

$$\frac{\mathsf{B}'q}{V}\,d\zeta = \frac{\mathsf{B}'q}{\zeta q + d}\,dq \leq \frac{\mathsf{B}'}{\zeta}\,dq.$$

There $\zeta \geq c/\varepsilon$. To integrate $\mathsf{A}'/(\mathsf{A}+\Phi)$, pair the decreasing and increasing branches of the convex quadratic at equal values of A. The later denominator is larger, so each paired contribution is nonpositive, including when $\mathsf{A} \leq 0$. Only a final increase above $\mathsf{A}(1) > c$ remains. At a fixed section $z = \xi$ this gives

$$\mathcal{R}_{\mathrm{out}} \leq C(1 + \xi^2/r^2), \qquad (q_0 + d)\partial_{q_0} T_+ \leq C(\kappa^2 + \varepsilon^2\xi^2).$$

The factor κ^2 converts q to the physical ordinate. The regular continuation in the E_b coordinate has derivative tending to one after ξ is fixed. With (5.3),

$$|(T_+)_{u_{\mathrm{up}}}| \leq C(r^2 + \xi^2).$$

Choose ξ small and then r small so that the last bound is less than $j/2$. The displacement is strictly increasing and has at most one zero, necessarily simple, on its returning interval.

For negative d, we use Lemma 4.1 after identifying the physical separator and estimating the departure map.

5.3. **Changing the incoming reference preserves the physical separator.** Write $d = -m < 0$ and $u = u_{\mathrm{up}}$. Lemma 5.2 has already placed every counted cycle in the wedge $a < 0$ with m exponentially small. To estimate incoming distortion, we may replace the selected incoming reference y_c^o by the solution y_c^n whose value at a fixed negative point is zero. The outgoing separator is still the trace $\beta = y_\sigma(0)/\kappa^2$, with source distance $s = q_0 - \beta > 0$. We verify that this change of reference preserves the complete derivative comparison.

Both incoming solutions obey the barriers $-4m/(1-\zeta) \leq q_c^i(\zeta) < 0$, $i = o, n$, on their common terminal negative interval. For $\zeta \in [-1/8, 0]$, $1 \leq \mathsf{A} \leq 2$, $\mathsf{B} > 0$, and $1/2 \leq V(\zeta, q_c^i) \leq 2$ after shrinking. Moreover $\zeta q_c^i \leq m/2$. Their exact scalar equations therefore give $(q_c^i)_\zeta \leq -m/4$, hence

$$-4m \leq c_i := q_c^i(0) \leq -m/32.$$

The old incoming graph's initial value lies between these same barriers by its fixed-negative-point estimate in Lemma 5.2. No equality of the graphs is assumed.

For the same fixed field and physical incoming value put $w_i = q_0 - c_i = s + w_{\beta,i}$, $w_{\beta,i} = \beta - c_i$, and $\lambda_i = -\varepsilon^2 \partial_u \log w_i$. The endpoint bounds and $q_0 = \beta + s \geq 0$ give

$$\frac{1}{128} \leq \frac{w_o}{w_n} \leq 128, \qquad \lambda_n = \lambda_o \frac{w_o}{w_n}, \qquad \lambda_i w_i = -\varepsilon^2 s_u.$$

Differentiating the exact identities, with c_i independent of u, gives

$$(\log \lambda_n)_u = (\log \lambda_o)_u + \varepsilon^{-2}(\lambda_n - \lambda_o). \tag{5.11}$$

The feedback calculation (5.6)–(5.8) applies to the zero-initial reference as well: it has the same $O(m)$ barriers and fixed-foot, taper and funnel bounds. Its moving-entry value is $L_i = -H(\Lambda, K\varepsilon^3)\Lambda_u/(K\varepsilon^3 - y_c^n(\Lambda))$, including the entry motion. The taper integrals have denominators $N_0 - 2$ and $2N_0 - 4$, $N_0 \asymp \varepsilon^{-2}$, so no $\log(1/r)$ or reciprocal radial loss is introduced. Thus the incoming constants are fixed before the outgoing sections.

If $c_n^\lambda \leq \lambda_n \leq C_n^\lambda$ and $|(\log \lambda_n)_u| \leq L_n \varepsilon^{-2}$, the identities give

$$c_n^\lambda/128 \leq \lambda_o \leq 128 C_n^\lambda, \qquad |(\log \lambda_o)_u| \leq (L_n + 129 C_n^\lambda)\varepsilon^{-2}.$$

These bounds hold for arbitrarily small $s > 0$. The outgoing map $T_+(s)$, its derivative $M = \partial_s T_+$, the departure derivative N, and the collision amplitude $A(s)$ still use the same physical source $\kappa^2(\beta + s)$, selected outgoing graph, and target. The derivative ratio is therefore

$$R = \frac{M|s_u|}{j} = \frac{M\varepsilon^{-2}\lambda_o w_o}{j} = \frac{M\varepsilon^{-2}\lambda_n w_n}{j}.$$

Equation (5.11) also gives $(\log \lambda_n)_u - \varepsilon^{-2}\lambda_n = (\log \lambda_o)_u - \varepsilon^{-2}\lambda_o$. Thus both references give the same logarithmic derivative of R, including its large terms. The outgoing distance is always $W_{\mathrm{out}}(0) = \kappa^2 s$, while $\kappa^2 w_n$ is used only to estimate incoming distortion. The separator and finite-event comparison are therefore preserved.

5.4. **Departure from the slow region.** We next estimate the derivative from a moving departure point to the fixed outgoing target. The logarithmic derivative of this map, combined with the departure equation, will give the curvature required by Lemma 4.1.

Write $d = -m$ and $\nu = \kappa^3 m$. First suppose that

$$P(z) \asymp r^2 + z^2 \quad (z \geq 0), \qquad -2 + \delta_a \leq a < 0. \tag{5.12}$$

There is a unique positive solution y_σ of $y_z = H = (zy - \nu)/F$ on this side with

$$y_\sigma(z) = \frac{\nu}{(1+\varepsilon^2)z} + O(\nu z^{-2}) \quad (z \longrightarrow \infty). \tag{5.13}$$

To construct it, use the backward Green kernel

$$K(\zeta, t) = \exp\left(-\int_\zeta^t \frac{x}{\mathsf{A}(x)}\, dx\right).$$

Writing $\mathcal{N}(q) = (\zeta q - m)(V^{-1} - \mathsf{A}^{-1})$, the integral equation is $q(\zeta) = m\int_\zeta^\infty K(\zeta,t)/\mathsf{A}(t)\, dt - \int_\zeta^\infty K(\zeta,t)\mathcal{N}(q)(t)\, dt$. The comparison $\mathsf{A} \asymp 1 + \varepsilon^2\zeta^2$ gives $\int_\zeta^\infty K/\mathsf{A} \leq C/(1+\zeta)$, by splitting at 1 and ε^{-1}. On the ball $\sup(1+\zeta)|q| \leq Cm$, the second integral has Lipschitz norm $O(m)$ in this weighted space: use $|V - \mathsf{A}| \leq C(1+\kappa\zeta)|q| + \kappa^2 q^2$ in the same two ranges. Contraction therefore gives the solution. The coefficient $1/(1+\varepsilon^2)$ is the balance at infinity. A first zero would cross downward and could not lead to the positive terminal behavior. Likewise, at a zero of $zy_\sigma - \nu$ its derivative with respect to z is $y_\sigma > 0$. Hence $y_\sigma > 0$ and $zy_\sigma - \nu < 0$ throughout. Scalar order shows that the value $\beta = y_\sigma(0)/\kappa^2$ separates the outgoing solutions from those hitting $y = 0$. The threshold itself does not reach the outgoing fast part of the fixed section. For its size, write $\sigma(\zeta) = y_\sigma(\kappa\zeta)/\kappa^2$. The strict sign above gives $0 < \sigma(1) < m$. On $0 \leq \zeta \leq 1$, $\mathsf{A} \geq 3/4$ after shrinking ε. The graphs zero and $(3 - 2\zeta)m$ are reverse-time lower and upper barriers: at the upper graph the scalar right-hand side is at least $-2m$, its derivative is $-2m$, and the denominator stays above $1/2$ for small m. Therefore $0 < \beta \leq 3m$, and Lemma 5.2 gives $0 < w_\beta = \beta - q_c(0) \leq 7m$.

Fix two positive sections $\xi_{\mathrm{in}} < \xi_{\mathrm{out}}$. Choose their ratio sufficiently large, then their physical positions, and finally the parameter neighborhood. Retained cycles meet the fast part of the inner section and continue through the outer section to the common target. For each pair fixed in this order, we will bound the logarithmic derivative of the outgoing continuation uniformly in the final parameter neighborhood; its right-tail version will also apply near a double root. For a small fixed $\theta > 0$, let $N(z) > 0$ be the derivative with respect to vertical height of this continuation from

$$y = y_\sigma(z) + \theta S(z), \qquad S = \varepsilon^2 P.$$

The outgoing logarithmic derivative. The key estimate comes from differentiating the exact energy defect as the starting point moves along the departure curve. The first assertion below remains under assumption (5.12); the last assertion requires that comparison only on its stated right tail. In both cases the continuation starts at z_0 and ends at the common target, with the field fixed. This right-tail estimate is what the collision through passage will use.

Lemma 5.3. *Assume $d = -m < 0$ lies in the exponential wedge* (5.4) *supplied by Lemma 5.2. Fix a sufficiently small $\theta > 0$. Choose the ratio $R_\xi = \xi_{\mathrm{out}}/\xi_{\mathrm{in}} > 1$ first (large enough for the post-hit contraction below), then choose the two physical sections, and only afterward choose the parameter neighborhood. With these choices there is an $L > 0$ such that, for every admissible parameter in that neighborhood and every starting point*

$$z_0 \in [L\kappa, \xi_{\mathrm{in}}], \qquad y_0 = y_\sigma(z_0) + \theta S(z_0),$$

the complete continuation from the inner section through the outer section to the common target is defined and its vertical derivative $N(z_0) > 0$ satisfies

$$|z_0\, \partial_{z_0} \log N(z_0)| \le C, \qquad L\kappa \le z_0 \le \xi_{\mathrm{in}}. \tag{5.14}$$

Here ∂_{z_0} differentiates the dependence on the starting-section coordinate z_0 while the vector field, normalized parameters, section positions, and target section are held fixed. In the proof, derivatives in the logarithmic phase variable are likewise field-fixed; whenever $t = \log\lambda$ is used, the radial derivative is taken at fixed λ (fixed t), as indicated by the subscript in $(z\partial_z)_\lambda$. The same estimate holds with this starting interval replaced by $[z_*, \xi_{\mathrm{in}}]$ whenever $z_* \ge rZ_b$ and the interval lies on that right tail (in particular, take $z_* = rZ_b$ when nonempty), where Z_b ranges in a fixed compact interval lying to the right of all possible positive roots and (5.12) *is required only on that tail. The constant in this right-tail assertion depends on the fixed tail and section data, but not on a positive lower bound for P (or F) in the collision strip to the left of the tail.*

Proof. Boundary graph. We estimate the flight from its moving starting point through the fixed outer section to the common target; the two distinct physical sections leave room for departure and subsequent contraction. Put $x = \log z$, $Z = z/r$, and $y_\sigma = \nu v/z$. For

$$\rho = \varepsilon^2 \frac{p_a(Z)}{Z^2}, \qquad \omega = \frac{\nu v(1 + bz - \nu v/z)}{z^3}, \qquad w = 1 - v,$$

the boundary equation becomes

$$w_x = (1 + c)w - 1, \qquad c = (\rho + \omega)^{-1}.$$

Here $0 < v < 1$, $\rho \asymp \varepsilon^2(1 + Z^{-2})$ and the first two Euler derivatives of ρ are $O(\rho)$. Also $\nu/(zS)$ and its first two Euler derivatives are $O(m)$ for $Z \ge L\varepsilon$. Although $1+bz$ is unbounded at infinity, its product with $\nu/(zS)$ is still $O(m)$. The bounded solution has the exact representation

$$w(x) = \int_x^\infty \exp\left(-\int_x^t (1 + c(s, w(s)))\, ds\right) dt.$$

At fixed w, $|\partial_x \log c| \le C$ and $|\partial_w \log c| \le Cm$. Since initially $|w_x| \le 1 + c \le C\varepsilon^{-2}$, the same bound for $|d \log c/dx|$ holds along the solution by (5.4). Splitting the integral after a fixed short interval gives $|w| \le C/(1 + c)$. Differentiating the equation,

$$(w_x)_x = (1 + c + c_w w)w_x + c_x w,$$

the first coefficient is at least $1 + c/2$ and the last term is bounded. The same backward kernel gives $|w_x| \le C/(1 + c)$, and the equation then gives $|w_{xx}| \le C$. We have proved

$$\sum_{j=0}^{2} \frac{|(z\partial_z)^j y_\sigma|}{S} + \frac{\nu}{zS} \le Cm. \tag{5.15}$$

Energy defect before departure. Let $H_0 = z/(1 + bz - y)$. The fast first integral of Lemma A.1 gives the exact cancellation, for $J_z = H_y J$,

$$\begin{aligned} \frac{d}{dz} \log((E_b)_y J) = \mathcal{C}, \qquad \mathcal{C} = \left(\partial_y + \frac{(E_b)_{yy}}{(E_b)_y}\right)(H - H_0), \\ H - H_0 = -\frac{\nu h + zS}{Fh}, \qquad h = 1 + bz - y. \end{aligned} \tag{5.16}$$

We estimate the change of this integral when its starting point moves. Set $W = y - y_\sigma$, $R = W/S$, $\lambda = W/P$, and $D = (\log R)_z$. Write $y_0(z_0) = y_\sigma(z_0) + \theta S(z_0)$ and let $y(z; z_0)$ be the solution starting there. With $\ell_0 = \log z_0$, the derivative $\mathcal{V} = \partial_{\ell_0} y(z; z_0)$ at fixed z satisfies

$$\mathcal{V}_z = H_y \mathcal{V}, \qquad \mathcal{V}(z_0) = z_0(y_0' - H(z_0, y_0)) = -z_0 \theta S(z_0) D(z_0).$$

Choose $c_f < 1/4$ and then decrease ε_0 so that $\varepsilon^2 + c_f < 1/3$. Before the first hit $\lambda = c_f$, exact subtraction gives, with $A_* = -a \geq 0$,

$$\mathfrak{a}(z, W) = \frac{zF_\sigma - G_\sigma(1 + bz - 2y_\sigma - W)}{F_\sigma F_{\sigma+W}}, \qquad D = \mathfrak{a} - \frac{P'}{P},$$

where $F_\sigma = F(z, y_\sigma)$, $F_{\sigma+W} = F(z, y_\sigma + W)$, and $G_\sigma = zy_\sigma - \nu$. Equation (5.15), $z \geq Lr\varepsilon$, and $W = \lambda P$ imply, at fixed λ,

$$F_\sigma = S(1 + e_\sigma), \qquad F_{\sigma+W} = P(\varepsilon^2 + \lambda)(1 + e_+),$$

where

$$|e_\sigma| + |z\partial_z e_\sigma| \leq Cm, \qquad |e_+| + |z\partial_z e_+| \leq C(z + W + m).$$

The quotient of $G_\sigma(1 + bz - 2y_\sigma - W)$ by zF_σ, and its fixed-λ Euler derivative, are likewise $O(m)$. The corresponding principal expression is

$$D_0 = \frac{z}{P(\varepsilon^2 + \lambda)} - \frac{P'}{P}, \qquad zD_0 = \frac{Z\{A_* v_* + (1 - 2v_*)Z\}}{v_* p_a(Z)}, \qquad v_* = \varepsilon^2 + \lambda.$$

Since $v_* < 1/3$ and $A_* v_* + (1 - 2v_*)Z \geq A_* v_* + Z/3$, the principal difference cannot be lost through cancellation. Dividing the exact error by D_0 therefore yields

$$zD = \frac{Z\{A_* v_* + (1 - 2v_*)Z\}}{v_* p_a(Z)}(1 + e_1), \qquad v_* = \varepsilon^2 + \lambda, \quad |e_1| + |\partial_x e_1| \leq C(z + W + m). \tag{5.17}$$

The Euler derivative holds at fixed λ. In particular $D > 0$, $D \geq cz/[P(\varepsilon^2 + \lambda)]$, and $|z\partial_z \log D| \leq C$. Division of (5.16), using $\partial_y = S^{-1}\partial_R$, gives, for $j = 0, 1$ at fixed R,

$$\left|(z\partial_z)^j \frac{\mathcal{C}}{D}\right| \leq C((1 + R)^{-1} + S + m). \tag{5.18}$$

For completeness, $H - H_0 = -z(1 + R)^{-1}\Phi_0$. The function Φ_0 has bounded derivatives in each of its four arguments

$$(1 + R)^{-1}, \qquad h, \qquad y_\sigma/S, \qquad \nu/(zS).$$

Vertical differentiation therefore bounds $|\mathcal{C}|$ by $Cz(1 + R)^{-1}\{[S(1 + R)]^{-1} + 1\}$, while $D \geq cz/[S(1 + R)]$. Division gives $C\{(1 + R)^{-1} + S\}$; derivatives of the two boundary ratios contribute the displayed additional Cm. The same calculation after one fixed-R Euler derivative proves the second estimate in (5.18).

The moving departure point. To control the moving starting point, use $t = \log \lambda$. The endpoints are $t_0 = \log(\theta\varepsilon^2)$ and $t_f = \log c_f$, independent of z_0. The equation for x is $x_t = B = (zD)^{-1}$; thus $\mathcal{X} = \partial_{\ell_0} x$ satisfies $\mathcal{X}_t = B_x \mathcal{X}$, $\mathcal{X}(t_0) = 1$. Here B_x is taken at fixed t. With $e_1 = 0$,

$$B_0 = \frac{v_* p_a(Z)}{Z\{A_* v_* + (1 - 2v_*)Z\}}, \quad (B_0)_x = \frac{v_*\{A_* Z^2(1 - v_*) - A_* v_* - 2Z(1 - 2v_*)\}}{Z\{A_* v_* + (1 - 2v_*)Z\}^2}.$$

If the numerator is positive, $A_* Z \geq c$ and $((B_0)_x)_+ \leq C v_*$. Equation (5.17) thus implies

$$(B_x)_+ \leq C v_* + C(z + W + m)B.$$

Both terms have bounded integral in t. Indeed $\int v_* dt = \varepsilon^2 \log[c_f/(\theta\varepsilon^2)] + O(c_f)$; also integration of the lower bound for D gives $z_f \leq C(r + z_0)$, $z_f/z_0 \leq C/\varepsilon$ at the first hit $\lambda = c_f$. Since $y_z \leq Cz$ and $y_\sigma > 0$, $W(z) \leq C\varepsilon^2 P(z_0) + Cz^2$. In particular

$$\int_{z_0}^{z_f} \frac{z + W + m}{z}\, dz \leq C\{z_f + z_f^2 + [\varepsilon^2 P(z_0) + m]\log(C/\varepsilon)\} \leq C.$$

This uses (5.4), but no comparison of r with ε. Consequently $0 < \mathcal{X} \le C$. The hit occurs before $C_*(r + z_0)$: otherwise the integral of z/P would exceed the available finite integral of $(\varepsilon^2 + \lambda)d\lambda/\lambda$. More explicitly, $D = (\log \lambda)_z$ and the lower bound for D give

$$c \int_{z_0}^{z_f} \frac{z\, dz}{P(z)} \le \int_{\theta\varepsilon^2}^{c_f} \frac{\varepsilon^2 + \lambda}{\lambda}\, d\lambda \le C.$$

Since $P(z) \asymp r^2 + z^2$ on this side, the left member controls $\log[(r + z_f)/(r + z_0)]$, proving the asserted bound with a constant independent of both small parameters. Choosing $\xi_{\text{out}}/\xi_{\text{in}} > 2C_*$ and then shrinking r puts the hit before the outer section. For the energy defect before the hit, the physical-coordinate Leibniz formula is exactly

$$\partial_{\ell_0} \int_{z_0}^{z_f} \mathcal{C}(z, y(z; z_0))\, dz = \mathcal{C}_f \partial_{\ell_0} z_f - z_0 \mathcal{C}_0 + \int_{z_0}^{z_f} \mathcal{C}_y \mathcal{V}\, dz.$$

Changing to t incorporates both endpoint terms and the variational integral into

$$\partial_{\ell_0} \int_{t_0}^{t_f} \frac{\mathcal{C}}{D}\, dt = \int_{t_0}^{t_f} \mathcal{X}(z\partial_z)_\lambda \left(\frac{\mathcal{C}}{D}\right) dt.$$

Indeed, along a fixed t level, $\mathcal{V} = -zWD\mathcal{X}$. By (5.18), the absolute value of the last integral is at most

$$C \int_{t_0}^{t_f} [(1 + R)^{-1} + S + m] dt \le C \log(1 + \theta^{-1}) + C[\varepsilon^2(r^2 + \xi_{\text{out}}^2) + m] \log \frac{c_f}{\theta\varepsilon^2} \le C.$$

We have therefore proved

$$|\partial_{\log z_0} \log z_f| \le C, \qquad \left|\partial_{\log z_0} \int_{z_0}^{z_f} \mathcal{C}\, dz\right| \le C.$$

After departure. At this hit $z_f \ge c_r r$. Otherwise its height is at least $cc_f r^2$, whereas $y_z \le Cz$ bounds it by $C\varepsilon^2 r^2 + Cz_f^2$, a contradiction. Now set

$$\lambda = \frac{y}{P}, \qquad \pi = \frac{P}{z^2} = \frac{p_a(Z)}{Z^2}, \qquad \delta_Z = \frac{\nu}{z^3} = m\varepsilon^3 Z^{-3}, \qquad \tau = \log(z/z_f).$$

Its initial value is $\lambda(0) = c_f + y_\sigma(z_f)/P(z_f)$, whose Euler derivative is $O(m\varepsilon^2)$ by (5.15). Direct division of $y_z = (zy - \nu)/F$ gives the exact equation

$$\lambda_\tau = \frac{\lambda - \delta_Z/\pi}{\pi(\lambda h + \varepsilon^2)} - \lambda \frac{Z p_a'(Z)}{p_a(Z)}, \qquad h = 1 + bz - \lambda P. \tag{5.19}$$

We first control the transition from bounded Z to a stable strip. Fix $\lambda_- = c_f/2$ and $\lambda_+ = 3/4$, with the earlier choice $c_f < 1/4$. After shrinking the parameter neighborhood, the hit value satisfies

$$\lambda(0) = c_f + O(m\varepsilon^2) \in (\lambda_-, \lambda_+).$$

At this hit $G = zy - \nu > 0$, since $G = zP\lambda - \nu = zP(\lambda - \delta_Z/\pi)$ and $\delta_Z/\pi \ll c_f$. If G had a first zero afterward, then at that point $G_z = y + zy_z = y > 0$, a contradiction; hence $y_z = G/F > 0$ as long as the orbit stays in the outgoing phase box. A small fixed outer section closes the bootstrap $h \in [1/2, 3/2]$: $y_z \le 2z$ gives $y \le C(r^2 + z^2)$, hence $0 < \lambda \le B$. Decrease that section and then r to obtain $h \in [1 - \eta, 1 + \eta]$, with $\eta < 1/10$. On $Z \ge c_r$, both π and Zp_a'/p_a are bounded and π is a positive unit. Choose $\lambda_* < \min(\lambda_-, c_f/2)$ so small that (5.19) points upward there, and then require $\varepsilon^2 + \delta_Z/\pi \ll \lambda_*$. Thus $\lambda \in [\lambda_*, B]$. For a fixed sufficiently large Q, $\pi \to 1$ and $Zp_a'/p_a \to 2$ give

$$\lambda_\tau \ge c_1 > 0 \quad (\lambda_* \le \lambda \le \lambda_-), \qquad \lambda_\tau \le -c_1 < 0 \quad (\lambda_+ \le \lambda \le B), \qquad Z \ge Q.$$

The band $[\lambda_-, \lambda_+]$ is therefore invariant once $Z \ge Q$ and is reached within a bounded additional time. Let τ_Q be the first time both conditions hold, or the outer-section time τ_o if that occurs first. Its preceding interval has length at most $\max(0, \log(Q/c_r)) + B/c_1$ and common coefficient, defect and variation bounds on $\lambda \in [\lambda_*, B]$.

For the subsequent variation estimate, separate the large-Z equation into its limiting part and an integrable remainder. Put

$$f_\varepsilon(\lambda) = \frac{\lambda}{\lambda+\varepsilon^2} - 2\lambda,$$

and define $\mathcal{R}_1$ to be the difference between the right-hand side of (5.19) and f_ε. Equivalently,

$$\mathcal{R}_1 = \frac{\lambda - \delta_Z/\pi}{\pi(\lambda h+\varepsilon^2)} - \frac{\lambda}{\lambda+\varepsilon^2} - \lambda\left(\frac{Zp_a'}{p_a} - 2\right).$$

For the energy defect put $e_b = (E_b)_{yy}/(E_b)_y$. In these variables the exact expression in (5.16) is

$$\mathcal{K} := H - H_0 = -z\frac{\varepsilon^2\pi + \delta_Z h}{\pi h(\lambda h+\varepsilon^2)}, \qquad \mathcal{C} = (P^{-1}\partial_\lambda + e_b)\mathcal{K}.$$

Thus, with $\mathcal{E} = Z\partial_Z + z\partial_z$ acting at fixed λ, define $\mathcal{R}_2$ by

$$\lambda_\tau = f_\varepsilon(\lambda) + \mathcal{R}_1, \qquad z\mathcal{C} = \frac{\varepsilon^2}{(\lambda+\varepsilon^2)^2} + \mathcal{R}_2. \tag{5.20}$$

On $\lambda_- \le \lambda \le \lambda_+$ and $Z \ge Q$, the denominator factors are positive units. The three elementary estimates

$$\pi - 1 = O_{\mathcal{E}}(Z^{-1}), \qquad \frac{Zp_a'}{p_a} - 2 = O_{\mathcal{E}}(Z^{-1}), \qquad h - 1 = O_{\mathcal{E},\partial_\lambda}(z)$$

follow directly from the exact formulas

$$\pi - 1 = \frac{a}{Z} + \frac{1}{Z^2}, \qquad \frac{Zp_a'}{p_a} - 2 = -\frac{aZ+2}{p_a(Z)}, \qquad h - 1 = bz - \lambda\pi z^2, \qquad \frac{\delta_Z}{\pi} = \frac{m\varepsilon^3}{Zp_a(Z)}.$$

The function e_b and its required derivatives are bounded because $(E_b)_y$ is a positive analytic unit on the fixed energy rectangle. Substitution in the displayed exact formulas for $\mathcal{R}_1$ and $\mathcal{K}$, followed by the two operations $P^{-1}\partial_\lambda$ and multiplication by z, gives, including all mixed derivatives actually used,

$$\begin{aligned} &\sum_{i=0}^{1}\sum_{j=0}^{2} |\mathcal{E}^i\partial_\lambda^j\mathcal{R}_1| \le Cq_1, \quad q_1 = Z^{-1} + z + m\varepsilon^3 Z^{-3}, \\ &\sum_{i=0}^{1}\sum_{j=0}^{2} |\mathcal{E}^i\partial_\lambda^j\mathcal{R}_2| \le Cq_2, \quad q_2 = \varepsilon^2(Z^{-1}+z) + m\varepsilon^3 Z^{-3}. \end{aligned} \tag{5.21}$$

These bounds do not expand in $\varepsilon^2\log Z$. Increasing Q and decreasing the fixed outer section makes the full field inward pointing at the two fixed band boundaries and gives

$$f_\varepsilon' + \partial_\lambda\mathcal{R}_1 \le -c_0 < 0$$

throughout the interval.

At fixed τ and fixed field, differentiation in $\ell_f = \log z_f$ acts on the coefficients as $\mathcal{E}$. Thus $Z_1 = \partial_{\ell_f}\lambda(\tau; z_f)$ solves

$$(Z_1)_\tau = (f_\varepsilon' + \partial_\lambda\mathcal{R}_1)Z_1 + \mathcal{E}\mathcal{R}_1.$$

Its initial value is $O(m\varepsilon^2)$. With τ_Q defined and truncated as above, the preceding transition interval has bounded length and bounded variational coefficients, so $\sup_{[0,\tau_Q]}|Z_1| + \int_0^{\tau_Q}|Z_1|\,d\tau \le C$. On the remaining interval, when nonempty, variation of constants gives explicitly

$$|Z_1(\tau)| \le Ce^{-c_0(\tau-\tau_Q)} + C\int_{\tau_Q}^{\tau} e^{-c_0(\tau-s)}q_1(s)\,ds.$$

Moreover, with $\tau_o = \log(\xi_{\text{out}}/z_f)$,

$$\int_0^{\tau_o} q_1\,d\tau \le \int_{Z_f}^{\infty}\frac{dZ}{Z^2} + \int_{z_f}^{\xi_{\text{out}}} dz + m\varepsilon^3\int_{Z_f}^{\infty}\frac{dZ}{Z^4} \le C,$$

because $Z_f \ge c_r$. Fubini's theorem in the preceding stable-kernel formula now yields

$$\sup_{[0,\tau_o]} |Z_1| + \int_0^{\tau_o} |Z_1|\, d\tau \le C. \tag{5.22}$$

Put $\psi = z\mathcal{C}$. The full post-hit derivative, including the moving terminal point, is

$$\partial_{\ell_f} \int_0^{\tau_o} \psi\, d\tau = -\psi(\tau_o) + \int_0^{\tau_o} (\psi_\lambda Z_1 + \mathcal{E}\psi) d\tau.$$

On the stable interval, $\psi_\lambda = -2\varepsilon^2(\lambda+\varepsilon^2)^{-3} + \partial_\lambda \mathcal{R}_2$ and $\mathcal{E}\psi = \mathcal{E}\mathcal{R}_2$. Equations (5.21) and (5.22) bound the integral; its truncated pre-stable part has bounded length and smooth coefficients. A moving intermediate endpoint appears with opposite signs in the two parts and cancels. At the terminal section the normalized ordinate has derivative $Z_1(\tau_o) - \lambda_\tau(\tau_o)$, and hence

$$\partial_{\ell_f} y(\xi_{\mathrm{out}}; z_f) = P(\xi_{\mathrm{out}})[Z_1(\tau_o) - \lambda_\tau(\tau_o)].$$

Completion on the returning interval. Finally (5.16) gives the exact factorization

$$N(z_0) = (E_b)_y(z_0, y_0(z_0)) \exp\left(\int_{z_0}^{\xi_{\mathrm{out}}} \mathcal{C}\, dz\right) U_{\mathrm{reg}}(y(\xi_{\mathrm{out}}; z_0), p),$$

where $U_{\mathrm{reg}} > 0$ is the derivative of the remaining regular holonomy in the E_b coordinate. The entrance factor has bounded logarithmic $\log z_0$ derivative. The pre-hit calculation bounds both the defect derivative and $\partial_{\log z_0} \log z_f$; the post-hit calculation above bounds the remaining defect derivative and terminal ordinate derivative. For the already fixed pair of physical sections, U_{reg} is a positive C^2 unit after the final parameter shrinking. These bounds prove (5.14). Constants may depend on the fixed sections; L and the final parameter neighborhood are chosen afterward, so no reciprocal section size is absorbed into the small factor used below.

Every integral in this proof starts at the current point and proceeds rightwards, except the boundary kernel, which also uses only its right-hand tail. If $z \ge rZ_b$ lies beyond all roots, the comparisons and Euler estimates used above hold on that tail. In particular $p_a' \le 2Z$ and the positive sum in (5.17) are unchanged. No margin to the left of rZ_b is required. This proves the second assertion. □

The curvature follows from a moving departure section, not from a uniform bound for N itself. Put $W = y - y_\sigma$, $K = sW_s/W$. The divided coefficient relative to this boundary solution is

$$\mathfrak{a}(z, W) = \int_0^1 H_y(z, y_\sigma + tW)\, dt, \qquad W_z = \mathfrak{a}(z, W)W.$$

Here $W(0) = \kappa^2 s$, so $K(0) = 1$. Before the first hit $W = \theta S$, exact differentiation gives

$$(\log K)_z = W\mathfrak{a}_W, \quad D = \mathfrak{a} - P'/P \asymp \frac{z}{\varepsilon^2 P}, \quad |W\mathfrak{a}_W| + |W^2\mathfrak{a}_{WW}| \le C(W/S)D$$

on $z \ge L\kappa$. The bounded initial ζ-interval has the same integrated bound. In particular $K = 1 + O(\theta)$ before the hit. Differentiating its integral equation gives the exact total derivative

$$s\partial_s \log K_g = s(z_g)_s[W\mathfrak{a}_W]_g + \int_0^{z_g} K(W\mathfrak{a}_W + W^2\mathfrak{a}_{WW})\, dz.$$

The integral is $O(\theta)$ since $(W/S)_z = (W/S)D$, and the endpoint term has the same bound by the differentiated hitting equation. Thus

$$K_g = 1 + O(\theta), \quad |s(\log K_g)_s| \le C\theta, \quad (z_g)_s = -\frac{K_g}{sD_g}, \quad M(s) = N(z_g)\theta S(z_g)\frac{K_g}{s}. \tag{5.23}$$

The last identity uses the tangent to the first-intersection curve, $y_g' - H(z, y_g) = -\theta SD$ and therefore includes the change of hit time. Lemma 5.3 now gives

$$-s(\log M)_s = 1 - s(\log K_g)_s + \frac{K_g}{D_g}\left((\log N)_z + \frac{P'}{P}\right)_{z_g}.$$

Here $(z_g D_g)^{-1} \le C(\varepsilon^2 + \varepsilon/L + L^{-2})$. First choose θ small, then L large and ε small. The last display is at least $3/4$. An inner hit $z_g \le L\kappa$ implies $s \ge c_L\theta$ by the bounded inner difference equation, so it is absent when $s \le s_0$ for a sufficiently small fixed s_0.

To obtain the weighted bound, write $q = \sigma + \eta$, $J = q_s$, $D_0 = \mathsf{B} - \mathsf{C}(q+\sigma) > c$. Direct subtraction gives

$$\frac{d}{d\zeta}\log\frac{sJV(\zeta,q)}{\eta V(\zeta,\sigma)} = \frac{\eta}{V(\zeta,q)}(\mathsf{B}' - \mathsf{C}\sigma') - \frac{D_0\eta}{V(\zeta,q)}(\log V(\zeta,\sigma))'. \tag{5.24}$$

The first term has bounded integral; the coefficient of the second belongs to $(0,1)$. Under (5.12) the negative variation of $\log V(\zeta,\sigma)$ is bounded, because A stays positive and σ decreases. The initial quotient is a positive bounded unit by $q_0 \le Q_0$. At the outer section,

$$sJ \le CV(\zeta,\sigma)/D_0 \le C(1+\xi_{\mathrm{out}}^2/r^2), \qquad sM \le C\varepsilon^2(r^2+\xi_{\mathrm{out}}^2).$$

The signed-cancellation identity therefore gives the size bound on the whole actual returning interval I. Fix the incoming and short-map constants and choose a tolerance $\delta = \delta_* > 0$ satisfying both inequalities in (4.3). Then choose the two sections sufficiently small with their fixed ratio and shrink r to make the size coefficient below δ_*. The short-map constants can be fixed in advance as explained in Lemma 4.2. After these sections fix L and s_0, use $w_\beta \le 7m$ and the exponential wedge (5.4) to shrink ε until (4.4) holds, so that $R \ge 1/2$ implies $s \le C\delta w_\beta < s_0$. Lemma 4.1 now applies with both sections fixed before s_0. It bounds the total isolated multiplicity by two in this separated root-free case.

5.5. **Two separated positive roots.** For separated slow roots, we use an exact saddle passage and estimate its complete complement. The comparison then counts every retained return of a fixed field on the common physical source interval.

Suppose $-M \le a \le -2 - \delta_a$. Put

$$A = |a|r, \quad c = a^{-2}, \quad n = Ac, \quad e = \varepsilon/\sqrt{c}, \qquad X = z/n, \quad Y = y/(n^2e^2).$$

Here c lies in a compact subinterval of $(0,1/4)$, and

$$X' = e^2(YQ + 1 - X + cX^2), \qquad Y' = XY - em, \qquad Q = 1 + bnX - n^2e^2Y. \tag{5.25}$$

Write $p_c(X) = 1 - X + cX^2$. Its smaller reference root X_s is a positive unit, $p_c'(X_s) = -\sqrt{1-4c}$ is a negative unit, and the other root X_2 is separated from it. The actual equilibria instead satisfy

$$Y = em/X, \qquad Q_e(X) := p_c(X) + \frac{em(1+bnX)}{X} - \frac{n^2e^4m^2}{X^2} = 0.$$

On this compact c-interval the implicit function theorem gives $X_i^{\mathrm{eq}} = X_i + O(em)$, with common finite phase bounds. Their separation is at least half the reference gap after shrinking, and $Q_e'(X_s^{\mathrm{eq}})$ remains a negative unit. All saddle-centred coordinates and eigenvalues below refer to this actual equilibrium. Fix a small collar width σ below one tenth of their minimum gap. Apply the graph construction in Lemma D.5 on this fixed saddle collar. The normal eigenvalue is a positive unit, all base derivatives contain e^2, and $p_c'(X_s)$ stays away from zero. Thus Lemma 3.1 and its simple-root scalar change give, at fixed incoming and outgoing sections, the exact C^j Dulac map

$$D(s) = \xi_0(s/\omega_0)^{\rho_s}, \qquad \rho_s = e^2\beta_s, \quad 0 < c_s \le \beta_s \le C_s. \tag{5.26}$$

The fixed spectral and root gaps give this adaptation of the small-c construction uniformly on the present compact c-interval.

The smallness of c used near a simple contact must here be replaced by smallness of A. For $x = -z \ge Ln$,

$$P(-x) = x^2 + Ax + A^2c \asymp x(x+A), \qquad T_A = \log\frac{x_g + A}{A(1+Lc)} = \log(1/A) + O(1).$$

On the upper barrier, the negative graph derivative satisfies $H_y \le -\gamma/[\varepsilon^2(x+A)]$. Its integrated contraction is at most $-\gamma T_A/(ce^2)$. By making A small, this absorbs any fixed loss C/e^2 from a bounded scaled interval. It absorbs polynomial scaling losses too: for every fixed j,

$$A^{-j}\left(\frac{A(1+Lc)}{x_g+A}\right)^{\gamma/(ce^2)}$$

is increasing on $0 < A \le A_0$ once e is sufficiently small. There is no lower bound on A. Since $d < 0$, the incoming scaled height is consequently less than e^{-B/e^2} for any fixed B, after A_0 is chosen. It decreases on the negative half and grows by at most e^{C_σ/e^2} up to $X_s - \sigma$. Thus every positive returning trajectory reaches the saddle section below its fixed positive normal outlet; an earlier hit of $Y = 0$ is terminal. There is no early bypass in this case. Only $s > 0$ in (5.26) reaches the positive outlet: $s = 0$ lies on the stable separatrix, whereas for $s < 0$ the expanding negative normal deviation crosses $Y = 0$ while the separatrix height remains $O(em)$.

The outgoing trajectory must still pass the second-root region. From the positive saddle outlet use Y as phase:

$$\frac{dX}{dY} = \frac{e^2(YQ + p_c(X))}{XY - em}, \qquad \frac{d(X^2)}{dY} \le Ce^2(1 + 1/Y) + Cce^2X^2/Y.$$

These inequalities bound X up to $Y = e^{-2}$. Before a fixed large Y_0, its change is $O(e^2(1 + |\log Y_0|))$, so it stays to the left of X_2. Choose Y_0 above four times the bound for X; then $YQ + p_c \ge Y/4$. A first zero of this speed is impossible: at such a zero its Y-derivative is $1 + bnX - 2n^2e^2Y > 0$. Thus the orbit reaches $y = n^2$ with z/n in a fixed positive interval. The invariant-interval argument for y/z^2 in (5.38) below gives its positive parabolic continuation, independently of either polynomial root.

Let C_+ be this whole outgoing map, and N_- the remaining fast arc, decreasing-height entrance, negative passage and incoming section conversion. The proof of Lemmas D.3 and D.1, with the replacements just given, yields

$$C'_+ > 0, \quad C'_+ \le Cn^{3/2}, \quad |C''_+/C'_+| \le C, \qquad N'_- > 0, \quad |N''_-/N'_-| \le C\varepsilon^{-2}.$$

Indeed the positive Y-phase derivatives through order three have majorant $Ce^2(1 + 1/Y)$; the later relative-normal equation is unchanged. On the negative side the first and second variations have a polynomial scaling loss times $e^{-\gamma T_A/(ce^2)}$, followed by at most e^{C/e^2}. The displayed negative action absorbs both. Hence, for the full complement $H = N_- \circ C_+$,

$$\rho_s|H''/H'| \le Ce^2 + CA^{3/2}\sqrt{c} = o(1).$$

Lemma D.5 now gives total isolated multiplicity at most two for $D \circ H$ on its complete return interval.

5.6. **The positive double root.** Near the positive double root, we need coordinates that remain uniform as the slow roots coalesce. We first select the boundary graph and control its phase derivatives, then straighten its normal flow. The resulting product retains the full passage multiplier for the weighted comparison.

It remains to treat a in a fixed small neighborhood of -2. In $z = rZ$, $y = r^2\varepsilon^2 Y$, put $w = Y - \varepsilon m/Z$ and rescale time by the positive factor r; this changes neither trajectories nor return maps. In this rescaled time variable,

$$\dot{Z} = \varepsilon^2 F_s(Z, w), \qquad \dot{w} = Zw + \frac{\varepsilon^3 m}{Z^2}F_s(Z, w), \qquad F_s = Q_\varepsilon + A_\varepsilon w - r^2\varepsilon^2 w^2, \tag{5.27}$$

where

$$Q_\varepsilon = p_a(Z) + \frac{\varepsilon m(1 + brZ)}{Z} - \frac{r^2\varepsilon^4 m^2}{Z^2}, \qquad A_\varepsilon = 1 + brZ - \frac{2r^2\varepsilon^3 m}{Z}.$$

Choose fixed bases $Z_- < 1 - 2\delta < 1 + 2\delta < Z_+$ in the positive half-plane, with fixed buffers on either side. All possible roots of Q_ε in this strip lie strictly between the bases. Write $Z_b = Z_+$ for the right exterior base below. Lemma 3.1 requires a chosen invariant graph with finite phase

bounds. On the no-root side we must obtain those bounds for the solution selected by the physical boundary condition, even when the slow speed has an arbitrarily small minimum.

Lemma 5.4 (No-root tail and collision continuation)**.** *Fix a finite phase order k (at least $2j+11$ for the subsequent full normal-product output of order $j+2$) and choose fixed numbers*

$$0 < Z_- < Z_b < Z_1 < Z_2$$

and a fixed $\delta_0 > 0$ with $Z_\ell := Z_- - \delta_0 > 0$, so that Q_ε is a positive unit on $[Z_\ell, Z_-]$, every possible root in the collision strip lies in (Z_-, Z_b), and $P(rZ)/(r^2Z^2)$ is a positive unit for $Z \geq Z_b$. There are constants $q_* > 0$, $K_k > 0$, and one parameter shrinking, independent of the minimum of Q_ε, such that on the no-root side

$$Q_\varepsilon > 0 \quad \text{on } [Z_\ell, Z_2], \qquad Q_\varepsilon \geq q_* \quad \text{on } [Z_b, Z_2],$$

the physical boundary solution y_σ from (5.13) induces a unique graph $w = h(Z)$ on $[Z_\ell, Z_1]$ and

$$-K_k\varepsilon^3 m \leq h(Z) < 0, \qquad \max_{0\leq i\leq k} \|\partial_Z^i h\|_{L^\infty[Z_\ell,Z_1]} \leq K_k\varepsilon^3 m.$$

At $m = 0$ the graph is $h \equiv 0$. No constant in this statement uses a positive lower bound for Q_ε on $[Z_-, Z_b]$.

Proof. The case $m = 0$ is immediate, so assume $m > 0$. Put $\tau = \varepsilon^2$, $n = \varepsilon^3 m$, and $\nu = r^3 n$. On the physical right tail $z \geq rZ_b$ consider

$$U(z) = \frac{\nu}{z}, \qquad L(z) = (1 - M\varepsilon^2)\frac{\nu}{z},$$

where M is fixed large before ε is decreased. Since $P(rZ)/(r^2Z^2)$ is a positive unit there, the strip $L \leq y \leq U$ satisfies, uniformly in r/ε,

$$c\varepsilon^2 z^2 \leq y(1 + bz - y) + \varepsilon^2 P(z) \leq C\varepsilon^2 z^2.$$

For the scalar equation $y_z = H := (zy - \nu)/F_{\rm ph}$, the reverse-flow boundary expression $-H + J'$ is negative at $J = U$ and positive at $J = L$ once $M > 2C$. Thus the strip is reverse invariant. Terminal solutions at $z = T$ with $y(T) = \nu/((1 + \varepsilon^2)T)$ remain in the strip; on it

$$(H)_y = \frac{z(y^2 + \varepsilon^2 P) + \nu(1 + bz - 2y)}{F_{\rm ph}^2} \geq \frac{c}{\varepsilon^2 z}.$$

The scalar mean-value equation therefore gives, for $T' > T$ and fixed $z < T$,

$$|y_{T'}(z) - y_T(z)| \leq \frac{\nu}{T}(z/T)^{c/\varepsilon^2}.$$

The terminal solutions converge on every compact tail interval to a unique positive solution. To identify its boundary condition, set $\ell = \log z$ and $v = zy/\nu$. The strip bounds v, and, with the field fixed, $F_{\rm ph}/z^2 = \varepsilon^2 + O(e^{-\ell})$ gives

$$v_\ell = (1 + \varepsilon^{-2})v - \varepsilon^{-2} + O(e^{-\ell}).$$

Variation of constants for the bounded solution yields $v = (1 + \varepsilon^2)^{-1} + O(e^{-\ell})$, so this limit is y_σ in (5.13). On the fixed right buffer $F_{\rm ph} \geq c\varepsilon^2 z^2$, so for each fixed finite k the differentiated scalar variational equations are regular and give C^k convergence with the finite-order majorants recorded below. Hence the terminal jets of y_σ exist and graph differentiation below is justified. Its pullback to the collision coordinates satisfies

$$h(Z) = \frac{y_\sigma(rZ)}{r^2\varepsilon^2} - \frac{\varepsilon m}{Z}, \qquad -Kn/Z \leq h(Z) < 0$$

on the right buffer. Enlarge K by the fixed factor $1/Z_\ell$ if necessary, so that this bound also implies $h \geq -Kn$ on the collision rectangle below.

It remains to continue the graph through the possibly small slow speed. In reverse time $s = -t$, the horizontal boundaries of the strip

$$Z_\ell \le Z \le Z_2, \qquad -Kn \le w \le 0$$

point inward while $F_s > 0$, and the base coordinate decreases towards Z_ℓ. Shrink the fixed tube once so that $F_w \ge c_* > 0$ there. At $w = 0$ one has $w_s = -nF_s/Z^2 < 0$, while at $w = -Kn$ one has $w_s \ge n(KZ_\ell - C/Z_\ell^2) > 0$. On $F_s = 0$ with $w < 0$,

$$(F_s)_s = -F_w Z w > 0.$$

The corner $F_s = w = 0$ is impossible because there $F_s = Q_\varepsilon > 0$. Thus the tail remains in the rectangle and $Z_s = -\tau F_s < 0$. If it did not reach Z_ℓ in finite reverse time, compactness and monotonicity would give a nonempty invariant limit set on $F_s = 0$; the equation for w then forces $w = 0$, contradicting $F_s = Q_\varepsilon > 0$. This proves continuation as a graph without any bound on the crossing time.

On the right buffer, where $Q_\varepsilon \ge q_*$ and $|h| \le Kn/Z_b$, shrink the parameter box once more so that

$$|A_\varepsilon h| + r^2\varepsilon^2 h^2 \le q_*/2.$$

Consequently $F_s \ge q_*/2$ throughout this buffer. The graph equation

$$\tau F_s h_Z = Zh + \frac{n}{Z^2} F_s$$

then gives the crude terminal bounds

$$|h^{(i)}(Z)| \le C_i n \tau^{-i} \qquad (Z \in [Z_1, Z_2],\ 0 \le i \le k):$$

the case $i = 1$ follows directly, and differentiating the equation successively gives the induction step because $F_s \ge q_*/2$ and all coefficients on the fixed buffer are bounded. Since F_s is also uniformly bounded above there, a reverse traversal of any fixed subinterval of the right buffer takes at least a fixed positive multiple of τ^{-1}. Fix the required finite order $k \ge 1$ and partition this buffer by

$$\zeta_j = Z_2 - \frac{j}{k}(Z_2 - Z_1), \qquad 0 \le j \le k.$$

Every subbuffer $[\zeta_j, \zeta_{j-1}]$ has reverse crossing time at least c_k/τ. For $p = h_Z$, differentiation and conversion to reverse time give

$$p_s = -\left(Z + \frac{nF_w}{Z^2} - \tau F_Z - \tau F_w p\right) p - h - n\left(\frac{F_Z}{Z^2} - \frac{2F}{Z^3}\right).$$

On the first subbuffer the crude bound gives $\tau F_w p = O(n)$. Thus the coefficient in parentheses is $Z + O(n+\tau) \ge Z_\ell/2$ after shrinking. Variation of constants across this subbuffer, whose crossing time is at least c_k/τ, gives $|p(\zeta_1)| \le K_1 n$. The interval $|p| \le K_1 n$ is then inward pointing in reverse time, so the good first-jet bound holds on $[Z_\ell, \zeta_1]$.

For $i \ge 2$, differentiate the graph equation i times and collect the highest derivative. With $p_i = h^{(i)}$ one obtains

$$\tau F_s (p_i)_Z = L_i p_i + R_i, \qquad L_i = Z + \frac{nF_w}{Z^2} - \tau\{iF_Z + (i+1)F_w h_Z\}.$$

Every term in R_i contains n, h, or a lower jet. Inductively, suppose the jets through order $i-1$ have their good bounds on $[Z_\ell, \zeta_{i-1}]$. Then $|R_i| \le C_i n$ throughout the next subbuffer $[\zeta_i, \zeta_{i-1}]$ and $L_i \ge Z_\ell/2$. At its right endpoint the crude estimate is still $|p_i| \le C_i n \tau^{-i}$. Variation of constants in reverse time over this subbuffer gives

$$|p_i(\zeta_i)| \le C_i n \tau^{-i} e^{-c_k Z_\ell/(2\tau)} + C_i' n \le K_i n.$$

The reverse barriers $|p_i| \le K_i n$ propagate this bound on $[Z_\ell, \zeta_i]$. Thus each jet order uses its own subbuffer after the lower-jet estimates have become available there. After finitely many steps all jets through order k have the claimed bounds on $[Z_\ell, Z_1]$. This proves the stated C^k estimate without dividing by the vanishing minimum of Q_ε. □

On the rooted side, normal hyperbolicity supplies a graph across the strip. For $m > 0$, dividing the normal variable by $\varepsilon^3 m$ retains the forcing factor in its phase bounds; the incoming separatrix selects the part needed by the return.

Lemma 5.5. *On the no-root side, the tail solution* (5.13) *continues through the fixed extended strip* $[Z_\ell, Z_1]$ *as a graph* $w = h(Z)$. *On the rooted side there is such an invariant graph whose incoming restriction is the stable separatrix of the first simple root, or the incoming parabolic separatrix at a double root. For every prescribed finite* k, *these graphs satisfy* $\|h\|_{C^k} \le K_k \varepsilon^3 m$, *uniformly as the roots coalesce. The bounds concern phase derivatives; joint parameter differentiability across the discriminant is not asserted.*

Proof. Put

$$\tau = \varepsilon^2, \qquad \alpha = \varepsilon m, \qquad n = \varepsilon^3 m = \tau\alpha, \qquad \varepsilon^4 m^2 = \tau\alpha^2.$$

The last identity is needed in the coefficient of Q_ε. Write also $\nu_s := n$ and abbreviate $F := F_s$. For $n > 0$ write $w = n\psi$. The exact auxiliary system is

$$\dot{Z} = \tau F(Z, n\psi), \qquad \dot{\psi} = Z\psi + Z^{-2} F(Z, n\psi), \tag{5.28}$$

where

$$Q_\varepsilon = p_a(Z) + \frac{\alpha(1+brZ)}{Z} - \frac{r^2\tau\alpha^2}{Z^2}, \qquad A_\varepsilon = 1 + brZ - \frac{2r^2\tau\alpha}{Z}.$$

Because $n = \tau\alpha$, (5.28) extends smoothly to $n = 0$. Freeze (α, a, b, r) in a compact box and let τ tend to zero. At $\tau = 0$ its critical graph is

$$\psi_0(Z) = -\frac{Q_0(Z)}{Z^3}, \qquad Q_0(Z) = p_a(Z) + \frac{\alpha(1+brZ)}{Z}.$$

With $u = \psi - \psi_0$, the equations on the working strip are exactly

$$\dot{Z} = \tau F_\tau, \qquad \dot{u} = Zu + Z^{-2}(F_\tau - Q_0) - \tau\psi_0' F_\tau, \qquad F_\tau = F(Z, n(\psi_0 + u)). \tag{5.29}$$

Thus their C^N distance from $(\dot{Z}, \dot{u}) = (0, Zu)$ is $O_N(\tau)$ on one fixed tube, with no inverse power of n and no root-separation constant.

Choose a fixed cutoff equal to one on a larger collision strip, extend ψ_0 by a fixed extension operator, and close the base into a circle. Choose a positive function a_0, equal to Z on the working strip and bounded below by a positive constant on the circle, and extend (5.29) there as

$$\dot{\theta} = \chi\tau F_\tau, \qquad \dot{u} = a_0 u + \chi\{Z^{-2}(F_\tau - Q_0) - \tau\psi_0' F_\tau\}.$$

At $\tau = 0$ the invariant circle $u = 0$ has zero tangential rate and a fixed normal gap. Fix the desired phase order k first and choose the source order at least $k+1$. Fenichel's finite-order persistence theorem [14, Theorem 3, p. 215] gives an invariant C^{k+1} graph $u = \phi$ close to zero. The common phase bounds are obtained from its invariance equation. Write the extended field as $\dot{\theta} = \mathcal{A}(\theta, u)$ and $\dot{u} = \mathcal{B}(\theta, u)$; then $\mathcal{A} = O_{C^{k+1}}(\tau)$ and $\mathcal{B} = a_0 u + O_{C^{k+1}}(\tau)$. At a maximum of $|\phi|$, the equation $\mathcal{A}(\theta, \phi)\phi' = \mathcal{B}(\theta, \phi)$ gives $\|\phi\|_{C^0} \le C\tau$. Differentiate successively through order k and evaluate at a maximum of each derivative's absolute value. The next derivative vanishes there; the coefficient of the highest retained derivative is $\mathcal{B}_u + O_k(\tau) \ge \min a_0/2$ after shrinking. The first step uses the close graph's bounded first derivative, and all later steps use the preceding bounds. The remaining terms are $O_k(\tau)$, so induction gives $\|\phi\|_{C^k} \le C_k\tau$. Hence $\psi = \psi_0 + \phi$ has a common C^k bound after one common shrinking of τ. Reconstructing $h = n\psi$ yields

$$\|h\|_{C^k} \le K_k n = K_k \varepsilon^3 m. \tag{5.30}$$

For $n = 0$ take $h = 0$, which solves the original equation exactly. The circle, the finite order, and K_k are chosen before the parameter shrinking, so (5.30) remains uniform when $\varepsilon^3 m$ is smaller than every exponential in $1/\varepsilon$. Restricting the graph back to the original strip removes all dependence on the auxiliary extension from the equations used below.

For any such graph, invariance gives the exact factorization

$$Zh = (\varepsilon^2 h_Z - \nu_s/Z^2)F(Z,h), \qquad F(Z,h) = UQ_\varepsilon,$$
$$U = \left(1 - \frac{(A_\varepsilon - r^2\varepsilon^2 h)(\varepsilon^2 h_Z - \nu_s/Z^2)}{Z}\right)^{-1} > 0. \tag{5.31}$$

In particular it contains the equilibria at the roots of Q_ε. Its incoming orbit approaches the first root in positive time. Every nonzero displacement from the graph grows at a uniform positive normal rate while it stays in the tube. Hence this incoming orbit is precisely the stable separatrix at a simple first root, and the incoming parabolic separatrix at a double root. This restriction is independent of the auxiliary extension. The rest of the graph is used only to define coordinates on the strip.

On the no-root side, apply Lemma 5.4 with the same right buffer and collision bases. It continues the physical tail (5.13) with its sign and prescribed phase jets, without dividing by the minimum of Q_ε. The factorization (5.31) therefore holds for this graph, which is the one used in the through passage. On the rooted side, the incoming stable or parabolic separatrix is intrinsic; the remaining weak graph may depend on the circle extension and serves only to define product coordinates. It need not agree with a separately prescribed outgoing tail. The rooted fibre-exit estimates use its finite phase jets and the same incoming threshold. □

On an incoming branch the selected graph has the exact formula

$$h(Z) = -\varepsilon m \int_Z^{Z_T} \exp\left(-\int_Z^t \frac{x}{\varepsilon^2 F_s(x)}\,dx\right)\frac{dt}{t^2}, \tag{5.32}$$

where Z_T is the first root or infinity and $F_s(x)$ is evaluated on the selected graph. The kernel is strictly between zero and one, so $-\varepsilon m/Z < h < 0$ there. The positive trace $Y_\sigma = h + \varepsilon m/Z$ pulls back uniquely to $q_0 = \beta > 0$. On the no-root side it is exactly the same trajectory as (5.13), not another choice of slow manifold. Hence the two definitions of β agree on their overlap. On the outgoing side of this threshold, the physical displacement is $W = y - y_\sigma > 0$, with $W(0;s) = \kappa^2 s$. On the collision strip, this is $W = \kappa^2(w - h)$, not the nullcline offset w.

Translate this graph by $u = w - h(Z)$. The base equation is $\dot{Z} = \varepsilon^2 v(Z) + \varepsilon^2 c(Z,u)u$, and the divided normal coefficient is

$$B(Z,u) = Z + \left(\frac{\varepsilon^3 m}{Z^2} - \varepsilon^2 h_Z\right) c(Z,u) \geq c_* > 0.$$

Fix nested compact intervals

$$[Z_-, Z_b] \Subset I_0 \Subset I_1 \Subset I_2 \Subset (Z_\ell, Z_1).$$

They and the physical bases Z_-, Z_b are fixed before the finite-order parameter shrinking. The selected graph and its phase jets are already defined on the larger interval $[Z_\ell, Z_1]$, so the product is taken over I_0 with both physical bases in its interior. Lemma 5.5 supplies the finite jet bounds required in Lemma 3.1, with $\tau = \rho = \varepsilon^2$. Fix the full output order $J = j + 2$ first; the selected graph is constructed once with $k \geq 2j + 11$, leaving C^{2j+10} coefficients after h_Z. The product is then taken at order J, with its section losses already reserved. The lower order $2j + 9$ sufficient for the fibre comparison alone is not used for this full normal construction. Its exact coordinates give

$$\dot{\theta} = \varepsilon^2 v(\theta), \qquad \dot{\eta} = b_n(\theta)\eta, \qquad b_n \geq c > 0. \tag{5.33}$$

The normal coordinate is tangent to the identity and both coordinate maps have uniformly bounded finite derivatives. At either exterior base v is a positive unit. The first section consequence of that lemma gives conversion to a fixed Z-section in time $O(\eta)$, with bounded derivatives, after cancelling the common ε^2 in the hitting equation. No such cancellation at a base root is needed.

To apply one return comparison, we must first place every counted cycle in the same small source interval before the collision. The next lemma obtains this from contraction along the preceding negative segment.

Lemma 5.6 (Localization of the collision source)**.** *Fix the collision coefficient interval, the exterior bases with their buffers, and a forcing bound $m_0 > 0$, with $0 < m \le m_0$. For every fixed $B > 0$ there are $r_0, \varepsilon_0 > 0$ such that every counted cycle satisfies*

$$0 < Y(Z_-) \le e^{-B/\varepsilon^2} \qquad (0 < r < r_0,\ 0 < \varepsilon < \varepsilon_0). \tag{5.34}$$

For a fixed sufficiently small fiber height η_0, all these cycles belong to one open interval I_{ret} of actual physical returning sources on which the canonical left-base amplitude A is defined and $0 < A < \eta_0$. They lie in its smaller part $A < \eta_0/2$.

Proof. Choose a fixed large L and keep the left exterior base and its buffer strictly to the left of the collision roots. After decreasing the fixed coefficient interval, $p_a \ge p_* > 0$ on $[-L, Z_-]$. The constants in the negative-entry estimate of Lemma 5.2 are common on this coefficient box and the fixed range $0 < m \le m_0$; thus

$$y(-Lr) \le K\varepsilon^3 (Lr/x_g)^{\gamma/\varepsilon^2}.$$

Here $K, \gamma, x_g > 0$ and all sections are fixed before r and ε are decreased. In the physical scaled variables the exact orbit equation is

$$Y_Z = \frac{ZY - \varepsilon m}{\varepsilon^2\{p_a(Z) + (1 + brZ - r^2\varepsilon^2 Y)Y\}}.$$

In a provisional fixed tube $0 < Y < h_*$ the denominator in braces is at least $p_*/2$. The height decreases for $Z \le 0$, and for $0 \le Z \le Z_-$ it satisfies $Y_Z \le CY/\varepsilon^2$. Consequently, before any first exit from the tube,

$$Y(Z) \le K\varepsilon r^{-2} \exp\left(-\frac{\gamma \log(x_g/(Lr)) - C_+}{\varepsilon^2}\right) \qquad (-L \le Z \le Z_-),$$

where C_+ is fixed. First choose $r_0 < x_g/L$ so that $\gamma \log(x_g/(Lr_0)) > C_+ + B + 1$. For $N = \gamma/\varepsilon^2 > 2$, the function $r^{-2}(Lr/x_g)^N$ increases with r. Hence the displayed bound is at most

$$K\varepsilon r_0^{-2} e^{-(B+1)/\varepsilon^2} \le e^{-B/\varepsilon^2}$$

after decreasing ε_0. Decrease it also so that this bound is less than $h_*/2$. This puts the entrance inside the tube and excludes a first upward exit. A counted cycle cannot hit $Y = 0$, by Lemma A.2. Since $\dot{Z} \ge \varepsilon^2 p_*/2$ in the rescaled time, it reaches Z_- in finite physical time at most $2(L + Z_-)/(r\varepsilon^2 p_*)$. This proves (5.34) without a relation between r and ε.

We now restrict the actual source interval. Let I_{phys} be the positive contact-passage interval supplied by Lemma A.2, with the prescribed upper source core and common outgoing target core. Every such passage has $F > 0$, so its physical height $Y_-(s)$ at $Z = Z_-$ is defined. The scalar variational equation gives $\partial_s Y_- > 0$ for the incoming height coordinate $s = q_0 - \beta$ and each fixed field. The selected graph has physical trace $Y_\sigma(Z_-) > 0$, and its pullback to $z = 0$ is $q_0 = \beta$. Thus $Y_- - Y_\sigma(Z_-) > 0$ is equivalent to $s > 0$. On the no-root side the selected tail stays below the outgoing fast core, and scalar order excludes sources at or below it. On the root side the selected incoming graph tends to the first equilibrium. The absolute value of a negative normal difference grows at a positive rate by (5.33), whereas the graph height is $O(\varepsilon m)$; such a positive passage hits $Y = 0$ before leaving the fixed normal tube. The equality source remains on the selected graph. Hence a counted cycle must have $s > 0$.

Use the same selected graph, exterior base, and buffers as in (5.33). The change from the physical section $Z = Z_-$ to the canonical section $\theta = \theta_-$ gives a function ϕ_p of the normal difference $u = Y_- - Y_\sigma(Z_-)$, with

$$A = \phi_p(u), \qquad \phi_p(0) = 0, \qquad c \le \phi_p' \le C$$

on one common interval about zero. Indeed, $u = \eta + \eta^2 L_2$ there; the phase shear is $O(\varepsilon^2\eta)$ and the exterior base speed is ε^2 times a positive unit. The first section consequence of Lemma 3.1 gives an $O(\eta)$ flight with bounded derivatives. It preserves the selected graph, so its normal multiplier is a positive unit. Choose η_0 small enough to lie in the common positive image of ϕ_p, and put $u_*(p) = \phi_p^{-1}(\eta_0)$. Define

$$I_{\mathrm{ret}} = \{s \in I_{\mathrm{phys}} : 0 < Y_-(s) - Y_\sigma(Z_-) < u_*(p)\}. \tag{5.35}$$

This is an open interval, possibly empty, because Y_- is strictly increasing; the target restriction is already included in I_{phys}. On it $0 < A < \eta_0$ and $A_s > 0$. Every counted cycle is on the outgoing side of the selected boundary and satisfies

$$0 < A \le C(Y_- - Y_\sigma(Z_-)) \le CY_- \le Ce^{-B/\varepsilon^2} < \eta_0/2$$

after one further decrease of ε_0. Thus all its source values lie inside I_{ret}. The artificial boundary $A = \eta_0$ contains no counted zero. The boundary $A = 0$ is the selected invariant trajectory and does not reach the outgoing fast core. □

On this actual source interval the difference equation before the left base, where p_a has a fixed positive margin, gives

$$K_A := sA_s/A = 1 + O(\eta_0), \qquad |s(\log K_A)_s| \le C\eta_0. \tag{5.36}$$

To see both bounds, use this positive displacement and its divided coefficient $\mathfrak{a}$. Integrate $(\log(sW_s/W))_z = W\mathfrak{a}_W$ and its $\log s$ derivative; their majorants are $C\eta\, d\log\eta$, whose integral is $O(\eta_0)$. The final amplitude coordinate is tangent to the identity, so it preserves these estimates.

A fiber exit near a root still has to reach the regular outgoing arc. We first follow the entire positive fiber to the height $y = r^2$, with bounds uniform as the roots coalesce. Only then does the parabolic coordinate y/z^2 enter a fixed contracting interval.

Lemma 5.7 (Finite fibre-to-parabolic transition)**.** *Fix a sufficiently small $\eta_0 > 0$ and a compact interval $I_Z = [Z_-, Z_+] \Subset (0,\infty)$ containing the collision bases. Let $\eta = \eta_0$ denote the fibre in the exact product coordinates* (5.33)*. Write $x = Z$ on its physical image and $\Theta(x)$ for the corresponding product phase. The physical fibre graph is*

$$w_0(x) = h(x) + \eta_0 + \eta_0^2 L_2(\Theta(x), \eta_0), \qquad \frac{\eta_0}{2} \le w_0(x) \le \frac{3\eta_0}{2},$$

after decreasing η_0 once, with $\|w_0 - \eta_0\|_{C^3} \le C(\eta_0^2 + \varepsilon^3 m)$. There are a coefficient neighbourhood $|a+2| < \delta_a$, a number $r_0 > 0$, and $\varepsilon_0 > 0$ such that, for

$$|a+2| < \delta_a, \quad 0 < r < r_0, \quad 0 < \varepsilon < \varepsilon_0, \quad 0 \le m \le m_0,$$

every solution of the collision equations starting on this fibre with $x = Z \in I_Z$ has a unique graph representation $Z = Z(Y)$ up to $Y = \varepsilon^{-2}$. On this interval

$$Y_{\mathrm{in}}(x) = w_0(x) + \frac{\varepsilon m}{x}, \qquad c_0 \le Z(Y) \le C_0, \qquad F_s(Z, Y - \varepsilon m/Z) \ge c_0\eta_0, \qquad ZY - \varepsilon m \ge c_0 Y.$$

The terminal graph $Z_t(x)$, parameterised by its initial fibre coordinate $x = Z(Y_{\mathrm{in}}(x))$, satisfies

$$0 < c_1 \le (Z_t)_x \le C_1, \qquad |\partial_x^j Z_t| \le C_1 \quad (j = 2, 3), \qquad c_1 \le Z_t \le C_1,$$

with constants independent of r, ε, m and of the distance from a to the double-root value.

Proof. Write Y for the phase and put

$$H = 1 + brZ - r^2\varepsilon^2 Y, \qquad N = YH + p_a(Z), \qquad D = ZY - \varepsilon m.$$

On the fibre, $w = Y - \varepsilon m/Z = w_0(Z)$. Since

$$p_a(Z) = (Z-1)^2 + (a+2)Z,$$

we may choose δ_a and then r_0 so that, uniformly on the fixed bootstrap strip, $H \geq 3/4$, $A_\varepsilon \geq 3/4$, and $F_y = 1 + brZ - 2r^2\varepsilon^2 Y \geq 1/2$ on the whole phase strip. Choose a fixed $B > Z_+$ with $B^2 - Z_+^2 > 3$. The same identity allows us to decrease δ_a once more so that $p_a(Z) \geq -\eta_0/8$ on the enlarged bootstrap interval $[Z_-/2, B]$. Then the exact cancellation after substitution of $w = Y - \varepsilon m/Z$ gives

$$F_s = N = YH + p_a(Z) \geq Y/2 \qquad (Y \geq \eta_0/2).$$

Choose ε_0 so that $\varepsilon m_0 \leq Z_-\eta_0/8$. Then $D = Zw_0(Z) \geq Z\eta_0/2$ at the initial fibre. In the region $Z \in [Z_-/2, B]$, $Y \in [\eta_0/2, \varepsilon^{-2}]$, and $D \geq ZY/2$, direct substitution in the collision equations gives

$$\frac{dZ}{dY} = f(Y,Z) := \varepsilon^2 \frac{N}{D}, \qquad \dot{Z} = \varepsilon^2 F_s, \qquad \mathcal{G} := D = ZY - \varepsilon m, \qquad G_{\text{original}} = r^3\varepsilon^2\mathcal{G}.$$

The preceding choices give $F_s = N \geq Y/2$ and $D = ZY - \varepsilon m \geq ZY/2$ on this region, so $Z_Y > 0$ and both physical speeds are positive.

For $0 \leq k \leq 3$, the rational equation has

$$|\partial_Z^k f(Y,Z)| \leq C_k \varepsilon^2 (1 + Y^{-1}),$$

with constants depending only on the fixed strip, coefficient box and η_0. Moreover

$$(Z^2)_Y = 2\varepsilon^2 \frac{H + p_a(Z)/Y}{1 - \varepsilon m/(ZY)} \leq \frac{2\varepsilon^2(1 + |b|rB)}{1 - \chi_m} + \frac{2P_B\varepsilon^2}{(1-\chi_m)Y}, \qquad \chi_m := \frac{4\varepsilon m_0}{Z_-\eta_0} < \frac{1}{2},$$

where $P_B = \sup_{|a+2|<\delta_a,\, Z\in[Z_-/2,B]} |p_a(Z)|$. Choose a margin $\sigma > 0$ with $2 + \sigma < B^2 - Z_+^2$. First decrease r_0 so that the contribution of $|b|rB$ to the first numerator is at most $\sigma/4$, and then decrease ε_0 so that χ_m is small enough and the logarithmic term below is at most $\sigma/2$. Before these final choices, integration up to $Y = \varepsilon^{-2}$ gives

$$Z^2(Y) \leq Z_+^2 + \frac{2(1 + |b|r_0B)}{1 - \chi_m} + \frac{2P_B\varepsilon^2}{1 - \chi_m} \log \frac{2\varepsilon^{-2}}{\eta_0}.$$

Choose first r_0 and then ε_0 so that the last bound is at most $Z_+^2 + 2 + \sigma < B^2$. This closes the upper boundary; $Z_Y > 0$ closes the lower one, and $D \geq c_0 Y$ follows from $\varepsilon m_0 \ll Z_-\eta_0$. This proves existence through $Y = \varepsilon^{-2}$.

It remains to control the fibre dependence. The initial graph is $Y_{\text{in}}(x) = w_0(x) + \varepsilon m/x$, so its first three derivatives are uniformly bounded on I_Z and $Y'_{\text{in}} = O(\eta_0^2 + \varepsilon^3 m + \varepsilon m)$. At fixed Y, the first variation obeys

$$V_Y = f_Z V, \qquad V(Y_{\text{in}}(x)) = 1 - f(Y_{\text{in}}(x), x) Y'_{\text{in}}(x).$$

The initial value lies in $[1/2, 3/2]$, while

$$\int_{\eta_0/2}^{\varepsilon^{-2}} \varepsilon^2 \left(1 + \frac{1}{Y}\right) dY \leq C.$$

Thus $c_1 \leq V \leq C_1$. The second and third differentiated equations have the same linear coefficient f_Z and forcing $f_{ZZ}V^2$ and $3f_{ZZ}VV^{(2)} + f_{ZZZ}V^3$, respectively. The displayed integrable majorant and bounded initial jets therefore give $|V^{(2)}| + |V^{(3)}| \leq C_1$. The terminal bounds follow. □

The terminal graph from the finite transition supplies the initial conditions for parabolic contraction. The next lemma continues the whole fiber to the target and controls its first two derivatives.

Lemma 5.8. *After η_0 is fixed, the coefficient neighborhood of $a = -2$ can be chosen so that every point of the entire fixed fiber $\eta = \eta_0$, $\theta_- \leq \theta \leq \theta_+$, reaches the same regular outgoing target. In the coordinate $x = Z = z/r$ restricted to this fiber, its target map Φ satisfies*

$$0 < |\Phi_x| \leq C_\xi r^\alpha, \qquad |\Phi_{xx}/\Phi_x| \leq C, \qquad \alpha > 0. \tag{5.37}$$

The target section ξ is fixed before shrinking r, ε.

Proof. Choose the coefficient interval and small parameters as in Lemma 5.7. Lemma 5.7, with fiber coordinate $x = Z$, gives $F, G > 0$, a fixed compact positive Z-interval, and three terminal derivatives at $Y = \varepsilon^{-2}$, uniformly up to the double root. Its phase equation is

$$Z_Y = \varepsilon^2 \frac{Y(1 + brZ - r^2\varepsilon^2 Y) + p_a(Z)}{ZY - \varepsilon m}.$$

The terminal bounds are therefore

$$z_t = rZ_t(x), \quad y_t = r^2, \qquad c \le (Z_t)_x \le C, \quad |\partial_x^j Z_t| \le C \ (j = 2, 3), \quad c \le Z_t \le C.$$

From there put $u_f = y/z^2$, $\ell = \log z$, $\pi = P/z^2$, and $\delta_z = \nu/z^3$. The exact equation is

$$(u_f)_\ell = \Psi(\ell, u_f) = \frac{u_f - \delta_z}{u_f(1 + bz - u_f z^2) + \varepsilon^2 \pi} - 2u_f. \tag{5.38}$$

The terminal height $Y = \varepsilon^{-2}$ gives $y_t = r^2$ and $u_f(\ell_t) = 1/Z_t^2$, uniformly separated from zero. This transition from the original fiber height $y = O(r^2\varepsilon^2)$ puts the orbit in the region where the parabolic contraction applies. For $z \ge cr$, the first two Euler derivatives of π are bounded; roots of P occur only in the additive term $\varepsilon^2\pi$. Choose a fixed interval $I = [u_-, u_+]$ containing all $1/Z_t^2$, with $0 < u_- < 1/2 < u_+$. After first fixing I and then choosing the target ξ small, the faces of I point inward, since uniformly on I

$$\Psi = \frac{1}{1 + bz - u_f z^2} + O(\varepsilon^2 + \varepsilon^3) - 2u_f.$$

Writing $D = u_f(1 + bz - u_f z^2) + \varepsilon^2\pi$ and retaining the u_f-dependence of its denominator gives the exact identity

$$\Psi_{u_f} = \frac{\varepsilon^2\pi + \delta_z(1 + bz - 2u_f z^2) + u_f^2 z^2}{D^2} - 2.$$

Here $1 + bz - u_f z^2 \ge c > 0$ and $D \ge c > 0$ on the forward invariant collar. Since π is bounded and $\delta_z = O(\varepsilon^3)$ there, we may shrink ξ and then the parameters so that $\Psi_{u_f} \le -c < 0$ uniformly. All derivatives needed through order three are bounded. Thus the orbit reaches $z = \xi$ with $F \ge cy > 0$.

If J is the vertical variation of (5.38), then $0 < J(\ell_\xi, \ell_t) \le (z_t/\xi)^\alpha$. The moving initial point must also be included. Its transverse factor is

$$(u_t)_x - \Psi(\ell_t, u_t)(\ell_t)_x = -\frac{u_t - \delta_{z_t}}{u_t(1 + bz_t - u_t z_t^2) + \varepsilon^2\pi(z_t)} \frac{(Z_t)_x}{Z_t}.$$

It is a negative unit with bounded two derivatives. The formula for the initial-time derivative of the flow, and $J_\ell = \Psi_{u_f} J$, therefore give the first inequality in (5.37). Logarithmic differentiation of the variational exponential has an endpoint term bounded by $|\Psi_{u_f}(\ell_t, u_t)|$ and an integral with factor $\Psi_{u_f u_f} J$; its absolute integral is bounded by the stable kernel. The first logarithmic derivative of the transverse factor is also bounded. This proves the second inequality. The final fixed holonomy in energy coordinates preserves both estimates. □

On the interval which exits through $\eta = \eta_0$, write $x = Z = z/r$ at this hit. The change from the canonical θ to x on that fixed fiber is a C^3 diffeomorphism with positive unit derivative. Write $\widetilde{v}$ for v multiplied by this derivative and $\widetilde{b}_n$ for b_n composed with the inverse change; both and their first derivatives remain bounded. Let x_- be the image of θ_- under this change. The backward incoming amplitude is

$$\mathcal{D}(x) = \eta_0 \exp\left(-\int_{x_-}^{x} \frac{\widetilde{b}_n(t)}{\varepsilon^2 \widetilde{v}(t)} \, dt \right).$$

The base speed is positive along the fibre-exit branch before its first root. Thus $\mathcal{D}$ decreases in the physical coordinate $x = Z$, and differentiation of $A(s) = \mathcal{D}(x(s))$ gives

$$|sx_s| = \varepsilon^2 |\widetilde{v}| K_A / \widetilde{b}_n, \qquad M = |\Phi_x| \varepsilon^2 |\widetilde{v}| K_A / (\widetilde{b}_n s).$$

By Lemma 5.8, $sM \le C_\xi \varepsilon^2 r^\alpha$ uniformly on the whole fiber-exit returning interval. Moreover

$$\begin{aligned} -s(\log M)_s &= 1 - s(\log K_A)_s - s x_s \partial_x \log|\widetilde{v}\Phi_x/\widetilde{b}_n| \\ &\ge 1 - C\eta_0 - C\varepsilon^2 \ge 1/2. \end{aligned}$$

There is no division by a root separation in this estimate: $s x_s\, \widetilde{v}_x/\widetilde{v} = O(\varepsilon^2 \widetilde{v}_x)$. With the target fixed, shrink r so that $C_\xi r^\alpha$ is below the tolerance required by Lemma 4.1. These are the size and curvature bounds for the fiber-exit piece in the common comparison below, uniformly up to the double root.

5.7. **Passage through the double-root region without a fiber exit.** We now treat the through passages across the collision strip and join them to the fiber exits on the localized source interval of Lemma 5.6.

On the no-root side, sufficiently small amplitudes cross the whole strip before reaching the fiber. Between the fixed bases, the exact product (5.33) gives

$$B = m_b A(s), \qquad m_b = \exp\left(\int_{\theta_-}^{\theta_+} \frac{b_n(t)}{\varepsilon^2 v(t)}\, dt\right). \tag{5.39}$$

The multiplier is finite for each such field, but has no uniform upper bound. Passage from θ_- to θ_+ occurs for $0 < A < \eta_0/m_b$. We keep (5.39) in the complete map. Figure 2 places this interval beside the fiber-exit interval. Their possible common boundary is included in the same complete return map.

At the right base $z_b = rZ_b$ all possible roots are to the left and $P \asymp r^2 + z^2$, $P' > 0$ for $z \ge z_b$. The canonical graph is the same positive decreasing solution y_σ constructed from the right-hand boundary condition. The change from the canonical base to $z = z_b$ is

$$t = \frac{y - y_\sigma}{\kappa^2} = \Psi_b(B), \quad \Psi_b(0) = 0, \quad c \le \Psi_b' \le C, \quad |\Psi_b''/\Psi_b'| + |\Psi_b'''/\Psi_b'| \le C.$$

The strong-fiber displacement is $O(\varepsilon^2 B)$ and the exterior base speed is ε^2 times a positive unit. Dividing the implicit hit equation by ε^2 proves these bounds without inverting the long passage.

Start (5.24) at z_b with initial difference t. Its initial quotient is a bounded positive unit. On this right tail $\mathsf{A} = P/r^2$ increases and the total variation of $\mathsf{B}\sigma - \mathsf{C}\sigma^2$ is bounded by $C\sigma(z_b)$. Hence its negative logarithmic variation is bounded independently of the earlier polynomial minimum. For the complete target derivative M_t this gives

$$tM_t \le C\varepsilon^2(r^2 + \xi_{\text{out}}^2).$$

Take η_0 sufficiently small so that the initial ratio W/S is less than $\theta/2$. Retained cycles have that ratio larger than θ at ξ_{in}, so they first meet $W = \theta S$ before the inner section. Lemma 5.3 applies on this exterior tail, with room to complete the flight before ξ_{out}. In (5.23), now with initial variable t, the last logarithmic term is bounded by $C/(z_g D_g) \le C\varepsilon^2$, because $z_g \ge rZ_b$. It follows that

$$-t(\log M_t)_t \ge 3/4.$$

This is why the two physical sections cannot be identified: a departure point arbitrarily close to its final target need not have a bounded logarithmic normal derivative.

Since $B\Psi_b'/\Psi_b = 1 + O(B)$, conversion to B yields

$$BM_B \le C\varepsilon^2(r^2 + \xi_{\text{out}}^2), \qquad -B(\log M_B)_B \ge 2/3.$$

For the complete map from s, $M = M_B(m_b A) m_b A_s$; the field is fixed when s is differentiated. Thus, exactly,

$$\begin{aligned} sM &= (BM_B)K_A, \\ -s(\log M)_s &= \{-B(\log M_B)_B\}K_A - (K_A - 1) - s(\log K_A)_s \ge 1/2. \end{aligned}$$

The unbounded multiplier has canceled. Choosing the fixed sections small and then r, ε small gives both estimates in (4.1) throughout $0 < A < \eta_0/m_b$. We now combine them with the fiber-exit estimates on the same returning source interval.

For a fixed field, the returning source has the following partition.

Lemma 5.9 (Source partition)**.** *On the actual source interval I_{ret} of Lemma 5.6, A is a diffeomorphism from the source coordinate s onto an interval $\mathcal{A}_{\mathrm{actual}} \subset (0, \eta_0)$. On the no-root side the ordered open pieces are*

$$\begin{aligned} \mathcal{I}_{\mathrm{through}} &= A^{-1}(\mathcal{A}_{\mathrm{actual}} \cap (0, \eta_0/m_b)) , \\ \mathcal{I}_{\mathrm{fiber}} &= A^{-1}(\mathcal{A}_{\mathrm{actual}} \cap (\eta_0/m_b, \eta_0)) . \end{aligned} \tag{5.40}$$

Either piece may be empty. There is at most one finite event value, $A = \eta_0/m_b$, when it belongs to the actual source interval. On the root side every source in I_{ret} exits through the fiber and there is no internal event value. All counted cycles are in this interval, and its artificial boundary $A = \eta_0$ contributes no counted zero.

Proof. The construction (5.35) and the positive section conversion give $A_s > 0$; (5.36) gives its quantitative form. On the no-root side the product equation (5.39) has $m_b > 1$, so the only change of exit is $m_b A = \eta_0$. Pulling back the two amplitude intervals gives (5.40). At equality the trajectory meets the right base and fiber at the same point, in finite time. If the threshold is outside the actual amplitude image, it is omitted. The target-core preimage remains an interval by scalar order and $F_y > 0$, as in Lemma A.2.

On the root side the scalar base tends to its first root in infinite time, whereas $\dot{\eta} = b_n \eta$, $b_n \geq c > 0$, takes every $A > 0$ to η_0 in finite time. Hence only the fiber exit occurs. The selected boundary $A = 0$ is excluded from the returning interval. Finally, Lemma 5.6 places every counted cycle strictly below $A = \eta_0/2$, so the upper artificial boundary introduces no counted event zero. The ordering in Figure 2 is increasing s, or equivalently increasing A; it is opposite to the original u_{up}-order because $s_{u_{\mathrm{up}}} < 0$. □

The complete return across the finite junction. On I_{ret}, write $X(u)$ for the upper-section point and ϕ_p for the fixed-field flow. The target hit $\pi_z \phi_p(t, X(u)) = \xi$ is transverse, with $F_p(\xi, y) \geq c_\xi > 0$. It defines one C^2 target map $\widehat{T}_+(u) = E_b(\pi_y \phi_p(\tau(u), X(u)))$. At $A = \eta_0/m_b$ the orbit meets the right base and fiber at their common point in finite time; there $v > 0$, $b_n \geq c > 0$ and $A_s > 0$. Thus the two formulas are restrictions of this same map, and the physical section diffeomorphisms give $M = \partial_s \widehat{T}_+ > 0$. Its first two derivatives agree at the event, which is counted once. A base root has infinite flight time and is never such a junction. These are the flow and continuity hypotheses of Lemma 4.2; on the root side there is only fiber exit.

The through and fiber estimates hold on their whole respective returning intervals. Fix the common positive curvature lower bound and the incoming and short-map constants first, and choose $\delta_* > 0$ satisfying both inequalities in (4.3) with these common bounds. The short map constants can be fixed before the outgoing section: its limiting energy map is independent of that section, and for each fixed section a parameter shrinking bounds its C^2 error. Likewise the through size constant is uniform in the fixed small energy rectangle; after fixing the section, its final regular holonomy derivative is bounded by two. Choose the fixed section ratio, then ξ_{out} sufficiently small, and then r so that the through coefficient $C(r^2 + \xi_{\mathrm{out}}^2)$ is below δ_*. With this target fixed, decrease r further so that the fiber coefficient $C_\xi r^\alpha$ is below the same tolerance. Their curvature lower constants have a common positive minimum, independently of m_b and the root gap. The localization lemma only decreases the final positive thresholds; it changes none of these physical sections or coefficient intervals.

Taking the maximum size constant and minimum curvature constant, the two passage estimates therefore give

$$sM \leq \varepsilon^2 \delta, \qquad -s(\log M)_s \geq c_M > 0$$

on both open pieces. The incoming and short-map estimates also hold on I_{ret}. These are precisely the hypotheses of Lemma 4.2; the continuity just proved includes the possible junction.

Its total actual multiplicity bound two therefore holds on this interval, which contains every counted cycle.

Completion of Proposition 5.1. For negative forcing, cover the coefficient set by the three intervals $a \geq -2 + \delta_a$, $a \leq -2 - \delta_a$, and $|a + 2| < 2\delta_a$. For each fixed field choose one applicable interval. In either of the first two, the preceding complete-return comparison gives the bound two. In the third, Lemma 5.6 places every counted cycle inside one actual source interval with $A < \eta_0/2$. On the root side only the fiber exit remains. On the no-root side, Lemma 5.9 gives the through and fiber pieces and at most one finite junction. Lemma 4.2 therefore gives total multiplicity at most two on this common interval.

Every comparison counts the complete return from the single signed source section to the common target. Fix the collision fiber before shrinking its coefficient interval; this also fixes the gaps in the separated-root regimes. Then fix the physical sections before the final parameter bounds. A compact coefficient set requires only finitely many such choices, whose minima give common positive thresholds r_0, ε_0 and a common positive Hausdorff radius. Together with the nonnegative-forcing comparison on the entire passing interval, this proves the proposition. □

6. Two-endpoint degeneration and the connecting return

Suppose the limiting polynomial has a simple zero at each end of the segment of equilibria. For nonnegative forcing, the long stay near the left endpoint makes the complete-period multiplier smaller than one; the attracting-annulus argument then bounds the number of cycles by two. For negative forcing, the lower passage produces strong contraction and a negative displacement. The intervening passage can amplify derivatives, especially when the positive saddle has a small characteristic ratio. We retain the contraction and signed displacement through this passage and use the flat source of the resulting return map to bound its zeros.

Proposition 6.1. *Suppose*

$$P_*(z) = C_* z^2 - A_* z, \qquad C_* < 0, \quad A_* > 0, \qquad C_* f_{b_*} = A_*.$$

Write the normalized forcing as $\delta = \varepsilon^3 D$. There are neighborhoods of b_* and $(P_*, D = 0)$, and positive constants ε_0, ρ, such that the normalized family has total actual minimal-return multiplicity at most two for isolated cycles at Hausdorff distance less than ρ from Γ_0^b for $0 < \varepsilon < \varepsilon_0$. For every field in this neighborhood with $D \geq 0$, there are at most two cycles and each counted cycle is hyperbolic attracting; for $D < 0$ there are at most two.

Throughout this section we write the nearby family exactly as

$$F = y(1 + bz - y) + \varepsilon^2 P(z), \qquad G = zy + \delta, \qquad P(z) = Cz^2 - Az + A^2 c, \tag{6.1}$$

where $0 < A_0 \leq A \leq A_1$, C belongs to a compact negative interval, c is small, and $\delta = \varepsilon^3 D$. All constants are uniform on a sufficiently small compact neighborhood of the limiting direction. The sections in Lemma A.2 will be kept fixed. In particular, every cycle under consideration lies in $y > 0$ and has a unique intersection with the same outgoing section $z = z_c > 0$.

For $D < 0$, Lemma 6.2 writes the lower passage as a strong power followed by a contracting signed map. Lemma 6.3 constructs outgoing orbits of the same field; their continuation to the physical section controls the source derivatives in the return map. Proposition 6.4 then continues the full signed power-output interval to the positive saddle. It preserves the negative baseline (6.43) and uses the lower contraction to absorb intermediate growth, giving the weighted bounds (6.31). These are the inputs to Lemma 4.3.

Figure 3 records this composition. There η_0 is the fixed incoming-cut height and $R > 1$ is the lower-saddle power exponent. The coordinate a is the outgoing stable amplitude at the positive saddle, $q = C_+(a)$ is its coordinate on the common physical section, and $u(a) = u(C_+(a), p)$ is the corresponding lower-saddle source coordinate. The quantity $t = \eta_0 u(a)^R$ is the strong power output, Ψ is the physical connector with value $H(a)$, and $\mathcal{D}$ is the positive-saddle Dulac map compared with a.

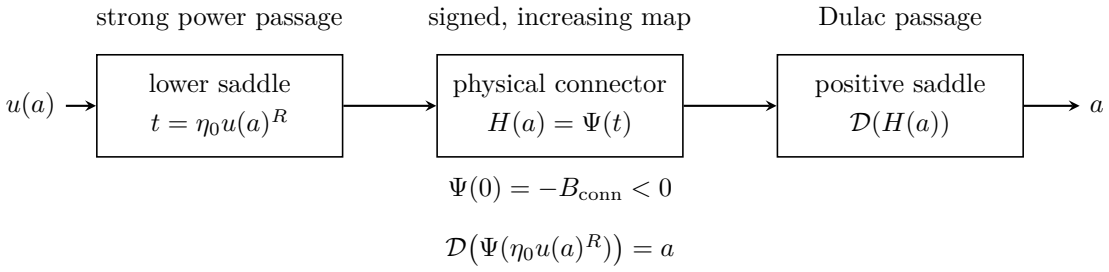


FIGURE 3. Schematic composition of the complete negative-forcing two-endpoint return. The arrows show the order of scalar maps, not phase-plane trajectories. The full signed connector Ψ, with $\Psi(0) = -B_{\mathrm{conn}} < 0$, is retained in $H(a) = \Psi(\eta_0 u(a)^R)$. The positive-saddle Dulac map is used on the returning domain $H(a) > 0$.

6.1. **Nonnegative forcing.** We prove attraction by estimating the complete-period multiplier. The annulus count then gives the bound two.

Let $\mathfrak{m}$ be the characteristic multiplier of a positive cycle. Since $z = (\log y)^{\cdot} - \delta/y$, integration of the divergence over a complete period gives the exact identity

$$\log \mathfrak{m} = \oint \left[by - \varepsilon^2 A - (1 + 2\varepsilon^2 C)\frac{\delta}{y} \right] dt. \tag{6.2}$$

For $\delta \geq 0$ the integrand is at most $by - \varepsilon^2 A$. The time spent near the left zero will dominate all positive contributions. We estimate this time along the orbit, allowing reversals in z.

Choose fixed small $q, z_c, h_c > 0$ such that the contact rectangle $[-q, z_c] \times (0, h_c)$ satisfies

$$1 + bz - y \geq \tfrac{1}{2}, \qquad F_y > 0, \qquad P(-q) > p_q > 0, \qquad |P(z)| \leq A/(8b_+), \tag{6.3}$$

where b_+ is an upper bound for b. The last two inequalities hold after shrinking the coefficient neighborhood. The regular fast arc and the descending passage to $y = K\varepsilon^3$ have uniformly bounded positive contribution to (6.2). Indeed, on a fixed negative interval, $G \leq -c_0 y$ for $y \geq K\varepsilon^3$, so a descent from a fixed height y_h contributes at most $(b_+/c_0)\int_{K\varepsilon^3}^{y_h} dy$. The moving-entry estimate (A.6) gives $\Lambda = \ell_b + O(\varepsilon^2(1 + |\log \varepsilon|))$, where ℓ_b is the regular fast-foot map.

Choose $H \geq K$ with $qH > \sup D$. Until the first subsequent hit of $z = -q$, the boundary $y = H\varepsilon^3$ points downward. Thus the positive cycle has $y \leq H\varepsilon^3$ on this entire part of its orbit, and the integrand in (6.2) is at most $-A_0\varepsilon^2/2$. This hit exists, since the same periodic orbit must return to its upper fast section.

Put $f_* = f_{b_*}$ and fix $a > 0$ with $f_* + a < -q$. For any sufficiently small fixed $\tau > 0$, shrink the coefficients and the fast source interval so that $|\Lambda - f_*| \leq \tau$ and

$$|P(z)| \leq M(z - f_* + \tau) \quad (f_* + 2\tau \leq z \leq f_* + a).$$

Here M can be chosen before τ, since $P_*(f_*) = 0$ and the polynomial derivatives are bounded. Along the low orbit, $|F| \leq M\varepsilon^2(z - f_* + \tau + \varepsilon)$ on this strip. To bound the crossing time even when z reverses, use the function which equals

$$\log \frac{z - f_* + \tau + \varepsilon}{3\tau + \varepsilon}$$

on the strip and is constant on each complementary half-line. Its absolute time derivative is at most $M\varepsilon^2$. Its net change between the descending entry and the first $-q$ hit is $\log[(a + \tau + \varepsilon)/(3\tau + \varepsilon)]$. Consequently the contribution of this whole low part is at most

$$-\frac{A_0}{2M} \log \frac{a}{4\tau}, \qquad \varepsilon \leq \tau. \tag{6.4}$$

Leftward excursions do not affect this estimate: the function is constant outside the strip, and its time derivative was estimated in absolute value.

At the first $-q$ hit the orbit enters the contact rectangle with $F > 0$. Its boundary clearances, positivity and $F_y > 0$ give the prescribed right exit with $F > 0$ throughout, by the invariant-quadrant argument in the proof of Lemma A.2.

Below $y_{\mathrm{th}} = \varepsilon^2 A/(2b_+)$ the integrand of (6.2) is nonpositive. Above this height, (6.3) gives $F \geq y/4$. Its positive contact contribution is therefore at most $4b_+(q+z_c)$. Fix all these positive costs first, and then choose τ so that (6.4) is smaller than their negative sum minus one. Only afterward shrink the coefficients, source interval and $\varepsilon_0 \leq \tau$. We obtain $\log \mathfrak{m} < -1$ for every counted cycle, uniformly in all ratios of c, D, ε. Choose $\mathcal{A}_- \Subset \mathcal{A}_+$ as in Section A.2, putting the larger annulus inside the multiplier and section clearances and the counted cycles and their intervening annuli in the smaller one. The estimate holds throughout the ambient coefficient window for $D \geq 0$. Section B.1 supplies nondegenerate perturbations with $D > 0$, also from $D = 0$. Lemmas A.5 and A.4 now give the bound two.

6.2. **The lower passage for negative forcing.** The lower weak separatrix lies below $y = 0$ for negative forcing. We construct the transition from this negative baseline across the whole power-output interval, with the first and second source derivatives needed for the return comparison.

Assume henceforth that $\delta = -\nu < 0$. At $y = 0$ we have $G = -\nu$; a trajectory which leaves $y > 0$ cannot reenter it. This elementary sign will also determine the displacement in the connection between the two saddles.

Write $\mu := \varepsilon^2$ for the slow scale and $\nu := |\delta|$ for the forcing magnitude. Below, q is the physical ordinate on the outgoing section, u is the regular source coordinate, and v is the normal amplitude after the lower saddle passage.

Lemma 6.2. *Let q denote the physical ordinate on the outgoing section $z = z_c$; we work on a fixed compact source interval in q, with the field parameters p held fixed. Fix a small negative section $z = -q_0$ to the right of the lower polynomial root. There are a regular source coordinate $u = u(q,p)$ and fixed constants $\eta_0 > 0$, $0 < \theta < 1$, such that every counted cycle has $0 < u < \theta$ and reaches $-q_0$ with ordinate*

$$Y(q) = \Psi_-(\eta_0 u(q)^R), \qquad R \asymp \varepsilon^{-2}. \tag{6.5}$$

Here p denotes the frozen field parameters,

$$0 < c_u \leq u_q \leq C_u, \qquad |u_{qq}| \leq C_u, \tag{6.6}$$

$$\Psi_-(v) = \beta_- + \kappa_- v[1 + E(v/\varepsilon^2, p)], \qquad -M\nu \leq \beta_- \leq -m\nu < 0, \tag{6.7}$$

$$e^{-M/\varepsilon^2} \leq \kappa_- \leq e^{-m/\varepsilon^2}, \qquad E(0,p) = 0, \qquad \|E\|_{C^1} \leq M. \tag{6.8}$$

The last norm is in the displayed amplitude variable, on a fixed small interval; no parameter derivative is asserted. The signed map is C^2 on the whole interval $0 \leq v \leq \eta_0\theta^R$. Its positive part is one source interval, possibly empty. On this interval, after shrinking ε_0,

$$0 < Y, Y_q \leq M\varepsilon^{-2}e^{-m/\varepsilon^2}, \qquad Y_{qq} \geq 0, \qquad \nu \leq Me^{-m/\varepsilon^2}\theta^R. \tag{6.9}$$

Proof. Let $\beta < 0$ be the lower root, so that $P'(\beta) > p_0 > 0$. Set $\mu = \varepsilon^2$, fix the required final order $j \geq 8$, and first construct the invariant graph with additional phase derivatives. For $\nu > 0$, the substitution $y = \nu\zeta$ gives

$$\dot{z} = \nu\zeta(1 + bz - \nu\zeta) + \mu P(z), \qquad \dot{\zeta} = z\zeta - 1.$$

These polynomial equations extend regularly to $\nu = 0$. The physical graph is recovered by $h = \nu\zeta$, which degenerates at that parameter value. At $\mu = \nu = 0$, the graph $\zeta = 1/z$ on the fixed negative interval is normally attracting with a fixed spectral gap. Embed this interval in a fixed compact boundaryless base and extend the graph and field there, using a cut-off independent of the frozen field parameters. Writing $w = \zeta - 1/z$, the extended critical field has $\dot{z} = 0$ and $\dot{w} = a_0(z)w$ on $w = 0$, with $a_0 \leq -\sigma < 0$. For a fixed order $r \geq 2j + 12$, on a common tube and uniformly in the frozen parameters, $\|\dot{z}\|_{C^r} + \|\dot{w} - a_0(z)w\|_{C^r} \leq C(\mu + \nu)$. The compact boundaryless graph therefore has zero tangential rate and a uniform normal gap. After one

common shrinking of the parameter box, Fenichel's compact normally-hyperbolic persistence theorem, [14, Theorem 3, p. 215], gives a persistent invariant C^r graph $\zeta = k(z;\mu,\nu,p)$ in a common tube. The differentiated-maximum argument in the proof of Lemma 5.5 applies with transport size $\mu+\nu$ and normal coefficient $a_0 \le -\sigma < 0$; reversing its sign does not change the absolute-value estimate. With $r \ge 2j+12$ it yields $\|k - z^{-1}\|_{C^{2j+11}} \le C_j(\mu+\nu)$ on the extended circle. Restrict the graph to the nested working interval, where the extension agrees with the original field. At $\nu = 0$ its physical reconstruction is $h = \nu k \equiv 0$. The invariance equation gives the quantitative estimate:

$$zk - 1 = k_z\{\nu k(1 + bz - \nu k) + \mu P(z)\}.$$

Thus $k - z^{-1} = O_{C^{2j+10}}(\mu+\nu)$ and $h = \nu k$ satisfies

$$h(z) = \nu\{z^{-1} + O_{C^{2j+10}}(\mu+\nu)\}, \qquad h < 0 \text{ for } \nu > 0, \tag{6.10}$$

with the regular limiting choice $h \equiv 0$ at $\nu = 0$. Only phase derivatives, with the field fixed, are used below. Notice also that $\nu/\mu = \varepsilon|D| \to 0$ already follows from the normalization; it does not require the later exponentially small survival estimate.

Translate the chosen graph by $w = y - h(z)$ and put $q_n(z,w) = 1 + bz - 2h(z) - w$. The exact equations are

$$\dot z = \mu\mathcal{V}(z) + q_n(z,w)w, \qquad \dot w = (z - h_z q_n(z,w))w, \qquad \mathcal{V} = P + \frac{h(1+bz-h)}{\mu}.$$

The coefficients have common C^{2j+8} bounds, $\mathcal{V} = P + O_{C^{2j+8}}(\nu/\mu)$, and the normal coefficient is bounded above by a negative constant. Lemma 3.1 applies with $\tau = \mu$ and $\rho = 1$, at order $j+1$. It gives an exact product, preserving this particular graph, with uniformly transverse normal coordinate.

The equilibrium on this graph lies at $z = \beta + O(\nu/\mu)$ and has a negative eigenvalue bounded away from zero and a positive eigenvalue comparable with μ. The graph through it agrees to its right with its weak unstable separatrix. Since $\mathcal{V}'$ is a positive unit at its simple zero, the scalar change following Lemma 3.1, after division by the absolute normal rate, gives

$$\dot\xi = \mu b_w \xi, \qquad \dot\eta = -\eta, \qquad 0 < m_w < b_w < M_w. \tag{6.11}$$

The coordinate ξ increases to the right of the equilibrium. This is an exact finite-order orbital normal form; its construction does not divide by the small eigenvalue.

Choose the incoming cut $\eta = \eta_0 > 0$ and an outgoing weak cut $\xi = \xi_{\text{out}} > 0$. The regular fast orbit reaches the incoming cut in bounded time. Its weak coordinate, divided by ξ_{out}, is $u(q)$. The Poincaré determinant formula gives (6.6), with positive orientation in physical ordinate q: both transverse determinants are positive. After fixing both weak exits, shrink the source so that their normalized absolute incoming coordinate is less than $\theta < 1$. The right local output is exactly $v = \eta_0 u^R$, with $R = (\mu b_w)^{-1}$. For $u = 0$ the orbit approaches the saddle below $y = 0$. For $u < 0$ its left output is at most $\eta_0\theta^R$. Choose a fixed physical line $z = z_{\text{left}}$ left of all lower roots, with $P(z_{\text{left}}) < -p_{\text{left}} < 0$. The still-positive left output has height $o(\mu)$ there. For a small fixed $h_{\text{left}} > 0$, the half-strip $z \le z_{\text{left}}$, $0 < y \le h_{\text{left}}\mu$ has inward flow on its right and upper sides. It cannot return to the positive outgoing section.

We next transfer the weak outgoing cut to a fixed physical vertical section $z = z_l$, retaining the derivative loss caused by the slow speed. Write the inverse coordinates as $(z,y) = (X(\xi,\eta), H(\xi,\eta))$ and choose $X(\xi_{\text{out}}, 0) = z_l$. The nearby occurrence on that physical line is determined by

$$X(\xi_{\text{out}} e^{\mu b_w t}, ve^{-t}) = z_l.$$

For $0 \le v \le c_0\mu$, with $c_0 > 0$ fixed and small, its time derivative is comparable with μ. The normal direction is uniformly transverse, and the weak exit stays a fixed distance from the base root. The $\rho = 1$ section change (3.5), with $v = \mu V$, therefore gives

$$\Phi(v) = h(z_l) + a_l v + (v^2/\mu)E_l(v/\mu), \qquad a_l = H_\eta - h_z X_\eta > 0, \qquad \|E_l\|_{C^{j-3}} \le M. \tag{6.12}$$

In particular Φ' is bounded above and below by positive constants, and $\Phi''/\Phi' = O(\mu^{-1})$. The actual argument range $v \le \eta_0\theta^R$ is exponentially smaller than μ.

On $[z_l, -q_0]$ the polynomial is uniformly positive. The signed tube about h has $F \asymp \mu$. For the scalar equation $y_z = \mathcal{R}(z,y) = (zy-\nu)/F$,

$$\mathcal{R}_y = \frac{z(y^2+\mu P)+\nu(1+bz-2y)}{F^2}, \qquad -M/\mu \le \mathcal{R}_y \le -m/\mu, \tag{6.13}$$

and $|\partial_y^j \mathcal{R}| \le M_j \mu^{-j}$ for $j \ge 2$. Positivity at the lower right cut requires $\nu \le M\theta^R$, in addition to $\nu/\mu \to 0$. To define the transition also for signed inputs, use the estimate on $\mathcal{R}_y$: it makes the tube $|y - h(z)| \le c_1\mu$ forward invariant, for fixed small $c_1 > 0$. After decreasing c_0, it contains the image of Φ on $0 \le v \le c_0\mu$. Thus the signed transition is defined on this whole interval, not only on the exponentially short range of counted orbits. Write its scalar part as $y(z,w)$ with $y(z_l,w) = w$, and put $J = y_w$ and $L = (\log J)_w$. Their variational equations are

$$(\log J)_z = \mathcal{R}_y, \qquad L_z = \mathcal{R}_{yy}J, \qquad J(z_l) = 1, \quad L(z_l) = 0. \tag{6.14}$$

Since $0 < J \le e^{-m(z-z_l)/\mu}$, integration gives $|y_{ww}/y_w| = |L| \le M\mu^{-1}$. Composition with (6.12) yields $|\Psi_-''/\Psi_-'| \le M\mu^{-1}$. Its baseline is $\beta_- = h(-q_0)$, between two fixed negative multiples of ν, and (6.13) gives the stated exponential bounds for $\kappa_- = \Psi_-'(0)$. Moreover $\Psi_-'(v)/\kappa_-$ is bounded above and below on $0 \le v \le c_0\mu$. For $V = v/\mu$, the exact formula

$$E(V,p) = \int_0^1 \left\{ \frac{\Psi_-'(t\mu V)}{\kappa_-} - 1 \right\} dt$$

therefore gives $E(0,p) = 0$ and a uniform C^1 bound on $0 \le V \le c_0$. This proves (6.7)–(6.8).

The map Ψ_- is increasing. Positivity at its target is equivalent to positivity throughout the preceding orbit, because $G(z,0) < 0$. Hence $Y > 0$ selects one upper interval and implies the last inequality of (6.9). Finally, with $v = \eta_0 u^R$,

$$\frac{Y_{qq}}{Y_q} = (R-1)\frac{u_q}{u} + \frac{u_{qq}}{u_q} + \frac{\Psi_-''}{\Psi_-'}Rv\frac{u_q}{u}.$$

The last term is bounded by $MR(v/\mu)u_q/u$ and is absorbed by the first term on the whole interval. This proves convexity and the remaining bounds in (6.9). □

6.3. **Positive outlets and local passages.** We first exclude contact constants bounded away from zero on the negative scaled side, then treat the saddle-node range. For the remaining positive outlets we construct reference orbits of the same field. Their continuation to the physical section supplies the source derivative bounds used in both the saddle-node and two-saddle returns.

Set

$$k = \frac{c}{\varepsilon^2}, \qquad \ell = -\frac{\nu}{A^3\varepsilon^6}. \tag{6.15}$$

The survival inequality in (6.9) implies $\ell \to 0$ uniformly whenever a counted cycle exists. For any fixed $k_0 > 0$, the fields with $k \le -k_0$ have no such cycle. Indeed, on the line $z_g = Ac/2 < 0$,

$$P(z_g) = A^2(c/2 + Cc^2/4) \le -A^2|c|/4.$$

At heights $0 \le y \le h_g = A^2\varepsilon^2|c|/16$ the field points left. Since $h_g \ge A_0^2 k_0 \varepsilon^4/16$, the output of Lemma 6.2 lies below this height for small ε. The strip $z \le z_g$, $0 < y \le h_g$ also points inward on its upper side, and a downward exit cannot be reversed. This argument includes $k \to -\infty$.

For $|k|$ small put $n = A\varepsilon^2$, $z = nU$, $y = n^2V$. The exact equations are

$$U' = V(1 + bnU - n^2V) + k - U + C\varepsilon^2U^2, \qquad V' = UV + \ell. \tag{6.16}$$

Fix a small $d_0 > 0$, and choose $|k| \le k_c < d_0/8$ and $|C|q_0 \le A_0/8$. Then

$$P(-x) \ge \tfrac{3}{4}Ax, \qquad d_0A\varepsilon^2 \le x \le q_0.$$

Every still-positive lower output decreases in height while $z < 0$. It is exponentially smaller than $\varepsilon^2 P$ down to $U = -d_0$, so $F \geq \varepsilon^2 P/2 > 0$ there. It either reaches this section in finite time or first hits $y = 0$. In the former case its ordinate and first derivative in the scaled coordinate satisfy

$$V = o(1), \qquad 0 < V_q \leq M\varepsilon^{-6} e^{-m/\varepsilon^2} =: \zeta_\varepsilon \longrightarrow 0, \qquad V_{qq} \geq 0. \tag{6.17}$$

Here convexity is preserved on the small negative tail. Indeed, direct differentiation gives, with $Q = 1 + bz$,

$$\mathcal{R}_{yy} = \frac{2\{z[y^3 + (3y - Q)\varepsilon^2 P] - \nu[\varepsilon^2 P + Q^2 - 3Qy + 3y^2]\}}{F^3} > 0,$$

whereas $\mathcal{R}_y < 0$ by (6.13). The forcing term is absorbed because $\nu/(|z|\varepsilon^2 P) \leq M\nu\varepsilon^{-6} \to 0$. Only this small contact interval, where $Q > 0$, is assigned a convexity sign; the remote negative interval was included in Ψ_-.

Apply the finitely differentiable product coordinates of Lemma D.4 to (6.16). Write them as $\dot{s} = W(s, p)$, $\dot{w} = -w$, with center value $\sigma(p)$. At $k = \ell = 0$ the strong line is $V = 0$, and $s = V\chi(U, V)$ with $\chi > 0$. Consequently the entries in (6.17) have

$$3d_0/4 < w < 5d_0/4, \qquad |s - \sigma| \leq M(V + |k| + |\ell|).$$

The fixed strong cut $w = w_0 = d_0/2$ is reached in time $\log(w/w_0)$, bounded above and below by positive constants. Its coordinate is an increasing C^3 function of V, with bounded logarithmic curvature. Thus the complete map $N(q)$ from the original physical section to this cut satisfies

$$N_q > 0, \qquad \sup N_q \leq M\zeta_\varepsilon, \qquad N_{qq}/N_q \geq -M\zeta_\varepsilon. \tag{6.18}$$

The next lemma supplies the reference graphs and transverse incidence required by Lemma D.3. Applying that transition lemma then gives the outgoing estimates on a fixed relative neighborhood of each reference.

Lemma 6.3. *Choose the fixed physical section $z = z_c > 0$ before the final parameter shrinking. For every field admitting a counted lower passage, the following three outgoing orbits continue to $z = z_c$:*

1. *the zero-stable-amplitude orbit from the right exit of the saddle-node product box, for $|k| \leq k_c$;*
2. *the outgoing unstable separatrix of the positive saddle, for $k \in [k_c, K_1]$;*
3. *the outgoing unstable separatrix of the small-ratio saddle, for $e = \varepsilon/\sqrt{c} \leq e_0$.*

After the corresponding compact local outlet change, put $z = nU$, $y = n^2 V$, with $n = A\varepsilon^2$ in the first two cases and $n = Ac$ in the third. Each reference is a graph $V = h(U)$ on a fixed positive scaled section and satisfies, up to $z = z_c$,

$$c_h(1 + U^2) \leq h(U) \leq C_h(1 + U^2), \qquad \mathcal{F}(U, h) \geq c_F h, \tag{6.19}$$

where $\mathcal{F}$ is the horizontal component of the scaled field. The local outgoing amplitude a can be oriented so that on the initial reference section

$$g(a) := \frac{V(a)}{h} - 1, \qquad g(0) = 0, \qquad 0 < c_g \leq g' \leq C_g, \qquad |g''| + |g'''| \leq C_g. \tag{6.20}$$

All constants are uniform on the stated ranges. Thus the reference and initial-incidence hypotheses of Lemma D.3 hold in its applications below.

Proof. All three cases reduce, after the local outlet change, to

$$\dot{U} = \mathcal{F}(U, V) = VQ(U, V) + p(U), \qquad \dot{V} = UV + \ell_*, \qquad Q = 1 + bnU - n^2 V. \tag{6.21}$$

For the first two cases

$$p(U) = k - U + C\varepsilon^2 U^2, \qquad n = A\varepsilon^2, \qquad \ell_* = -\nu/n^3,$$

whereas in the small-ratio case

$$p(U) = e^2(1-U) + C\varepsilon^2 U^2, \qquad n = Ac, \qquad \ell_* = e^3\eta = -\nu/n^3.$$

The survival estimate (6.9) implies $\ell_* \to 0$ in all three cases and $\eta \to 0$ in the last, since the denominators lose at most a fixed power of ε. Fix a small $\varrho > 0$. After choosing z_c and then shrinking the parameters,

$$1 - \varrho \le Q \le 1 + \varrho \tag{6.22}$$

whenever $0 \le z \le z_c$ and $y \le C(n^2 + z^2)$. The upper barriers below give this height bound and close the first-exit argument on the chosen physical rectangle.

At the saddle-node organizer $k = \ell_* = \varepsilon = 0$, the outgoing center graph satisfies

$$h_0(U) = U + U^2 + O(U^3), \qquad \mathcal{F}(U, h_0) = U^2 + O(U^3).$$

Fix a small $\delta > 0$ and choose the right product exit so that its zero-stable-amplitude orbit reaches $U = \delta$. Finite smooth dependence of the product and section coordinates gives

$$\begin{aligned} h(\delta) &= \delta + \delta^2 + O(\delta^3 + |k| + |\ell_*| + \varepsilon^2), \\ \mathcal{F}(\delta, h(\delta)) &= \delta^2 + O(\delta^3 + |k| + |\ell_*| + \varepsilon^2). \end{aligned} \tag{6.23}$$

Decrease the previously chosen k_c, if necessary, so that $k_c \le \kappa_0\delta^2$, and then shrink ε and the forcing. For $m_0 = 1 + \delta/4$, the line $L_1 = m_0 U$ starts below the reference. Its exact inward-crossing expression is

$$\begin{aligned} (UL_1 + \ell_*) - L_1'\mathcal{F}(U, L_1) = m_0\{&(1 - C\varepsilon^2 - m_0 bn + m_0^2 n^2)U^2 \\ &- (m_0 - 1)U - k\} + \ell_* > 0, \end{aligned} \tag{6.24}$$

for $U \ge \delta$. A fixed sufficiently small $L_2 = \gamma(1 + U^2)$ is also a lower barrier: at a first meeting,

$$(UL_2 + \ell_*) - L_2'\mathcal{F}(U, L_2) \ge UL_2\{1 - 2\gamma(1 + \varrho + |C|\varepsilon^2/\gamma)\} - |\ell_*| - 2\gamma U|k| > 0.$$

A fixed sufficiently large $H_2 = M(1 + U^2)$ is an upper barrier, since $\mathcal{F}(U, H_2) \ge H_2/2$ and, at a first upward meeting,

$$\frac{UH_2 + \ell_*}{\mathcal{F}(U, H_2)} < H_2'.$$

The two lower barriers give

$$\frac{\mathcal{F}(U,h)}{h} \ge 1 - \varrho - \frac{1}{m_0} - \frac{|k|}{m_0\delta} - \frac{|C|\varepsilon^2}{\gamma} \ge c_F > 0. \tag{6.25}$$

The barriers and positive speed prove (6.19) in the saddle-node range and continue the reference to $z = z_c$. The product coordinates may be oriented at the organizer by $s = V + O(2)$ and $w = V - U + O(2)$. Increasing the outgoing amplitude is then positively transverse to the flow; the Poincaré determinant formula and the compact transition to $U = \delta$ give (6.20).

For $k \in [k_c, K_1]$, let (U_s, V_s) be the positive saddle and let h be its actual outgoing unstable separatrix. Choose a new fixed $\delta_b \ll k_c$. Uniform hyperbolicity gives on $U = U_s + \delta_b$

$$h = (1 + k)\delta_b + O(\delta_b^2 + \varepsilon^2 + |\ell_*|), \qquad \mathcal{F} = k\delta_b + O(\delta_b^2 + \varepsilon^2 + |\ell_*|). \tag{6.26}$$

Put $m_1 = 1 + k_c/2$ and $t = U - U_s$. The line $L_1 = m_1 t$ begins below h, and its exact crossing expression is

$$(UL_1 + \ell_*) - L_1'\mathcal{F}(U, L_1) = m_1 t[U + 1 - m_1 Q] - m_1(k - U_s) - m_1 C\varepsilon^2 U^2 + \ell_*. \tag{6.27}$$

On the compact k-interval the first term has a fixed positive lower bound proportional to $t(1+U)$; it absorbs all remaining terms after the parameter shrinking. The same parabolic lower and upper barriers as above give

$$\gamma(1 + U^2) \le h(U) \le M(1 + U^2), \qquad \frac{\mathcal{F}(U,h)}{h} \ge 1 - \varrho - \frac{1}{m_1} - \frac{|k - U_s|}{m_1\delta_b} - \frac{|C|\varepsilon^2}{\gamma} \ge c_F.$$

Continue first from the parameter-dependent line $U = U_s + \delta_b$ to a fixed $U_0 > K_1 + 2\delta_b$. This is a uniformly transverse compact transition, after which the barriers continue to $z = z_c$. The saddle separatrices and these fixed section maps have common finite derivatives; orienting the outgoing amplitude by the positive transverse determinant proves (6.20).

It remains to construct the small-ratio reference. In the variables of (6.29), its selected local outlet is the actual outgoing unstable separatrix. Continue it by a bounded local map to the fixed horizontal level $Y = d_1 > 0$, where X lies in a fixed positive compact interval and $e|\eta| \ll d_1$. Using Y as phase until $Y = e^{-2}$, one has

$$\frac{dX}{dY} = \frac{e^2\mathcal{F}_s}{XY + e\eta}, \qquad \mathcal{F}_s = Y(1 + bnX - n^2e^2Y) + 1 - X + CcX^2.$$

The initially positive $\mathcal{F}_s$ cannot first vanish: at such a point the term containing X_Y vanishes and

$$\frac{d\mathcal{F}_s}{dY} = 1 + bnX - 2n^2e^2Y > \tfrac{1}{2}.$$

Moreover $XY + e\eta \geq cXY$ and

$$\frac{d(X^2)}{dY} \leq Ce^2(1 + Y^{-1}) + Cce^2X^2/Y.$$

Gronwall's inequality gives positive uniform upper and lower bounds for X at $Y = e^{-2}$. In the common variables this is the fixed horizontal level $V = e^2Y = 1$. Continue to a fixed U_0 to its right; on that compact piece $\mathcal{F}$ and $UV + \ell_*$ are positive units. For $U \geq U_0$, the barriers $\gamma(1 + U^2)$ and $M(1 + U^2)$ give

$$\gamma(1 + U^2) \leq h(U) \leq M(1 + U^2), \qquad \frac{\mathcal{F}(U, h)}{h} \geq 1 - \varrho - \frac{e^2U}{\gamma(1 + U^2)} - \frac{|C|\varepsilon^2}{\gamma} \geq c_F.$$

This proves (6.19) in the third case.

Finally orient its local amplitude so that increasing a decreases the outlet X by a positive unit. The Y-phase map has positive-unit first variation and bounded derivatives through order three, since the relevant coefficients have the integrable majorant $Ce^2(1 + Y^{-1})$. If $x(a)$ is the abscissa at $V = 1$, then $x'(a) < -c < 0$. Differentiating the scalar flow from this moving initial point to fixed U_0 gives

$$\partial_x V(U_0; x) = -\frac{x + \ell_*}{\mathcal{F}(x, 1)} \exp\left(\int_x^{U_0} \partial_V \frac{UV + \ell_*}{\mathcal{F}(U, V)}\, dU\right) < 0.$$

All factors are positive units; hence $V_a(U_0) > 0$ is a unit, and the first three differentiated equations give (6.20). As the scale and coefficient parameters approach the organizer, the three references and their fixed relative collars converge on compact subsets of $0 < z \leq z_c$ to its outgoing fast orbit. The small-ratio limit here includes $c \to 0$; for fixed nonzero small c, the fast limit may start at the displaced positive root. In the joint limit the parabolic bounds give zero height at $z = 0$, and uniqueness for the regular fast scalar equation identifies the limit. Thus, after first making the coefficient neighborhood small enough, all references and collars reach the same fixed transverse target inside its regular physical flow box. This supplies the common target and its positive clearance for the return construction. □

The saddle-node return. Return to the saddle-node passage and choose the right product exit $s = s_+$ with $W(s_+) > 0$. The only entries which reach it have $[s, s_+] \subset \{W > 0\}$; they form one interval, to the right of the last root if roots are present. The local map and its exact differential identity are

$$\mathcal{D}(s) = w_0 \exp\left[-\int_s^{s_+} \frac{d\tau}{W(\tau)}\right], \qquad W\mathcal{D}' = \mathcal{D}. \tag{6.28}$$

Let $C_+(a)$ be the continuation from this exit to the physical section. Part (1) of Lemma 6.3 verifies the actual reference and its positive-unit incidence. Lemma D.3 therefore gives $C_+' > 0$

and bounded C_+''/C_+' and C_+'. For $H = N \circ C_+$,

$$aH''/H' \geq -Ma - Ma\zeta_\varepsilon \geq -M_1 a.$$

Fix the product box with $W_s \leq 1/4$, then a_0 with $M_1 a_0 \leq 1/4$. The passage estimate in Lemma D.4 places every local output below a_0 once its entry is sufficiently close to σ; choose k_c and then ε_0 accordingly. Thus (D.18) holds with margin $1/2$. Lemma D.4 gives total isolated return-zero multiplicity at most two, including an exactly quadratic stationary zero. The finite-passage domain is one interval, obtained from increasing maps: positive lower output, positive entry at the strong cut, the unique component of $W > 0$ reaching s_+, and the original physical section interval. The comparison therefore applies throughout this actual domain.

The positive Dulac map. We now prepare the positive-saddle passage for Lemma 4.3. It suffices to consider fields admitting a counted cycle, so that the survival bound in (6.9) holds. In the range $k \in [k_c, K_1]$, equation (6.16) gives a positive saddle with eigenvalues near -1 and k, and ordinate $-\ell/U_s > 0$. For large k, put

$$e = \varepsilon/\sqrt{c}, \quad n = Ac, \quad z = nX, \quad y = n^2e^2Y, \quad \eta = -\nu/(n^3e^3). \tag{6.29}$$

Here $c \geq K_1\varepsilon^2$ and $A \geq A_0$, so the survival bound implies $|\eta| \leq M\varepsilon^{-6}e^{-m/\varepsilon^2} \to 0$. The equations become

$$X' = e^2\mathcal{F}, \quad \mathcal{F} = Y(1 + bnX - n^2e^2Y) + 1 - X + CcX^2, \qquad Y' = XY + e\eta.$$

The saddle near $(1, 0)$ has characteristic ratio $r = e^2\beta_s$, where β_s is bounded above and below by positive constants. Lemma D.5 applies with the present bounded negative C and the chosen stable graph. Its use of Lemma 3.1 has $\tau = \rho = e^2$ and a fixed positive normal gap, or a negative gap in reversed time. It gives the exact C^j orbital normal form

$$\dot{\xi} = -r\xi, \qquad \dot{\omega} = \omega, \qquad \mathcal{D}(s) = \xi_0(s/\omega_0)^r. \tag{6.30}$$

The stable graph is $Y = -e\eta/X + O_{C^j}(e^3|\eta|) > 0$ and the strong-fiber slope in X is $O_{C^{j-1}}(e^2)$.

6.4. **The signed connector and the two-saddle return.** For $k \geq k_c > 0$, the connector must carry the lower power output to the positive saddle. When its characteristic ratio is small, this passage can amplify height and derivatives by an exponential factor. The lower contraction must dominate that factor on the whole signed input interval. The next proposition proves this domination, preserves the negative baseline, and bounds the weighted logarithmic curvature needed by Lemma 4.3.

Proposition 6.4 (The complete signed connector)**.** *Fix the compact coefficient window in* (6.1)*, the finite phase order, and the selected lower graph and positive-saddle coordinates constructed above. The lower weak unstable branch is the zero source; the incoming unstable coordinate increases with physical height from the positive saddle's stable branch. Fix $k_c > 0$, a small-ratio threshold $e_0 > 0$, and $K_1 \geq \max\{k_c, e_0^{-2}\}$. For each final choice of the lower power-output cut and compatible canonical incoming cuts within the fixed saddle cores, take $\eta_0 > 0$, $0 < \theta < 1$, $R \asymp \varepsilon^{-2}$ and the survival constants $C_s, m_s > 0$ from Lemma 6.2 and* (6.9)*. There are $\bar{c}, \varepsilon_0 > 0$ such that every field in this window satisfying*

$$\delta = -\nu < 0, \qquad 0 < \varepsilon < \varepsilon_0, \quad 0 < c < \bar{c}, \quad k = c/\varepsilon^2 \geq k_c, \qquad 0 < \nu \leq C_s\theta^R e^{-m_s/\varepsilon^2}$$

has the following connector. Put $t_* = \eta_0\theta^R$. The forward flow from the entire lower power-output interval $0 \leq t \leq t_*$ reaches the selected canonical incoming section of the positive saddle in finite positive time. Its unstable coordinate defines $\Psi \in C^2([0, t_*])$ with $\Psi(0) = -B_{\mathrm{conn}} < 0$ and $\Psi' > 0$, and

$$\eta_0 \sup_{[0,t_*]} \Psi' \leq 1, \qquad \left| t\frac{\Psi''(t)}{\Psi'(t)} \right| \leq 1 \quad (0 < t \leq t_*). \tag{6.31}$$

The estimates are uniform for these fields; all source derivatives are taken with the vector field fixed.

Proof. Continue the whole signed interval $0 \le v \le \eta_0\theta^R$ from Lemma 6.2 to the canonical incoming section of the positive saddle. Put $\mu = \varepsilon^2$. The signed lower transition and the assumed survival bound give, with $0 < \theta < 1$,

$$|Y(-q_0)| \le M(\theta^R + \nu), \quad \nu \le M\theta^R e^{-m/\mu}, \quad R \ge r_-/\mu, \tag{6.32}$$

where $Y(-q_0) = \Psi_-(v)$. On this v-interval, $0 < \Psi'_-(v) \le Me^{-m/\mu}$, and $|\Psi''_-(v)/\Psi'_-(v)|$ is bounded by a fixed power of ε^{-1}. With the finite phase order, e_0, K_1, the saddle cores and root gap fixed, choose the cuts and tube widths before the upper bound $\bar{c}$ for c. The restrictions

$$LA_1\bar{c} < q_0, \qquad LA_1\varepsilon_0^2 < q_0 \tag{6.33}$$

are imposed before shrinking ε_0.

Choose q_0 smaller, if necessary, so that

$$P(-x) \ge \frac{7}{8}Ax + A^2c \qquad (0 \le x \le q_0). \tag{6.34}$$

In the bounded-ratio range set $n = A\varepsilon^2$ and in the small-ratio range set $n = Ac$. Thus $n \ge c_n\varepsilon^2$. With

$$\omega_\varepsilon = K\{\theta^R + \nu/n\},$$

choose K larger than twice the constant in (6.32), and then require $LK > 2$, $n < 1$, and $\omega_\varepsilon < 1/2$. Let B_1, C_Q bound respectively $|1 + bz - y|$ and $|1 + bz - 2y|$ on the contact box. Since the right side of (6.34) is bounded below by a fixed multiple of n, polynomial losses in ε^{-1} are dominated by the exponential factors in (6.32); hence we may impose

$$B_1\omega_\varepsilon \le \frac{\varepsilon^2}{4}\min_{Ln\le x\le q_0} P(-x), \qquad C_Q\nu \le \frac{x\varepsilon^2 P(-x)}{4} \quad (Ln \le x \le q_0). \tag{6.35}$$

For $z = -x$ and $|y| \le \omega_\varepsilon$, (6.35) therefore gives

$$\frac{1}{2}\varepsilon^2 P(z) \le F(z, y) \le \frac{3}{2}\varepsilon^2 P(z). \tag{6.36}$$

The upper horizontal face has $G = -x\omega_\varepsilon - \nu < 0$, while the lower face satisfies

$$G = x\omega_\varepsilon - \nu \ge xK\theta^R + (LK - 1)\nu > 0.$$

Thus the signed interval reaches $z = -Ln$ inside this strip. For $\mathcal{R} = G/F$ an exact differentiation gives

$$\mathcal{R}_y = \frac{z(y^2 + \varepsilon^2 P) + \nu(1 + bz - 2y)}{F^2} < 0. \tag{6.37}$$

If $J = y_{y_0}$ and $L_2 = y_{y_0y_0}/J$, then

$$J_z = \mathcal{R}_y J, \qquad (L_2)_z = \mathcal{R}_{yy} J. \tag{6.38}$$

Since $0 < J \le 1$ and the coefficients of $\mathcal{R}$ are bounded by fixed powers of ε^{-1}, integration bounds L_2 by such a power. For the terminal map $H(y_0) = y(-Ln; y_0)$ and $g = H \circ \Psi_-$,

$$\frac{H''}{H'} = L_2, \qquad \frac{g''}{g'} = (L_2 \circ \Psi_-)\Psi'_- + \frac{\Psi''_-}{\Psi'_-}.$$

Thus passage to the inner scale preserves exponential contraction of the first derivative and introduces only polynomial growth in the relative second derivative.

For $k \in [k_c, K_1]$, use $n = A\varepsilon^2$, $z = nU$, $y = n^2V$ and $\ell = -\nu/(A^3\varepsilon^6)$. Write

$$F_b(U, V) = V(1 + bnU - n^2V) + k - U + C\varepsilon^2U^2$$

for the horizontal component of (6.16). On $[-L, 0]$ and $[0, U_s - \sigma]$, where U_s is its positive zero at $V = 0$, one has $F_b(U, 0) \ge c_* > 0$ after shrinking ε. Writing $D_\varepsilon = |V(-L)| + |\ell|$, a first-exit tube $|V| \le K_{\mathrm{Gr}}D_\varepsilon$ (with K_{Gr} larger than the fixed Gronwall constant) closes because the V contribution to F_b is at most $c_*/2$. The scalar equation and its first two V derivatives

are uniformly bounded on this fixed interval; hence Gronwall and the analogous variational recursions in (6.38) give

$$0 < \Psi' \le M\varepsilon^{-N} e^{-m/\varepsilon^2}, \qquad \left|\frac{\Psi''}{\Psi'}\right| \le M\varepsilon^{-N}. \tag{6.39}$$

The final passage to the canonical incoming cut is a compact hyperbolic flow-box change with positive unit derivative and bounded C^2 jets.

It remains to treat $c \ge K_1\varepsilon^2$. Put

$$e = \frac{\varepsilon}{\sqrt{c}}, \quad n = Ac, \quad z = nX, \quad y = n^2e^2Y,$$

so that

$$X' = e^2\mathcal{F}, \qquad Y_X = \frac{XY + e\eta}{e^2\mathcal{F}}, \qquad \mathcal{F} = Y(1 + bnX - n^2e^2Y) + 1 - X + CcX^2, \tag{6.40}$$

with $\eta = -\nu/(n^3e^3)$. Decrease $\bar{c}$ so that, for a fixed $\sigma > 0$, $1 - X + CcX^2 \ge 3c_\sigma$ on $[-L, 1-\sigma]$; then choose δ_0 so that $\mathcal{F} \ge 2c_\sigma$ whenever $|Y| \le \delta_0$. At $X = -L$, the preceding strip estimate gives

$$|Y| + |\eta| \le M\varepsilon^{-N_0} e^{-a_0/\varepsilon^2}.$$

On the provisional tube the first two Y derivatives of the right side in (6.40) are bounded by M/e^2. Gronwall gives a factor $\exp(M_*/e^2)$. After these constants are fixed, shrink $\bar{c}$ so that, with $a_\theta = r_-|\log\theta|$,

$$M_{\mathrm{ht}}\bar{c} < a_0/3, \qquad M_*\bar{c} < m/3, \qquad M_*\bar{c} < a_\theta/3. \tag{6.41}$$

These choices have three separate uses: the first closes the height tube, the second preserves the lower-saddle decay of Ψ', and the third makes the curvature bound small after multiplication by θ^R. For $J = Y_{Y_0}$ and $L = Y_{Y_0Y_0}/J$, the exact equations

$$J' = f_Y J, \qquad L' = f_{YY} J$$

with f the right side of the scalar equation yield

$$J \le e^{M/e^2}, \qquad |L| \le Me^{-2}e^{M/e^2}.$$

Consequently, after the physical connector, the complete signed map satisfies

$$0 < \Psi' \le \varepsilon^{-N} e^{-m/\varepsilon^2 + M_*/e^2}, \qquad \left|\frac{\Psi''}{\Psi'}\right| \le \varepsilon^{-N} e^{M_*/e^2}. \tag{6.42}$$

We finally make the section conversion explicit. In finite-order product coordinates near the positive saddle, write

$$\dot{\xi} = -r_s\xi, \qquad \dot{\omega} = \omega, \qquad r_s = e^2\beta_s, \qquad 0 < \beta_- \le \beta_s \le \beta_+,$$

with $-X_\xi \asymp 1$ and $X_\omega = O_{C^2}(e^2)$ in the physical coordinate $X = X(\xi, \omega)$. Fix the canonical cut $\xi = \xi_b$ and $T_0 > 0$, and set $x_v = X(\xi_b e^{r_sT_0}, 0)$. The preceding vertical cut is specified by

$$X(\chi(\omega), \omega) = x_v, \qquad \chi^{(j)}(\omega) = O(e^2), \quad 1 \le j \le 2.$$

The canonical hitting time is

$$T(\omega) = r_s^{-1} \log \frac{\chi(\omega)}{\xi_b}.$$

The common factor e^2 cancels in its first two derivatives. Choosing its fixed normal half-width d so that

$$K_X d < \tfrac{1}{4}\beta_-\xi_b T_0, \qquad d \sup|T'| < \tfrac{1}{2},$$

ensures positive hitting time and makes $\omega \mapsto \omega e^{T(\omega)}$ increasing; a further fixed reduction keeps its image in the canonical collar. Since

$$Y = Y_s(\xi) + \omega u_1(\xi, \omega), \qquad u_1 \ge u_* > 0,$$

we have on this vertical line

$$\frac{d}{d\omega}Y(\chi(\omega),\omega) = u_1 + O(e^2) + O(d) > 0,$$

with bounded second derivative. Thus conversion from physical height to ω, followed by the hitting map, is an increasing C^2 change preserving (6.42). The whole signed input interval therefore reaches the canonical incoming cut in finite time, with these estimates.

The baseline $v = 0$ starts at the negative-height weak separatrix of the lower saddle and stays negative, because $G(z, 0) = -\nu < 0$. The incoming stable separatrix of the positive saddle lies in $y > 0$: a nonpositive point could not converge to its positive equilibrium. Physical height on the incoming section increases with its unstable coordinate, whose zero is this stable separatrix. Therefore

$$\Psi(0) = -B_{\mathrm{conn}} < 0. \tag{6.43}$$

This compares the two actual separatrices at the canonical target and fixes the sign of the connector offset.

It remains to prove (6.31). For these fixed final saddle cuts, $t_* = \eta_0\theta^R$. Equations (6.39) and (6.42), with $1/e^2 = c/\varepsilon^2$, give

$$\eta_0 \sup_{[0,t_*]} \Psi' \le M\eta_0\varepsilon^{-N}e^{-(m-M_*\bar{c})/\varepsilon^2},$$

$$\sup_{0<t\le t_*} \left| t\frac{\Psi''}{\Psi'} \right| \le M\eta_0\varepsilon^{-N}e^{-(a_\theta-M_*\bar{c})/\varepsilon^2}.$$

Here $M_* = 0$ in the bounded-ratio range. After fixing the cuts, choose $\bar{c}$ as in (6.41) and then shrink ε_0. These bounds give (6.31) on the full signed input interval. □

The return equation and its zeros. Use the connector of Proposition 6.4. Let a be the outgoing stable amplitude at the positive saddle. Parts (2) and (3) of Lemma 6.3 verify, respectively, the bounded-ratio and small-ratio reference graphs and their positive-unit incidence. Lemma D.3 therefore gives the positive physical continuation $q = C_+(a)$, with

$$0 < C'_+ \le Mn^{3/2}, \qquad |C''_+/C'_+| \le M.$$

Together with (6.6), this implies that $u(a) := u(C_+(a), p)$ satisfies

$$0 < u_a \le Mn^{3/2}, \qquad \frac{u_{aa}}{u_a} = \frac{u_{qq}}{u_q}C'_+ + \frac{C''_+}{C'_+} = O(1).$$

The exact complete complement is

$$H(a) = \Psi(\eta_0 u(a)^R). \tag{6.44}$$

The actual returning domain is the intersection of the fixed outgoing amplitude interval, the inverse image of the physical source interval, $0 < u < \theta$ and $0 < H < s_0$. It is one interval because all these maps are increasing. The lower positivity condition is an upper interval; it is already forced by $H > 0$, since the target then lies above the positive stable branch and no earlier negative height can reach it.

Every counted cycle has this passage. Its lower branch was selected in Lemma 6.2, and the preceding estimates keep its entire approach inside the positive saddle's incoming tube. Starting at its positive $-q_0$ hit, the contact-rectangle argument above again gives $F > 0$ up to the outgoing physical section, using $P(-q_0) > 0$, $F_y > 0$, positivity of the orbit and the boundary clearances. In the bounded-ratio saddle a negative unstable input leaves through the left side with $F < 0$, contradicting this contact speed; a zero input has infinite flight time. In (6.30), a negative normal input reaches negative physical height before the fixed negative normal exit, whereas a zero input is the stable separatrix. Only $s > 0$ can complete the required passage.

By Proposition 6.4, on the actual returning interval,

$$0 < H = \Psi(t) < \Psi(t) + B_{\mathrm{conn}} = \int_0^t \Psi'(v)\,dv \le u^R.$$

The lower product gives $R \le M\varepsilon^{-2}$, so

$$\begin{aligned} &n = A\varepsilon^2 : && R \sup u_a \le M\varepsilon, \\ &n = Ac, \quad r = e^2\beta_s : && Rr \sup u_a \le M\sqrt{c}. \end{aligned}$$

The first bound controls $R \sup u_a$ in the bounded-ratio comparison; the second controls $Rr \sup u_a$ in the small-ratio comparison. Together with the bounded logarithmic curvature of u, they supply the source hypotheses of Lemma 4.3.

For bounded k, Lemma C.1 supplies (4.6) uniformly through ratio one. Moreover $H' = \eta_0 \Psi'(t) R u^{R-1} u_a$ tends uniformly to zero, and $Rr \ge 2$ for small ε. The second case of Lemma 4.3 therefore gives at most two return zeros. For small e, (6.30) is exact and

$$Rr \ge r_- \inf \beta_s / c \ge 2$$

after shrinking $\bar{c}$. Fix the outgoing normal cut ω_0 and a weak amplitude interval first, and bound the regular source derivatives on that interval. The power prefactor $c_D = \xi_0 \omega_0^{-r}$ then has a fixed upper bound for $r \le r_0 < 1$ and for every subsequent choice of ξ_0 in this interval. With this bound and $L_\Psi = 1$, the first case of Lemma 4.3 gives a_0 and η_*. Keeping ω_0 unchanged, choose the incoming weak cut $\xi_0 \le a_0$ and a separate incoming normal interval of width $d_{\mathrm{in}} \ll \xi_0$. Moving its preceding vertical cut by $O(e^2 d_{\mathrm{in}})$ gives a forward matching time $O(d_{\mathrm{in}}/\xi_0)$. For these final cuts, apply Proposition 6.4: first shrink $\bar{c}$ as in (6.41), and then ε_0, to obtain (6.31). Every local output satisfies $\mathcal{D}(s) \le \xi_0 \le a_0$, and decreasing $\bar{c}$ also makes $Rr \sup u_a \le M\sqrt{c} \le \eta_*$. The first case of the comparison applies on the whole interval and gives the same bound two. Section 2 identifies these simple or quadratic zeros with the corresponding actual minimal-return multiplicities.

Completion of the proof of Proposition 6.1. The sign $D \ge 0$ was treated by (6.2). For $D < 0$, the lower survival inequality supplies every required small ℓ uniformly. Fix the central constant k_c first. The three alternatives $k < -k_c$, $|k| \le k_c$ and $k > k_c$ have bounds zero, two and two, respectively. To cover the last alternative, obtain a small-ratio threshold e_0 and choose the final finite $K_1 \ge \max(k_c, e_0^{-2})$; take the bounded-ratio constants on this final interval $[k_c, K_1]$. The two alternatives overlap because $e = 1/\sqrt{k}$. There are finitely many cuts, coefficient neighborhoods, and positive ε and Hausdorff thresholds; take a common shrinking compatible with them. Each fixed field has one sign of D and one value of k. Thus every counted cycle of that field is controlled by the same regional argument, and the maximum of the regional bounds gives the bound on the whole coefficient neighborhood. Every sufficiently close cycle has its unique physical section point in the selected source interval by Lemma A.2. This proves the proposition. □

6.5. **The common two-endpoint window.** For a normalized direction $q = (E_0, E_1, E_2, D) \in S^3$ with $E_1, E_2 < 0$, use the single chart

$$A = -E_1, \qquad C = E_2, \qquad c = E_0/A^2.$$

The limiting two-endpoint curve over J is

$$q_b^* = (0, -A_b, C_b, 0), \qquad A_b = -f_b/\sqrt{1+f_b^2}, \quad C_b = -1/\sqrt{1+f_b^2}.$$

It is compact and $A_b, -C_b$ have positive lower bounds. The negative-forcing proof fixes its final central threshold, small-ratio threshold, and bounded-ratio interval before choosing the open coefficient window. The nonnegative proof fixes the larger forcing bound, the bounds on all positive contributions, and the logarithmic time tolerance before its strict joint window. Both windows contain the same curve and use the same physical field and cores.

Pull the ambient nonnegative window back to the normalized chart and intersect it with the negative window near each q_b^*. Choose smaller balls with closed balls contained in these effective intersections. Finitely many cover the limiting curve. Their union contains a tube of positive radius about it, by compactness. On their closures the two signs' terminal radial and Hausdorff thresholds have a positive finite minimum. This supplies one open two-endpoint window allowing

$Cf_b \neq A$, $c \neq 0$, and both signs of D. For a fixed field the sign selects one entire estimate; the cycle bound is $\max(2,2) = 2$.

7. A common upper bound in the full parameter neighborhood

The preceding propositions bound all counted cycles of a fixed field on open coefficient windows. To obtain Theorem 1.1(i), we must cover every raw parameter direction and choose common phase, parameter, and Hausdorff neighborhoods. The difficulty is that the strict simple-contact margins vanish at the positive-quadratic and two-endpoint directions. We cover these boundary directions first, then take minima on the remaining compact set.

Use the radius ε and direction $(E_0, E_1, E_2, D) \in S^3$ defined in Section 2. The root-collision and two-endpoint estimates are supplemented by the exclusions and positive-value estimate of Appendix B, the simple-contact comparison of Appendix E, and the forcing estimates of Appendix F.

7.1. **The positive quadratic direction.** Near $(E_0, E_1, E_2, D) = (0, 0, 1, 0)$, a second weighted radius separates the forcing, constant, and linear terms from the quadratic term. Replacing

$$\varepsilon \mapsto \varepsilon\sqrt{E_2}, \qquad (E_0, E_1) \mapsto (E_0, E_1)/E_2, \qquad D \mapsto D/E_2^{3/2}$$

normalizes $E_2 = 1$ without changing the field. Write

$$R = \max(|E_1|, \sqrt{|E_0|}, |D|^{1/3}), \quad P = z^2 + Raz + R^2c, \quad D = R^3d. \tag{7.1}$$

At $R = 0$ there is no positive periodic orbit by Lemma B.2. Otherwise $\max(|a|, \sqrt{|c|}, |d|^{1/3}) = 1$. Use the six faces in the order $d = 1, d = -1, c = -1, c = 1, a = 1, a = -1$, assigning ties once. Proposition F.2 bounds the first face by one, and Lemma F.1 excludes the next two. Proposition 5.1 treats $c = 1$. On $a = 1$, $c \leq 1/8$ is excluded; for $c \geq 1/8$ the substitution

$$r = R\sqrt{c}, \qquad a_* = 1/\sqrt{c}, \qquad d_* = d/c^{3/2}$$

again gives the compact positive-constant family.

The exclusion on $a = 1$ also holds on an open coefficient set. For $a > 15/16$ and $c < 5/32$, take $\alpha = 1/2$ in Lemma F.1, since

$$R^{-2}P(-R/2) = \tfrac{1}{4} - \tfrac{a}{2} + c < -\tfrac{1}{16}.$$

For $c > 1/16$ use the positive-constant substitution above, with a in a fixed neighborhood of one. These two open sets overlap and cover the whole $a = 1$ face.

These face estimates extend to open coefficient windows. For $d > 0$ put

$$\widehat{R} = Rd^{1/3}, \qquad \widehat{a} = a/d^{1/3}, \qquad \widehat{c} = c/d^{2/3}, \qquad \widehat{d} = 1.$$

On a neighborhood with d bounded below, the hatted coefficients range in a fixed compact set. Proposition F.2 therefore treats a neighborhood of the $d = 1$ face, including its ties with a or c. If $c > 0$ is bounded below, the substitution $\widehat{R} = R\sqrt{c}$, $\widehat{a} = a/\sqrt{c}$, and $\widehat{d} = d/c^{3/2}$ gives the positive-constant family. The strict negative margins defining the $d = -1$ and $c = -1$ faces persist on their respective neighborhoods. Ties are assigned after these overlapping open regions have been chosen.

For a neighborhood of the remaining face $a = -1$, take $a < 0$ bounded away from zero and put

$$A = -Ra > 0, \qquad \widehat{c} = \frac{c}{a^2}, \qquad \widehat{d} = \frac{d}{(-a)^3}.$$

After the change $z = A\,\widehat{z}$, the polynomial and forcing become $P = A^2(\widehat{z}^2 - \widehat{z} + \widehat{c})$ and $D = A^3\widehat{d}$. On the face itself $A = R$. Choose numbers $0 < c_{\mathrm{in}} < c_{\mathrm{out}}$ and $0 < d_{\mathrm{in}} < d_{\mathrm{out}}$, all sufficiently small, so that Proposition E.1 applies on the inner box $|\widehat{c}| < c_{\mathrm{out}}$, $|\widehat{d}| < d_{\mathrm{out}}$, with $C = 1$. The outer reductions start at the smaller thresholds c_{in} and d_{in}, so their windows overlap this box. If $|\widehat{d}| > d_{\mathrm{in}}$, change radius to $A|\widehat{d}|^{1/3}$. The new forcing is 1 or -1; the other coefficients, $-|\widehat{d}|^{-1/3}$

and $\widehat{c}|\widehat{d}|^{-2/3}$, range in fixed compact sets, to which the forcing results apply. If $|\widehat{d}| < d_{\text{out}}$ and $\widehat{c} < -c_{\text{in}}$, change radius to $A\sqrt{|\widehat{c}|}$ and apply the negative-constant exclusion. If $|\widehat{d}| < d_{\text{out}}$ and $\widehat{c} > c_{\text{in}}$, put

$$r = A\sqrt{\widehat{c}}, \qquad a_* = -1/\sqrt{\widehat{c}}, \qquad d_* = \widehat{d}/\widehat{c}^{3/2},$$

and use Proposition 5.1, including its separated positive roots and their collision. Each new coefficient range is compact and fixed before the corresponding radial scale is made small. Each substitution reaches one of the stated regional estimates, so the six-face decomposition ends here.

Each parameter therefore belongs to an open window whose estimate bounds all its counted cycles by two. At a positive-root collision, localization and finite-event gluing count the through and fiber-exit pieces together on one source interval. Elsewhere the bound is no larger. Fix the physical sections and coefficient thresholds, then take the finitely many required shrinkings of R, ε, the source interval, and the Hausdorff neighborhood. This gives an open-neighborhood bound of two at the positive-quadratic direction, with arbitrary relative rates of the coefficients.

7.2. **The remaining normalized directions.** Classify the other directions first by the signs of the polynomial and forcing, then by the endpoint-root configuration. Fix b_* and let P_*, D_* denote a direction on S^3. Lemma B.1 excludes an open neighborhood unless $P_* \geq 0$ on $[f_{b_*}, 0]$ and $D_* \geq 0$. If $P_*(0) > 0$, Proposition B.3 gives a neighborhood bound of one. If $P_*(0) = 0$ and $D_* > 0$, Proposition B.4 gives a bound of two. The remaining directions satisfy

$$E_{0*} = D_* = 0, \qquad P_*(z) = z(-A_* + C_* z), \qquad A_*^2 + C_*^2 = 1. \tag{7.2}$$

For $x = -z \in [0, -f_{b_*}]$, $P_*(-x) = x(A_* + C_* x) \geq 0$. Hence $A_* \geq 0$. If $A_* = 0$, then $C_* = 1$, the positive-quadratic direction treated above. If $A_* > 0$ and $P_* > 0$ on $[f_{b_*}, 0)$, the negative preceding interval has a fixed positive polynomial margin. In a neighborhood use $A = -E_1$, $c = E_0/A^2$, $d = D/A^3$ in Proposition E.1; here A is bounded away from zero and $C = E_2$ may have either sign. This gives at most two. The only other possibility is

$$A_* > 0, \qquad C_* < 0, \qquad C_* f_{b_*} = A_*.$$

The roots f_{b_*} and zero are simple, with derivatives A_* and $-A_*$. Proposition 6.1 supplies a neighborhood bound of two. This list is exhaustive. In particular, an interior double root or a double root at the lower endpoint must have $P_*(0) > 0$ and has already been treated. If $P_* \equiv 0$, normalization forces $D_* = 1$ or -1, also already treated.

7.3. **Necessary directions and the order of the residual cover.** Fix a compact interval $J \Subset (0, 2)$. Let $\mathcal{R}_J^-$ be the compact set of possible limiting directions, including the modulus $b \in J$, of negative-forcing whole-graphic cycles. The gate and damping identities give

$$D = 0, \qquad E_1 \leq 0, \qquad P(z) \geq 0 \quad (f_b \leq z \leq 0).$$

In the zero-constant subset $E_0 = 0$ these reduce to $P = z(-A + Cz)$, $A \geq 0$, $A^2 + C^2 = 1$. The only non-strict simple-contact boundaries are $A = 0, C = 1$ and $A > 0, C < 0, Cf_b = A$.

More precisely, for every relatively open $N \supset \mathcal{R}_J^-$ there exist $\varepsilon_N, \rho_N > 0$ such that every $D < 0$ field with $0 < \varepsilon < \varepsilon_N$ and a cycle within ρ_N of the whole graphic has direction in N. If not, choose $\varepsilon_n, \rho_n \to 0$ and counterexamples outside N. Compactness gives a direction subsequence $(b_n, q_n) \to (b_*, q_*)$. A strictly negative limiting D_* or a negative value of P_* is excluded by Lemma B.1; continuity of the positive half-plane damping restriction gives $E_{1*} \leq 0$. Their whole-Hausdorff convergence supplies the descending crossing required by the gate argument. Thus the limit belongs to $\mathcal{R}_J^- \subset N$, contradicting the complement's closedness. The two thresholds depend on the chosen N.

Choose effective smaller positive-quadratic and two-endpoint windows first. Their complements in the compact zero-constant necessary set form a compact strict simple-contact remainder. Only on that remainder fix the positive minima of A and the negative preceding polynomial margin. Finitely many strict simple-contact windows cover it. The resulting open union contains the zero-constant necessary set. Compactness then gives an $\eta > 0$ such that the remaining

necessary directions satisfy $E_0 \geq 2\eta$. Cover these by finitely many positive-value windows with fixed E_0 margin. Now apply the preceding necessary-direction statement to this effective open union. Directions with $D \geq 0$ are covered directly by the negative-polynomial gate, positive-value, positive-forcing, simple-contact, and two-endpoint windows. Thus every strict minimum is taken on a fixed compact remainder after the boundary windows have been chosen.

Lemma 7.1 (Compactness of the graphic family)**.** *For every compact $J \Subset (0,2)$, the set*

$$\mathcal{G}_J := \bigcup_{b\in J} \Gamma_0^b$$

is compact. More precisely, let $J_\nu \Subset (0,2)$ be finitely many compact subintervals and, for each ν, let $\mathcal{O}_{\nu i}$ ($i \in \mathcal{I}_\nu$) be the finitely many local phase neighbourhoods used by the coefficient charts over J_ν. If $\mathcal{G}_{\nu i}$ denotes the union of the corresponding graphic family and the finitely many section cores for chart (ν, i), and $\mathcal{G}_{\nu i} \Subset \mathcal{O}_{\nu i}$, then

$$\delta_{\nu i} := \operatorname{dist}(\mathcal{G}_{\nu i}, \mathbb{R}^2 \setminus \mathcal{O}_{\nu i}) > 0, \qquad \delta_* := \min_{\nu,i} \delta_{\nu i} > 0.$$

The global compact phase neighborhood K contains the union of the graphic families and section cores. Each chart uses its own local neighborhood $\mathcal{O}_{\nu i}$ and the common clearance δ_*.

Proof. The explicit fast orbit in (A.1) and the implicit-function construction of t_b and f_b in Section A make $b \mapsto t_b$ and $b \mapsto f_b$ continuous on J. Hence the two maps

$$(b,s) \longmapsto \gamma_b(st_b), \qquad (b,s) \longmapsto (sf_b, 0), \qquad (b,s) \in J \times [0,1],$$

are continuous and have compact images. Their union is exactly $\mathcal{G}_J$. For each fixed pair (ν, i), the same argument applies to the compact set $\mathcal{G}_{\nu i}$, so its distance to $\mathbb{R}^2 \setminus \mathcal{O}_{\nu i}$ is positive. There are only finitely many pairs, and hence their minimum δ_* is positive. □

Proof of assertion (i) of Theorem 1.1. Fix $J \Subset (0,2)$. At each $b_0 \in J$, take a finite subcover of S^3 by the open coefficient windows above, intersect their modulus intervals and take the minima of their radial and Hausdorff thresholds. Choose each chart's phase neighborhood to contain the graphic and closed enlarged section cores. Their continuity (Lemmas 7.1 and A.2) allows a common shrinking of this modulus interval preserving these inclusions over its closure. Now take a finite cover of J by such intervals I_ν, and compact subintervals $J_\nu \Subset I_\nu$ still covering J. The corresponding finite chart sets $\mathcal{I}_\nu$ and fixed phase neighborhoods satisfy $\mathcal{G}_{\nu i} \Subset \mathcal{O}_{\nu i}$, with the notation of Lemma 7.1.

By Lemma 7.1, choose a compact phase neighborhood K of $\mathcal{G}_J$ containing the finitely many required transverse sections. The same lemma gives a common positive clearance δ_* for the chartwise inclusions $\mathcal{G}_{\nu i} \Subset \mathcal{O}_{\nu i}$. Choose the common Hausdorff radius below δ_*, all local phase clearances, and all section-core radii. Let ε_* be the minimum of the positive radial thresholds, and take

$$V = \left\{ \mu : \frac{\mu_1^2 + \mu_2^2 + \mu_3^2}{\varepsilon_*^4} + \frac{\mu_4^2}{\varepsilon_*^6} < 1 \right\}. \tag{7.3}$$

Equation (2.2) puts every nonzero $\mu \in V$ below the common radial threshold.

For fixed $b \in J$ and nonzero raw parameter, select a compact core J_ν containing b and an applicable coefficient window. Its regional result bounds the total actual multiplicity of all isolated cycles at the chosen Hausdorff distance: Lemma A.2 places them on the same signed source interval. Every regional bound is at most two, so selecting any applicable window gives the bound two for this field. Each regional result already counts multiplicities on all its transition intervals and finite seams; in particular, a quadratic late zero cannot be accompanied by an early or threshold zero. At $\mu = 0$, Lemma B.2 excludes positive periodic orbits. Finite transition endpoints are included; equilibrium limits and infinite-time passages are not periodic orbits. Thus one K, V, ρ_{H} works for the whole interval J. □

8. Two actual orbits following the whole graphic

Fix $b \in (0,2)$ throughout this section. We use the subfamily

$$F = y(1+bz-y)+\varepsilon^2(z^2-z+c), \qquad G = zy-\nu, \qquad c,\varepsilon>0, \quad \nu \geq 0. \tag{8.1}$$

The weak saddle passage has a power-law exponent below one. We combine it with a strongly contracting complement whose splitting changes sign. The complement must be defined even when its output lies below the stable branch: only its positive outputs enter the saddle passage, and the boundary between them supplies the first displacement sign. We therefore construct it on a full signed rectangle, including the unstable outlet, before choosing periodic roots.

8.1. **The signed complement.** Set

$$z = cX, \qquad y = c\varepsilon^2 Y, \qquad e = \varepsilon/\sqrt{c}, \qquad \lambda_+ = \nu/(c^2\varepsilon^2), \qquad \tau = ct_{\mathrm{phys}}.$$

Then the exact equations are

$$X' = e^2\mathcal{F}(X,Y), \quad Y' = XY-\lambda_+, \quad \mathcal{F} = Y(1+bcX-c^2e^2Y)+1-X+cX^2. \tag{8.2}$$

Near the smaller positive equilibrium, we select the invariant graph containing its weak stable branch. The finite-order product construction of Section 3, applied in the proof of Lemma 8.1, gives coordinates (ξ,ω) and a positive time variable s such that

$$\frac{d\xi}{ds} = -r_s\xi, \qquad \frac{d\omega}{ds} = \omega, \qquad r_s = e^2\beta_s, \qquad 0<\beta_0\leq\beta_s\leq\beta_1. \tag{8.3}$$

For each fixed field, r_s is its constant characteristic ratio. The stable branch is $\omega = 0$ and the unstable branch is $\xi = 0$. Fix an incoming cut at $\theta = \theta_b$ and an outgoing cut

$$\Sigma_{\mathrm{out}} = \{\omega = \omega_0,\ \xi = a,\ 0 \leq a \leq a_2\}.$$

The value $a = 0$ is the actual unstable outlet. The incoming weak coordinate $\xi_0 = \xi(\theta_b)$ is a positive unit. The local positive-side passage is exactly

$$D_p(s) = \xi_0(p)(s/\omega_0)^{r_s(p)}, \qquad 0<s<\omega_0. \tag{8.4}$$

Here the argument s is the incoming ω-coordinate; at $s = 0$ the saddle flight is infinite.

Lemma 8.1. *There are fixed cuts, $a_2,\omega_0>0$, $m_N,K_0>0$, and a sufficiently small $c_0>0$ with the following property. The negative cut is $z=-q_g$, with $q_g>0$ fixed. For each fixed $0<c<c_0$ and all sufficiently small $\varepsilon>0$, put*

$$\nu_+ = 4q_gK_0\varepsilon^3e^{-m_N/\varepsilon^2}. \tag{8.5}$$

The forward complement of (8.1), from Σ_{out} to the incoming cut, exists in finite physical time on the entire closed rectangle $[0,a_2]\times[0,\nu_+]$. Its signed output $H_{c,\varepsilon,\nu}(a)$ is continuous there, has a strictly positive source derivative, and satisfies

$$|H_p(a)| < \omega_0/4, \qquad 0<H_p'(a)\leq M_\varepsilon := e^{-m_N/(2\varepsilon^2)}. \tag{8.6}$$

Moreover,

$$H_{c,\varepsilon,0}(0)>0, \qquad H_{c,\varepsilon,\nu_+}(0)<0. \tag{8.7}$$

The images of the whole rectangle lie in one fixed neighborhood of Γ_b. As $c,e,\lambda_+\to 0$, the outgoing passage and regular fast arc converge to the corresponding fast pieces of Γ_b, uniformly in a and ν.

Appendix G proves the lemma by following every outgoing source through the height-phase exit, parabolic passage, fast arc, and moving descent. A fixed negative slow interval contracts both height differences and source derivatives. Its contraction dominates the subsequent positive amplification and the finite powers of ε^{-1} incurred upstream. A final positive-time adjustment gives the signed output on the incoming cut, including negative outputs, which are outside the domain of D_p. Constants with a subscript c are fixed after c is chosen and absorbed by decreasing ε afterward.

8.2. **A homoclinic boundary and two roots of one field.** Fix c in the lemma's range and a constant $r_0 \in (0,1)$. Before selecting the splitting parameter, choose ε in that range so that $e^2\beta_1 < r_0$. Then $0 < r_s(p) < r_0$ holds throughout $0 \le \nu \le \nu_+$. Keep c, ε fixed and set $\alpha(\nu) = H_{c,\varepsilon,\nu}(0)$. By continuity and (8.7),

$$\nu_* = \inf\{\nu \in [0, \nu_+] : \alpha(\nu) < 0\}$$

belongs to $(0, \nu_+)$, satisfies $\alpha(\nu_*) = 0$, and is approached by parameters in the negative set. At this parameter the unstable outlet follows the finite complement to the stable branch of the same saddle, giving a homoclinic orbit. The sign change and continuity suffice for this choice of ν_*.

At $\nu = \nu_*$, write $h_1 = H'(0) > 0$. As $a \downarrow 0$,

$$\frac{D_p(H_p(a))}{a} \sim \xi_0 \omega_0^{-r_s} h_1^{r_s} a^{r_s - 1} \longrightarrow \infty. \tag{8.8}$$

Choose $a_1 \in (0, a_2)$ where the return displacement is positive. On the other hand, $H_p(a_2) \le M_\varepsilon a_2$, so

$$D_p(H_p(a_2)) \le C_D e^{-\beta_0 m_N/(2c)} < a_2/4. \tag{8.9}$$

The unit bound for ξ_0 and the fixed cuts determine C_D. The last inequality is imposed in the initial choice of c, before ε and a_1 are chosen.

Choose a single $\nu > 0$ in the negative set sufficiently close to ν_* that $H_p(a_1) > 0$ and the positive middle displacement persist. Since H_p is strictly increasing and $H_p(0) < 0$, there is a unique $a_{\min} \in (0, a_1)$ with $H_p(a_{\min}) = 0$. The actual returning interval is

$$J_p = (a_{\min}, a_2], \qquad \Pi_p(a) = D_p(H_p(a)). \tag{8.10}$$

Its left displacement limit is $-a_{\min} < 0$. Its middle displacement at a_1 is positive, and its displacement at a_2 is negative by (8.9). Thus there are distinct roots

$$a^- \in (a_{\min}, a_1), \qquad a^+ \in (a_1, a_2), \qquad \Pi_p(a^-) = a^-, \quad \Pi_p(a^+) = a^+. \tag{8.11}$$

For $\alpha < 0$ the estimate $H_p(a) \le M_\varepsilon a$ still holds, hence every output of (8.10) lies below $a_2/4$ in the same outgoing collar. Every root has $H_p(a) > 0$ and finite canonical saddle time $T_s = \log(\omega_0/H_p(a))$. Since $ds = b_N(\theta)d\tau$ and $b_N \ge b_0 > 0$, its physical time is at most $T_s/(cb_0)$.

8.3. **Minimal return and isolation.** The complement and D_p are passages of the same field on the same cuts. To show that their composition is the first return on J_p, we exclude an earlier intersection with the outgoing cut. Indeed ω increases after the outgoing hit until the horizontal positive cut is reached, then Y increases to e^{-2}. Thereafter the physical height is at least c^2 before descent, well above the $O(c\varepsilon^2)$ outgoing collar. The regular outer tube is separated from that collar, and the descending and negative pieces have negative z. The signed connector has $z \le z_v < c\theta_b$. Its incoming adjustment has $\theta \le \theta_b$, while the outgoing cut has $\theta \ge \theta_b + \sigma$. Finally D_p increases ω to ω_0 exactly once. These height and spatial clearances exclude every earlier hit.

To prove isolation, transfer this return to the physical analytic section

$$S_h = \{y = c\varepsilon^2\delta_1\}$$

inside the positive outgoing tube, with coordinate $q = -z/c$. Along the positive local passage D_p, put $u = Y - h(X) > 0$, where h is the selected stable graph. The incoming speed $\mathcal{F}$ is positive. At a first zero of $\mathcal{F}$, the equation for $u = Y - h(X)$ gives $Y' > 0$, so $\mathcal{F}' = \mathcal{F}_Y Y' > 0$, since $\mathcal{F}_Y > 0$ in the tube. Thus $\mathcal{F} > 0$ throughout this passage. At a zero of $Y' = XY - \lambda_+$ its derivative is $e^2\mathcal{F}Y > 0$, so Y has at most one critical point, a minimum. Both local endpoint heights are below δ_1; this leg therefore cannot meet S_h before the intended continuation beyond Σ_{out}. The intended hit has positive vertical speed. The finite map $L_p(a) = -x_1(a, p)$ from the outgoing cut to S_h is a C^j orientation-preserving diffeomorphism. Its actual first return is $R_p = L_p \circ \Pi_p \circ L_p^{-1}$. At every interior returning point, the polynomial flow and its transverse hitting time are analytic. Thus R_p is analytic on the connected physical returning interval. The displacement is not identically zero because it is positive at $L_p(a_1)$. Its interior zeros are

therefore isolated. The two roots in (8.11) give distinct isolated orbits: a single orbit with two outgoing hits would contradict the established first return. They lie in $y > 0$: on the axis $G = -\nu < 0$, so an orbit through the positive outgoing cut could cross downward only and could not close after doing so. Simplicity will follow from the upper bound in Section 9.

8.4. **Each orbit converges to the entire graphic.** Fix an open neighborhood $\mathcal{U}$ of Γ_b. All regular tubes, foot boxes, and axis strips are selected inside it before the realizing parameters are chosen. The whole complement rectangle has the following image bounds:

saddle and initial exit:	$\lvert z\rvert \le Cc, \quad \lvert y\rvert \le Cc\varepsilon^2,$
height-phase exit:	$\lvert z\rvert \le Cc, \quad 0 < y \le c^2,$
positive parabolic passage:	$\gamma(c^2+z^2) \le y \le M(c^2+z^2),$
negative slow passage:	$\lvert y\rvert \le K_0\varepsilon^3 + C_h\nu,$
signed connector:	$\lvert y\rvert \le C_c\varepsilon e^{-(m_N-2I_\sigma c)/\varepsilon^2}.$

The parabolic and outer passages converge on every compact fast subarc avoiding the endpoints. Their moving descending foot $\Lambda_p(a)$ converges to f_b, uniformly on the source rectangle. The low slow passage has $F > 0$ and covers all abscissae from $\Lambda_p(a)$ to $z_v = O(c)$. The final descending tail from $y = \rho$ to $K\varepsilon^3$ has abscissa displacement $O(\rho) + O(\varepsilon^2(1 + |\log\varepsilon|))$, so it stays in the foot's $O(\rho) + o(1)$ ball.

Choose $c_n \downarrow 0$ first, with $c_n < c_{n-1}/2$ and all fixed c inequalities satisfied. For each c_n choose ε_n small enough to satisfy its finite-power smallness conditions, $e_n < 1/n$, $\varepsilon_n < 1/n$, the common local windows, and the uniform image error required at that stage. Also require $\nu_{+,n}/(c_n^2\varepsilon_n^2) < 1/n$. The joint fast convergence fixes the required smallness of c_n before ε_n is chosen. Then choose one ν_n and both roots as above.

For either root separately, the local and initial parabolic pieces lie in the origin's $O(\rho)+O(c_n)$ ball until $z = \rho$. The remaining compact fast piece converges to γ_b, and the descending tail lies in the foot's $O(\rho)+o(1)$ ball. The slow graph has uniformly vanishing height and covers $[\Lambda_n, z_v]$. These facts give $\sup_{x\in L_n^\pm} \operatorname{dist}(x, \Gamma_b) \to 0$.

For the opposite inclusion, every compact fast subarc avoiding the endpoint balls is covered by each actual fast passage, with uniform parametrized convergence. Every point of $[f_b + \rho, -\rho]$ is covered in abscissa by each monotone slow passage, with height tending to zero. The remaining graphic tails lie in the endpoint balls; each orbit has a cutoff point tending to the foot and a slow point at $z = -\rho$. Their distance from those tails is $O(\rho) + o(1)$. First take ρ small and then n large. This gives (1.6) for each orbit individually and proves the existence and isolation part of assertion (ii).

9. Multiplicity and exactness

Fix $b \in (0,2)$ and a compact $J \Subset (0,2)$ containing b. The upper assertion supplies common K, V, ρ_{H} for J. The two orbits constructed in Section 8 belong to the same field, are distinct and isolated, and each converges to the whole graphic. For all sufficiently large n, both are contained in K, both satisfy the same Hausdorff condition, and $\mu_n \in V$. Hence

$$1 \le m(L_n^-), \qquad 1 \le m(L_n^+), \qquad m(L_n^-) + m(L_n^+) \le 2.$$

Both actual multiplicities are one, and there is no third isolated whole-graphic cycle in the counted neighborhood.

Let $a_n^- < a_n^+$ be the two roots in (8.11), and write $\Pi_n = D_{p_n} \circ H_{p_n}$ for their actual minimal return. On its returning interval, $D'_{p_n} > 0$ and $H'_{p_n} > 0$, so $\Pi'_n > 0$. Every zero gives an isolated cycle by the analytic-section argument of Section 8. The image estimates on the full source rectangle place each such cycle in the same counted neighborhood for all sufficiently large n. A further zero would therefore contradict the total multiplicity bound. Hence these are the only two zeros on the returning interval. Conjugacy to the physical analytic section preserves the

derivative at a fixed point, so multiplicity one gives $\Pi_n'(a_n^{\pm}) - 1 \neq 0$. The negative–positive–negative displacement signs in (8.10) now imply

$$(\Pi_n - \mathrm{id})'(a_n^-) > 0, \qquad (\Pi_n - \mathrm{id})'(a_n^+) < 0.$$

Consequently

$$\Pi_n'(a_n^-) > 1, \qquad 0 < \Pi_n'(a_n^+) < 1. \tag{9.1}$$

These are the characteristic multipliers of the planar positive-time return. They are preserved by changes of transverse section or section orientation. Thus L_n^- is hyperbolic repelling and L_n^+ is hyperbolic attracting, with the labels ordered by the coordinate a.

The realizing sequence gives $C_{\mathrm{H}}(b) \geq 2$ at each fixed b. Assertion (i) gives cyclicity at most two at that same b, and uniformly on compact modulus intervals. This completes Theorem 1.1. The realizing sequence is chosen separately for each fixed $b \in (0, 2)$.

10. Full coefficient space and open realization

Identify the space $\mathcal{Q}_2$ of real planar quadratic vector fields with $\mathbb{R}^{12}$ by their coefficients. The affine and positive-time normalization of Dumortier and Rousseau transfers the whole-graphic count to this full coefficient space.

Corollary 10.1. *Fix $b_0 \in (0, 2)$. There are an open coefficient neighborhood $\mathcal{W} \subset \mathcal{Q}_2$ of $X_{b_0,0}$ and $\eta > 0$ such that every $Q \in \mathcal{W}$ satisfies*

$$\sum_{\substack{L \text{ isolated periodic orbit of } Q \\ d_{\mathrm{H}}(L,\Gamma_{b_0})<\eta}} m_Q(L) \leq 2.$$

There is a sequence $Q_n \to X_{b_0,0}$ in $\mathcal{Q}_2$ for which each Q_n has two distinct isolated periodic orbits converging separately to Γ_{b_0} in two-sided Hausdorff distance. Both orbits are eventually hyperbolic. Thus the whole-graphic cyclicity of Γ_{b_0} in the full quadratic coefficient space is exactly two.

Proof. We verify the local normalization of [13, Proposition 2.1] by computing its differential. For $d \in \mathbb{R}^2$, $A \in GL(2, \mathbb{R})$ and $c > 0$, set

$$\Phi(d, A, c, b, \mu)(u) = cAX_{b,\mu}(A^{-1}(u - d)).$$

Its domain and target both have dimension twelve. At $(0, I, 1, b_0, 0)$, an affine variation $V = (p + az + \ell y, q + ez + fy)$ contributes $DVX - DXV$, where $X = X_{b_0,0}$; write κ, β, ν for the variations of c, b, μ. If the total variation vanishes, its second component is

$$\nu_4 - qz + (e - p)y - ez^2 + (b_0 e - a + \kappa)zy - (e + \ell)y^2 = 0.$$

Hence $q = e = p = \ell = \nu_4 = 0$ and $\kappa = a$. The y and y^2 coefficients of the first component now give $f = 2a$ and $2a = 0$. Its remaining coefficients give $\beta = \nu_1 = \nu_2 = \nu_3 = 0$. The differential is therefore invertible. The analytic inverse-function theorem supplies, for all Q in one coefficient neighborhood, analytic choices $H_Q(u) = A_Q u + d_Q$, $c_Q > 0$, b_Q and μ_Q such that

$$Q(H_Q(u)) = c_Q A_Q X_{b_Q,\mu_Q}(u), \qquad (H_Q, c_Q, b_Q, \mu_Q) \longrightarrow (\mathrm{id}, 1, b_0, 0).$$

Choose a compact interval $J \Subset (0, 2)$ with b_0 in its interior, and let K, V, ρ_{H} be given by Theorem 1.1(i). The fast terminal intersection satisfies $y' = f_b < 0$, so continuous dependence and transversality give Hausdorff continuity of $b \mapsto \Gamma_b$ (Appendix A). Consequently

$$E_Q := d_{\mathrm{H}}(H_Q \Gamma_{b_Q}, \Gamma_{b_0}) \longrightarrow 0.$$

After shrinking the coefficient neighborhood, let $M \geq 1$ bound both $\|A_Q\|$ and $\|A_Q^{-1}\|$. For $\widetilde{L} = H_Q^{-1} L$, the two triangle inequalities give

$$d_{\mathrm{H}}(\widetilde{L}, \Gamma_{b_Q}) \leq M(d_{\mathrm{H}}(L, \Gamma_{b_0}) + E_Q),$$
$$d_{\mathrm{H}}(L, \Gamma_{b_0}) \leq M d_{\mathrm{H}}(\widetilde{L}, \Gamma_{b_Q}) + E_Q.$$

Choose $\kappa_0 > 0$ so that the κ_0-neighborhood of $\bigcup_{b\in J}\Gamma_b$ is contained in K, and set $\theta = \min(\kappa_0, \rho_{\mathrm{H}})$. Shrink $\mathcal{W}$ so that $b_Q \in J$, $\mu_Q \in V$ and $E_Q < \theta/(4M)$, and take $\eta = \theta/(4M)$. Every orbit counted in the corollary then gives $\widetilde{L} \subset K$ and $d_{\mathrm{H}}(\widetilde{L}, \Gamma_{b_Q}) < \theta/2$. These inequalities also show that fixed-base and moving-graphic Hausdorff convergence are equivalent as $Q \to X_{b_0,0}$.

The positive constant time factor and affine map preserve isolated periodic orbits, their minimal returns and their multiplicities, by Section 2. Theorem 1.1(i) gives the upper bound.

Finally, the full coefficient space contains $Q_n = X_{b_0,\mu_n}$ from Theorem 1.1(ii). These fields eventually belong to $\mathcal{W}$ and realize two separate whole-graphic limits at the fixed modulus b_0. This proves sharpness in $\mathcal{Q}_2$. □

Corollary 10.2 (Open realization)**.** *Fix $b_0 \in (0,2)$, and let $\mathcal{W}$ and η be as in Corollary 10.1. There are fields $Q_n \to X_{b_0,0}$ and nonempty open coefficient balls U_n centered at Q_n, with*

$$Q_n \in U_n \subset \mathcal{W} \cap B(Q_n, 1/n) \subset \mathcal{Q}_2$$

such that, for every $Q \in U_n$, the isolated periodic orbits satisfying $d_{\mathrm{H}}(L, \Gamma_{b_0}) < \eta$ consist exactly of two hyperbolic cycles $L_n^-(Q)$ and $L_n^+(Q)$. Their characteristic multipliers satisfy

$$\lambda_Q(L_n^-(Q)) > 1, \qquad 0 < \lambda_Q(L_n^+(Q)) < 1,$$

and each cycle converges to the whole graphic uniformly on its ball:

$$\max_{\sigma\in\{-,+\}} \sup_{Q\in U_n} d_{\mathrm{H}}(L_n^\sigma(Q), \Gamma_{b_0}) \longrightarrow 0.$$

Moreover, $\sup_{Q\in U_n}\|Q - X_{b_0,0}\| \to 0$. For the local normalization $Q \mapsto (b_Q, \mu_Q)$ constructed above, the balls may be chosen so that $\nu(Q) := -(\mu_Q)_4 > 0$ throughout each U_n.

Proof. Take $Q_n = X_{b_0,\mu_n}$ from Theorem 1.1(ii) and delete finitely many terms so that $Q_n \in \mathcal{W}$ and the two cycles have the multipliers in (9.1) and each lies within Hausdorff distance $\eta/2$ of Γ_{b_0}. Put

$$d_n = \max_{\sigma\in\{-,+\}} d_{\mathrm{H}}(L_n^\sigma, \Gamma_{b_0}) \longrightarrow 0.$$

For each fixed n, hyperbolic persistence gives distinct continuations $L_n^\sigma(Q)$ in disjoint tubes. Indeed the finite minimal returns are analytic and their displacement derivatives are $\lambda - 1 \neq 0$, so the implicit-function theorem applies. The same flow-box clearances preserve minimality, and finite-orbit continuous dependence gives Hausdorff continuity; the multipliers remain on their respective sides of one.

Shrink an open coefficient ball U_n about Q_n so that $U_n \subset \mathcal{W} \cap B(Q_n, 1/n)$ and, for every $Q \in U_n$,

$$d_{\mathrm{H}}(L_n^\sigma(Q), L_n^\sigma) < 1/n, \qquad d_{\mathrm{H}}(L_n^\sigma(Q), \Gamma_{b_0}) < \eta \quad (\sigma \in \{-,+\}).$$

The two hyperbolic cycles each have multiplicity one. Corollary 10.1 excludes a third counted isolated cycle, proving the asserted exact count. The triangle inequality gives

$$\max_{\sigma\in\{-,+\}} \sup_{Q\in U_n} d_{\mathrm{H}}(L_n^\sigma(Q), \Gamma_{b_0}) \le d_n + 1/n \longrightarrow 0,$$

and $\sup_{Q\in U_n}\|Q - X_{b_0,0}\| \le \|Q_n - X_{b_0,0}\| + 1/n \to 0$.

The local inverse defining the normalization gives $\mu_{Q_n} = \mu_n$, whose fourth component is $-\nu_n < 0$. By continuity, a further shrinking of U_n ensures $(\mu_Q)_4 < -\nu_n/2 < 0$ throughout the ball. The radii are chosen separately for each n, so finite flight time for each realizing cycle suffices even if their periods increase. □

Appendix A. The graphic, sections, and complete displacement

All estimates below are uniform on a sufficiently small compact interval $I_b \Subset (0,2)$ about a prescribed b_*.

We use the physical displacement (2.1) from Section 2. The table records the regional coordinates and domains before they are transferred to those sections.

TABLE 1. Sections, coordinates, maps, and actual domains.

Object	Source and target	Map	Actual domain and orientation
Regular branch	$\Sigma_u(u_{\mathrm{up}})$ to $\Sigma_\xi(E_b)$	T_-	fixed regular core; $(T_-)_{u_{\mathrm{up}}} = -j < 0$
Contact branch	$\Sigma_u(u_{\mathrm{up}})$ to $\Sigma_\xi(E_b)$	T_+	open finite-passage components common with T_-
Displacement	same source and target	$\Delta = T_+ - T_-$	zeros are periodic orbits; isolated zeros are limit cycles
Negative passage	prescribed physical source to $z = -Ln$	J_-	positive right-going finite passage; local source coordinate u is not u_{up}
Positive complement	positive-saddle outgoing cut to $z = z_c$	C_+	relative neighborhood of a same-field positive reference; oriented so $C_+' > 0$
Saddle collars	fixed incoming to fixed outgoing cuts	$\mathcal{D}$	open finite-time side; a zero-amplitude separatrix is a boundary
Root collision	left base to through-passage or fiber-exit target	branch maps	localized actual interval $0 < A < \eta_0$; at most one finite event value
Lower connector	lower power output to the positive-saddle incoming cut	Ψ	increasing physical passage interval; $\Psi(0) = -B_{\mathrm{conn}} < 0$ is retained

A.1. **Fast orbits and transverse sections.** We first identify the limiting fast arc and then choose sections that persist for the perturbed fields. This supplies the geometric input for all subsequent transition maps. Set $\omega_b = \sqrt{4 - b^2}/2$. The solution of (1.2) through the origin is

$$z(t) = \frac{e^{bt/2}}{\omega_b} \sin(\omega_b t), \qquad y(t) = 1 + e^{bt/2} \left(-\cos(\omega_b t) + \frac{b}{2\omega_b} \sin(\omega_b t) \right). \tag{A.1}$$

The ordinate increases until $t = \pi/\omega_b$ and then decreases until $2\pi/\omega_b$, where it is negative. Its first positive zero t_b is therefore unique and belongs to $(\pi/\omega_b, 2\pi/\omega_b)$. At that zero $z(t_b) < 0$. The implicit function theorem shows that $f_b = z(t_b)$ and the whole arc depend smoothly on b. In particular $-f_b$ has a positive lower bound on I_b. The arc is simple: it is part of the trajectory of the linear focus (1.2), before its first return to the horizontal axis. Thus (1.3) is a Jordan curve. Continuous dependence, together with the transverse return at f_b, also proves $\Gamma_x^b \to \Gamma_0^b$ in Hausdorff distance as $x \downarrow 0$, uniformly on I_b.

The upper intersection of the arc with $z = 0$ has height

$$y_b = 1 + \exp\left(\frac{\pi b}{\sqrt{4 - b^2}} \right).$$

We use the section $\Sigma_u = \{(0, y_b + u_{\mathrm{up}}) : |u_{\mathrm{up}}| < u_0\}$ and a small outgoing section Σ_ξ on $z = \xi > 0$, near the lower intersection with the fast arc. The coordinate u_{up} ranges over a fixed interval containing zero, independently of every later rescaling. The endpoint parameter x indexes the unperturbed family of graphics. At the contact endpoint $x = 0$, every regional construction is pulled back to the regular physical coordinate u_{up}; x is not a return coordinate.

Lemma A.1. *On a fixed rectangle about the contact point where $1 + bz - y > 0$, there is an analytic first integral E_b of (1.2) such that*

$$(E_b)_y > 0, \qquad (E_b)_z + \frac{z}{1 + bz - y} (E_b)_y = 0, \qquad E_b(0, y) = y. \tag{A.2}$$

Its y derivative is a positive unit on a smaller rectangle. The transition along any fixed compact part of the fast arc, written in this first-integral coordinate at its two ends, converges to the

identity in C^k on the prescribed compact source interval, uniformly for $b \in I_b$, for every fixed finite k, as $\mu \to 0$.

Proof. Since the z component of (1.2) is nonzero there, each orbit meets $z = 0$ once locally. Define E_b to be the ordinate of that intersection. The analytic dependence theorem gives (A.2) and $(E_b)_y(0,0) = 1$. Continue this first integral along the required regular fast arc. The perturbed transition between fixed transverse sections depends smoothly on the parameters; at $\mu = 0$ its energy coordinate is constant. This proves the last assertion. Sections must be fixed before taking this limit. □

For $\mu \neq 0$ we shall write the family exactly as

$$F := \dot{z} = y(1 + bz - y) + \varepsilon^2 P(z), \qquad G := \dot{y} = zy + \varepsilon^3 D, \qquad P(z) = E_0 + E_1 z + E_2 z^2, \tag{A.3}$$

with bounded coefficients. The normalization covering every raw parameter is defined in Section 2; Section 7 completes its finite directional cover.

Lemma A.2. *After shrinking the parameter and Hausdorff neighborhoods, every cycle counted in Theorem 1.1 lies in $y > 0$ and meets Σ_u and every member of any prescribed finite collection of regular fast sections exactly once, inside prescribed smaller section intervals. The backward transition T_- from Σ_u to Σ_ξ, measured by E_b, satisfies*

$$(T_-)_{u_{\mathrm{up}}} = -j(u_{\mathrm{up}}, p), \qquad c \le j \le C, \qquad |\partial_{u_{\mathrm{up}}} \log j| \le C. \tag{A.4}$$

Here and below p denotes the fixed parameters of a transition.

Moreover, if P_p is the transition between two fixed vertical sections contained in a compact regular flow tube on the fast arc, then, after one common shrinking of the parameter neighborhood,

$$0 < c \le |P_p'| \le C, \qquad \left|\frac{P_p''}{P_p'}\right| + \left|\left(\frac{P_p''}{P_p'}\right)'\right| \le C. \tag{A.5}$$

If the transverse speeds on the two sections have the same fixed sign, as for the holonomies on the outgoing regular arc, then $P_p' > 0$. The same assertion holds when either section is measured in the energy coordinate E_b.

For bounded coefficients in (A.3)*, the forward orbit from the same signed section reaches $y = K\varepsilon^3$ at $z = \Lambda(u_{\mathrm{up}}, p)$ in a fixed negative interval about f_b, where*

$$-C \le \Lambda_{u_{\mathrm{up}}} \le -c < 0, \qquad |\Lambda_{u_{\mathrm{up}} u_{\mathrm{up}}}| \le C. \tag{A.6}$$

Suppose in addition that $P \ge p_0 > 0$ on a fixed interval $[z_{\mathrm{ent}}, -q_g]$, where $z_{\mathrm{ent}} < -q_g$, from these entries to $-q_g < 0$. Then every counted cycle reaches $z = -q_g$ at $0 < y \le K_1\varepsilon^3$, and its subsequent arc to Σ_ξ has $F > 0$. The union of admissible passages is represented on one open interval of u_{up} by a forward transition T_+ to the same target. Its intersections with T_- are precisely the periodic orbits obtained by finite passages there; isolated intersections represent limit cycles.

Proof. At each regular marker the organizing field is nonzero and transverse to its section. A uniformly small flow box implies that any orbit meeting a smaller ball also meets the section inside its prescribed core. One Hausdorff inclusion provides such a point on every counted cycle. Choose the sections so that their portions outside the prescribed cores have positive distance from the graphic. The other inclusion then confines every intersection to those cores. A periodic orbit is a Jordan curve. As a transverse arc is traversed, its intersections with that curve alternate between entering and leaving the bounded component. Their oriented intersection signs alternate, whereas the field has a fixed transverse sign on the section. Hence there can be only one intersection.

One marker has a fixed positive ordinate. If $\mu_4 \neq 0$, every crossing of $y = 0$ has the same sign, which is impossible for a periodic ordinate. If $\mu_4 = 0$, that line is invariant; a cycle close to the graphic cannot cross it, and Hausdorff closeness to the positive fast arc excludes a cycle in

$y < 0$. Thus the entire cycle lies in $y > 0$. The compact regular arc from Σ_ξ to Σ_u persists. Fix a closed flow tube around this arc, with slightly larger section cores. On the limiting field the tube contains no zero and the transverse speeds on its vertical sections are bounded away from zero. Compactness in b and continuity give the same bounds, a common flight-time bound, and common C^4 bounds for the normalized field after shrinking the parameter neighborhood. The target event equation has time derivative bounded below, so the implicit-function theorem and the first three variational equations give a common C^3 bound for every such holonomy map.

Writing $X_p = (F_p, G_p)$ for the normalized vector field, a transition $P_p(y_0) = y_1$ between vertical sections satisfies the planar determinant identity

$$F_p(z_1, y_1)P_p'(y_0) = F_p(z_0, y_0) \exp\left(\int_0^{T(y_0,p)} \operatorname{div} X_p(\phi_t)\, dt \right). \tag{A.7}$$

The event-time correction is parallel to the target vector field and does not alter this identity. The uniform speed and flight-time bounds give a positive lower bound for $|P_p'|$ as well as an upper bound. On the outgoing regular arc the two transverse speeds have the same fixed sign, hence $P_p' > 0$. Combining these facts with the C^3 estimate yields (A.5). In the energy coordinate, the section change has derivative $(E_b)_y$, a positive C^3 unit on the fixed tube, so the composition rules preserve the same bounds. Taking the two sections to be Σ_u and Σ_ξ, the signs $F < 0$ on Σ_u and $F > 0$ on Σ_ξ give the negative orientation $(T_-)_{u_{\mathrm{up}}} = -j(u_{\mathrm{up}}, p)$ with j a positive unit, as in (A.4). Its limiting energy derivative does not depend on the final position of Σ_ξ on that arc.

Follow the descending regular arc to a fixed small height y_h. For $K\varepsilon^3 \le y \le y_h$ and z in the negative entry interval, choose K so that $zy + \varepsilon^3 D \le -cy$. With y as independent variable, the difference between the perturbed and divided fast equations is exactly

$$\frac{\varepsilon^2 z P(z) - \varepsilon^3 D(1 + bz - y)}{z(zy + \varepsilon^3 D)}. \tag{A.8}$$

This function and its first two z derivatives are bounded by $C\varepsilon^2/y$. Integration and the first two variational equations give a C^2 error of size $O(\varepsilon^2(1 + |\log \varepsilon|))$ in the entry abscissa. The unperturbed map from the upper section to the lower fast foot has negative derivative bounded away from zero, by the determinant formula on the divided regular field. This proves (A.6) without a sign assumption on P.

Under the additional positive-polynomial hypothesis, put $h_s = -\varepsilon^3 D/z$ and $w = y - h_s$. Then

$$\begin{aligned} F &= \varepsilon^2 P + h_s(1 + bz - h_s) + (1 + bz - 2h_s)w - w^2, \\ \dot{w} &= zw - h_s' F. \end{aligned}$$

On $|w| \le K_1\varepsilon^3$, the horizontal speed is $\frac{1}{2}p_0\varepsilon^2 \le F \le C\varepsilon^2$. The two w boundaries point inward, because $z \le -q_g$ and $h_s' = O(\varepsilon^3)$. The tube contains the entry and reaches $z = -q_g$ in finite time. Its transition is strictly monotone for each positive ε; no uniform lower derivative is required.

Choose a small rectangle $[-q_g, \xi] \times (0, \eta)$ with $F_y = 1 + bz - 2y > 0$. Its top and the part of its right side outside the outgoing section core have positive distance from the graphic. At the left side $F > 0$. A counted cycle cannot leave through the bottom, left, or top, and must leave in finite time to return to Σ_u. It therefore exits at the prescribed right section. To exclude a zero of F before that exit, use the exact identities

$$\dot{F} = F_z F + F_y G, \qquad \dot{G} = yF + zG.$$

The quadrant $F \le 0, G \le 0$ is positively invariant in the rectangle. At a first zero of F reached from $F > 0$, one has $G \le 0$; equality would give an equilibrium. Entering that quadrant prevents the required right exit. Thus $F > 0$ on the whole arc.

Finally, use z as independent variable there. Between two passing solutions of $y' = G/F$, scalar uniqueness orders every intermediate solution. It stays between their graphs and has positive speed, since $F_y > 0$. Its target remains in the interval between their targets. The domain

of such passages is therefore an open interval. Taking its inverse image under the monotone preceding transition gives the asserted interval of u_{up}. Equality $T_+(u_{\mathrm{up}}) = T_-(u_{\mathrm{up}})$ joins two finite pieces of the same orbit; conversely every counted cycle gives that equality. Uniqueness identifies distinct zeros with distinct periodic orbits. □

For uniform regional estimates, fix the finite output order, the physical sections and enlarged cores first. Then choose the selected reference orbits, fibers and saddle collars, followed by compact coefficient blocks, root gaps, barriers and tube widths. The final parameter and Hausdorff shrinkings respect these choices; the actual finite-passage interval is established before any return comparison.

A.2. **An explicit graphic-anchored annulus.** Fix a compact interval $J \Subset (0,2)$ for this geometric construction. Put $o = (0,1)$, $v = y - 1$, and use the divided fast field $z' = bz - v$, $v' = z$ only for this geometric construction. Along its nonzero solutions the continuous polar angle about o satisfies

$$\theta' = \frac{z^2 - bzv + v^2}{z^2 + v^2} \geq 1 - \frac{\max J}{2} > 0.$$

The explicit fast orbit starts with angle $-\pi/2$, passes through the upper $z = 0$ point, and reaches the foot with angle $\theta_f(b) \in (\pi, 3\pi/2)$. Its first and second parts have $z > 0$ and $z < 0$, respectively, so it has not made an additional turn. Along the axis segment from the foot to the origin the angle continues strictly upward to $3\pi/2$, since $\det((z,-1),(1,0)) = 1$. Thus the whole graphic is a single positive radial graph

$$g_b(\theta) = o + R_b(\theta)e_\theta, \qquad e_\theta = (\cos\theta, \sin\theta).$$

On the fast part its radius is obtained by the strictly monotone angle-to-time inverse. On the axis part it is $-1/\sin\theta$. The two expressions agree at the foot and at the circular endpoint. The function $R_b(\theta)$ is jointly continuous, piecewise smooth, and has common positive lower and upper bounds on J. No derivative matching is claimed at the foot.

Choose $0 < d < \min R_b/2$, also smaller than the upper-source core. The explicit map

$$\Phi_b(\theta, t) = o + (R_b(\theta) + t)e_\theta, \qquad |t| \leq d$$

is globally injective. Different angles give different nonzero radial rays and each radius is strictly increasing in t. Its continuous inverse is $\theta = \arg(x - o)$, $t = |x - o| - R_b(\arg(x - o))$. It therefore embeds the entire cylinder, including both graphic endpoints. Its fibers are anchored exactly on Γ_b and $|\Phi_b(\theta, t) - g_b(\theta)| = |t|$. At $\theta = \pi/2$ the fiber is precisely $(0, y_b + t)$, the original vertical source, whose transverse speed has a fixed sign. Only joint continuity of R_b is needed, so the radial construction also applies at the singular endpoints.

Let $\mathcal{A}_b(r) = \Phi_b(S^1 \times (-r, r))$. An essential curve in this annulus meets every fiber: otherwise cutting at a missed fiber puts it in a rectangle and makes it contractible. Hence it is within two-sided Hausdorff distance r of the graphic. Conversely, for any fixed smaller annulus, joint continuity of the inverse gives a common Hausdorff radius placing every cycle within that distance of the graphic in the annulus. Lemma A.2 makes it cross the upper fiber exactly once. A contractible Jordan curve would leave both ends of this crosscut in its exterior and have an even number of transverse intersections. Thus the cycle is essential.

For two nested essential counted cycles in the smaller annulus, their intervening closed annulus remains there. The inner boundary connects to o without meeting either cycle and lies inside both; the outer boundary connects to infinity and lies outside both. Choose $\mathcal{A}_- \Subset \mathcal{A}_+$ with the larger annulus within the phase and Hausdorff bounds of the regional multiplier estimate, and place the final counted cycles in the smaller one. Their disks and intervening annuli are in $y > 0$, since a Jordan disk's lowest ordinate is attained on its boundary.

A.3. **An annulus argument.** When attraction can be proved over a whole period, we can count cycles without constructing a return map across every intervening separatrix. The annulus argument below turns that multiplier estimate into a bound in terms of the number of equilibria.

Lemma A.3. *Suppose a planar C^1 field has at most M equilibria in an annular neighborhood, and every periodic orbit essential in that annulus is hyperbolic attracting. Then any collection of such cycles whose intervening closed annuli remain in the neighborhood has at most $M+1$ members.*

Proof. Take any finite collection of nested cycles and let E be the closed annulus between two consecutive cycles. Suppose E has no equilibrium. Both boundary cycles are invariant; uniqueness prevents crossing them, and compact continuation gives a complete two-sided flow on E. The basin of the inner cycle relative to E is nonempty and open, while an attracting neighborhood of the outer cycle is disjoint from it. Its nonempty relative boundary is compact, invariant, and separated from both cycles. The omega-limit set of a point on this boundary is nonempty and contains no equilibrium. The Poincaré–Bendixson theorem [18, Section 7.3, Lemma 7.13] makes it a periodic orbit. If that orbit were contractible in the original annular neighborhood, its Jordan disk would lie inside E. Its index is one, whereas a zero-free disk has index zero [17, Section 3.12, Theorems 1–3]. Thus it is essential and, by hypothesis, attracting. The chosen boundary point belongs to its open basin, which is disjoint from the inner cycle's basin. This contradicts the definition of that boundary. Each intervening annulus therefore contains an equilibrium. There are at most M such disjoint annuli. Applying the argument to any finite subcollection proves the result also against a proposed infinite collection. □

Lemma A.4. *Let X_p be a planar family continuous in the C^1 topology at p_0. Suppose the counted cycles of X_{p_0} are hyperbolic attracting, lie in $y>0$, and are essential in an annular neighborhood $\mathcal{A}_- \Subset \mathcal{A}_+$, with their intervening closed annuli contained in $\mathcal{A}_-$. Suppose there are admissible parameters $p_k \to p_0$ such that every periodic orbit of X_{p_k} contained in $\mathcal{A}_+ \cap \{y>0\}$ and essential in $\mathcal{A}_+$ is hyperbolic attracting, and X_{p_k} has at most four equilibria in $y>0$, all nondegenerate. Then there are at most two counted cycles of X_{p_0}. The admissible sequence may lie in a one-sided parameter set.*

Proof. Fix a finite collection of n counted cycles and disjoint compact tubes in $\mathcal{A}_- \cap \{y>0\}$. Their finite minimal returns have $P'-1 \neq 0$, so hyperbolic persistence gives one nearby cycle in each tube [18, Section 12.2, Lemmas 12.6–12.7]. Transverse end hits and separated middle arcs preserve minimality. The persisted cycles remain positive, distinct, attracting, and nested; their intervening annuli remain in $\mathcal{A}_+$ and their bounded interiors in $y>0$. The required perturbation neighborhood may depend on the finite collection. For all sufficiently large k, apply the following argument to X_{p_k}. The vector-field index along a periodic orbit in the positive geometric orientation is $+1$, whether the flow traverses it clockwise or counterclockwise. The innermost disk therefore contains an equilibrium. Lemma A.3, applied to each intervening annulus using the attracting hypothesis for X_{p_k}, shows that it too contains an equilibrium. Its index sum is $1-1=0$. A nonempty collection of nondegenerate equilibria with zero index sum has at least two members, because the index of each member is $+1$ or -1. All these disjoint regions lie in $y>0$, where the total equilibrium bound is four. Hence

$$1+2(n-1)\le 4,$$

and $n \le 2$. The argument for arbitrary finite collections excludes an infinite collection as well. □

For the present polynomial family the nondegenerate parameters required in this lemma form a dense set. When $\delta=\mu_4\neq 0$, equilibria correspond to roots of

$$Q(z)=z^2(\mu_1+\mu_2 z+\mu_3 z^2)-\delta z-b\delta z^2-\delta^2,$$

and $Q'(z) = z \det D(F, G)$ at an equilibrium. The multiple-root condition is a proper algebraic condition in the four raw parameters. One exact witness, valid for each fixed b, is

$$\kappa = 6/625, \quad \delta = 12/25, \quad \mu_3 = -6/625, \quad \mu_2 = -12/125, \quad \mu_1 = (12/25)b - 42/125.$$

It gives $Q = -\kappa(z+1)(z+2)(z+3)(z+4)$ and discriminant $144\kappa^6 > 0$. A nonzero real polynomial cannot vanish on an open box, by the one-variable polynomial identity theorem applied successively to its coefficients. Its nonzero set is therefore dense in every open raw window. The witness proves properness and need not belong to the small parameter window. If a multiplier estimate holds only for $\delta \geq 0$, at a field with $\delta = 0$ choose perturbations with $\delta > 0$: every coefficient neighborhood has a nonempty open part on that side, which the algebraic exceptional set cannot contain. For each fixed positive ε, the map from (E_0, E_1, E_2, D) to the four raw coefficients is an invertible diagonal linear map. Thus the generic perturbations can be chosen within the same coefficient chart before radial normalization. The attracting hypothesis on $\mathcal{A}_+$ must be supplied by the regional multiplier estimate; it does not follow from persistence of the original finite collection.

Lemma A.5. *For sufficiently small raw parameters, system* (1.1) *has at most four equilibria in* $y > 0$.

Proof. Write $\delta = \mu_4$. If $\delta \neq 0$, then $y = -\delta/z$ and $z \neq 0$, so the abscissa satisfies

$$z^2(\mu_1 + \mu_2 z + \mu_3 z^2) - \delta z - b\delta z^2 - \delta^2 = 0.$$

This is a nonzero polynomial of degree at most four. Each root determines at most one positive ordinate. If $\delta = 0$, a positive equilibrium has $z = 0$ and $y - y^2 + \mu_1 = 0$, giving at most two. □

By Lemma A.2, every counted cycle lies in $y > 0$. Its Jordan disk also lies there, since the lowest ordinate of a bounded Jordan disk is attained on its boundary. Thus the intervening annuli lie in $y > 0$, and Lemma A.5 supplies the equilibrium bound used with Lemma A.3.

Appendix B. Global exclusion and multiplier reduction

With the coefficients of P and D in fixed compact sets, we exclude limiting directions with a negative polynomial value or strictly negative forcing. We then treat two open regions used in Section 7: a positive contact value gives at most one attracting cycle by the common-cycle theorem, and positive forcing gives at most two by the complete-period multiplier and the annulus count. Neither attracting argument requires P to be positive along the whole critical segment. The remaining contact-scale directions are treated in Sections 5 and 6, and Appendix E.

Lemma B.1. *A sufficiently small neighborhood of a normalized direction is free of cycles close to* $\Gamma_0^{b_*}$ *if either*

(1) $P(z_g) < 0$ *for some* $z_g \in [f_{b_*}, 0]$, *or*
(2) $D < 0$.

In each assertion the inequality is imposed on the limiting direction.

Proof. In the first case continuity permits an interior point $f_{b_*} < z_g < 0$ and a neighborhood on which $P(z_g) \leq -p_0 < 0$. The descending entry in Lemma A.2 has $y = K\varepsilon^3$ and $z = \Lambda < z_g$, with a fixed gap. Choose $H > K$ so that $z_g H + \sup|D| < 0$. On the whole half-strip $z \leq z_g$, the boundary $y = H\varepsilon^3$ points downward. Thus the orbit remains below that height until its first crossing of $z = z_g$ from the left. Such a crossing is required to complete the cycle. There, however,

$$F(z_g, y) \leq CH\varepsilon^3 - p_0\varepsilon^2 < 0,$$

a contradiction. Leftward excursions do not affect this argument.

In the second case take $D \leq -d_0 < 0$ on a neighborhood. Starting at the same descending entry, positivity can last at most K/d_0 units of time, because $G \leq -d_0\varepsilon^3$ while $z < 0$. On a fixed negative interval about the entry, $0 < y \leq K\varepsilon^3$ and $|F| \leq C\varepsilon^2$. A stopping-time argument

shows that neither vertical boundary can be reached within that time for small ε. The orbit must therefore reach $y = 0$, contrary to Lemma A.2. □

The gate argument excludes a strictly negative limiting forcing. When the forcing instead tends to zero, an exact Liénard transformation gives the following necessary restriction.

Lemma B.2. *Assume ε is sufficiently small that $1 + 2\varepsilon^2 E_2 > 0$ on the fixed coefficient set. If $D \le 0$, a positive periodic orbit of* (A.3) *requires $E_1 < 0$. If $D < 0$, it also requires*

$$|D| < \frac{\varepsilon E_1^2}{4b(1 + 2\varepsilon^2 E_2)}. \tag{B.1}$$

In particular there is no positive periodic orbit at $\mu = 0$, or when $E_0 = E_1 = D = 0$.

Proof. Put $a = \mu_3$, $\ell = \mu_2$, $k = \mu_1$, $\delta = \mu_4$, $v = \log y$, and $w = z + \delta e^{-v} = \dot{v}$. Direct substitution gives

$$\dot{w} = aw^2 + \big(be^v + \ell - (1 + 2a)\delta e^{-v}\big)w + e^v - e^{2v} - b\delta + k - \ell\delta e^{-v} + a\delta^2 e^{-2v}.$$

Define $q = (1 - e^{-av})/a$ when $a \neq 0$, and $q = v$ when $a = 0$; set $p = e^{-av}w$. Since $dq/dv = e^{-av} > 0$, this is a diffeomorphism of $y > 0$ onto an open strip. The transformed equation is $\dot{q} = p$, $\dot{p} = g(q) + f(q)p$, where

$$f(q) = by + \ell - (1 + 2a)\delta/y.$$

The energy $p^2/2 - \int g(q)\,dq$ has derivative $f(q)p^2$. The energy returns to its initial value on a nonconstant periodic orbit. Thus f must take both signs, since $yf = by^2 + \ell y - (1 + 2a)\delta$ is not the zero polynomial. For $\delta \le 0$ this requires $\ell < 0$; for $\delta < 0$ its discriminant must be strictly positive. Substitution of the normalized coefficients gives (B.1). □

Proposition B.3. *Every limiting direction with $P(0) > 0$ has an open parameter neighborhood on which at most one cycle is close to the whole graphic. If present, it is hyperbolic attracting and has actual minimal-return multiplicity one.*

Proof. Fix a limiting polynomial P_* with $P_*(0) > 0$ and a modulus b_*. If $P_* < 0$ somewhere on $[f_{b_*}, 0]$, use Lemma B.1. Otherwise $P_* \ge 0$ on this segment. Embed the actual field in the smooth family $X_{h,\lambda} = (yQ + hP, zy + h\kappa)$, where $Q = 1 + bz - y$, $h \ge 0$ and $\lambda = (b, E_0, E_1, E_2, \kappa)$ are independent parameters in a compact ambient block. Equation (A.3) is exactly its restriction $h = \varepsilon^2$, $\kappa = \varepsilon D$; the base value is $\kappa = 0$.

Here the contact is nilpotent of order two. Indeed, set $Y = E_b(z, y)$ and $\phi_b(z) = E_b(z, 0)$ in Lemma A.1; then $\phi_b''(0) = -1$. The analytic unit $d = (E_b(z, y) - \phi_b(z))/y$ is positive. Multiplication by $d/Q > 0$ gives $\dot{z} = Y - \phi_b(z) + hA$, $\dot{Y} = hB$, where

$$A = \frac{d}{Q}P, \qquad B = \frac{d}{Q}\big((E_b)_z P + (E_b)_y \kappa\big).$$

The shear $Y \mapsto Y + hA$ removes the first perturbation; its leading added term $A_z(Y - \phi_b)$ vanishes on the critical curve. Using $(E_b)_z = -z(E_b)_y/Q$ and $(E_b)_y(0, 0) = 1$, the slow term on that curve at $h = \kappa = 0$ is

$$g_0(z) = -\left(\frac{(E_b)_y(z, 0)}{1 + bz}\right)^2 zP_*(z), \qquad g_0'(0) = -P_*(0) < 0.$$

Thus the contact singularity is simple and has intrinsic index $+1$ in the convention of [5, pp. 175–178]. In the original family's slow time $s = ht$, the slow flow is $dz/ds = P_*(z)$, directed from f_{b_*} toward zero between its zeros, and the normal eigenvalue is $z < 0$. Since $P_* \not\equiv 0$, it has finitely many zeros on $[f_{b_*}, 0)$; each lies on a normally attracting hyperbolic arc. If none are present, the slow divergence integral is the finite, strictly negative integral $\int_{f_{b_*}}^{0} z/P_*(z)\,dz$.

Apply [5, Theorem 1, p. 173], prescribing a small neighborhood W of the contact disjoint from a fixed regular fast marker. It supplies a phase neighborhood T, a threshold $h_0 > 0$, and a neighborhood of the independent parameters. Continuity of Γ_0^b in b and a common two-sided

Hausdorff clearance ensure that every counted cycle lies in T and visits the marker neighborhood, so is not contained in W. Choose the coefficient window with $P(0) \geq p_* > 0$ and $|D| \leq M_D$; then sufficiently small ε puts $h = \varepsilon^2$ and $\kappa = \varepsilon D$ in the supplied product window. There is therefore at most one counted cycle, and, if present, it is hyperbolic attracting. Its actual minimal-return multiplier lies in $(0, 1)$, so its return zero is simple. All thresholds are chosen after the positive margin $p_* > 0$ has been fixed. □

Proposition B.4. *Every limiting direction with $P(0) = 0$ and $D > 0$ has an open parameter neighborhood on which there are at most two cycles close to the whole graphic, and every such cycle is hyperbolic attracting. This includes the direction $P \equiv 0$ with $D > 0$.*

Proof. Work with $d_0 \leq D \leq D_0$, $|E_1| + |E_2| \leq M$, and $|E_0| \leq \eta$, where η will be fixed small. Write $Q = 1 + bz - y$ and $\delta = \varepsilon^3 D$. Take a low contact rectangle on which $3/4 \leq Q \leq 5/4$, and fix a small $a > 0$ with $2aQ \leq 1/2$. On the curve $y = az^2$, for $z = -x < 0$,

$$G - 2azF = az^3(1 - 2aQ) + \delta - 2az\varepsilon^2 P(z) \leq -\tfrac{1}{2}ax^3 + D_0\varepsilon^3 + Ca\varepsilon^2 x < 0 \tag{B.2}$$

whenever $x \geq L\varepsilon$, for a sufficiently large fixed L. Choose $q > 0$ so small that the descending entry of Lemma A.2 lies at $\Lambda < -2q$, $y = K\varepsilon^3$. For $z \leq -q$ use instead $y = H\varepsilon^2$; this upper boundary points downward since $G \leq -qH\varepsilon^2 + C\varepsilon^3 < 0$. At their connecting vertical boundary $z = -q$, $H\varepsilon^2 \leq y \leq aq^2$, choose H so that $F \geq y/2 > 0$. These two strips form a forward invariant upper bound up to the first hit of $z = -L\varepsilon$. The lower boundary points inward because $G|_{y=0} = \delta > 0$. On the remote strip, bounded coefficients give

$$\operatorname{div}(F, G) = z + by + \varepsilon^2 P'(z) \leq -q/2, \qquad |F| \leq C\varepsilon^2. \tag{B.3}$$

Every such passage traverses $[-2q, -q]$; even with reversals in z, its time there is at least $q/(C\varepsilon^2)$. Its divergence contribution is therefore at most $-c_0\varepsilon^{-2}$, where $c_0 = q^2/(2C) > 0$. The remaining remote contribution is nonpositive. In the parabolic strip, after reducing q and then ε,

$$z + by + \varepsilon^2 P'(z) \leq -x + bax^2 + C\varepsilon^2 \leq -x/2.$$

The cycle reaches the inner scale with $y \leq aL^2\varepsilon^2$ and accumulated divergence at most $-c_0\varepsilon^{-2}$. Reversals on the negative side only increase the time with negative divergence.

Set $z = \varepsilon U$, $y = \varepsilon^2 V$, and $d\tau = \varepsilon\, dt$. The exact equation is

$$U' = V(1 + b\varepsilon U - \varepsilon^2 V) + P(\varepsilon U), \qquad V' = UV + D. \tag{B.4}$$

Choose $v > 0$ with $(L+1)v < d_0/4$, then η sufficiently small, and finally ε so that $|P(\varepsilon U)| \leq 2\eta$ for $|U| \leq L + 1$. On $0 \leq V \leq v$, $V' \geq d_0/2$. This portion lasts at most $2v/d_0$; by reducing v and η, its displacement in U is less than $1/2$. Once $V \geq v$, the boundary $V = v$ points upward and $U' \geq V/2 \geq v/2$. Thus $U = L$ is reached in a bounded τ-time, and Gronwall's inequality gives $v \leq V(L) \leq B_1$ for a fixed B_1. Continuation proves these estimates on a fixed box, after choosing $\varepsilon^2 B_1$ below the physical height bound. The bounded τ-time and scaled divergence $U + b\varepsilon V + \varepsilon P'(\varepsilon U)$ give a bounded core divergence integral.

For $U \geq L$, take $\nu = v/4$ and fix

$$0 < \gamma \leq \min\{1/8,\, v/(4L^2)\}.$$

Then $\nu + \gamma L^2 \leq v/2 < v$. Shrink the fixed physical rectangle so that $Q = 1 + b\varepsilon U - \varepsilon^2 V \leq 5/4$. Thus $1 - 2\gamma Q \geq 11/16$. The curve $V = \nu + \gamma U^2$ is a lower barrier. Indeed, after the ratio below is made less than $1/8$,

$$V' - 2\gamma UU' = UV(1 - 2\gamma Q) + D - 2\gamma U P(\varepsilon U) \geq (11/16 - \gamma/4)UV + D > 0.$$

The required ratio is controlled because

$$\frac{|P(\varepsilon U)|}{\nu + \gamma U^2} \leq \frac{\eta}{\nu} + \frac{M\varepsilon}{2\sqrt{\nu\gamma}} + \frac{M\varepsilon^2}{\gamma}$$

can be made less than $1/8$, choosing γ, η, and ε in that order. It follows that

$$y \geq \nu\varepsilon^2 + \gamma z^2, \qquad F \geq cy. \tag{B.5}$$

Choose the fixed positive section only after these constants have been fixed. Take $\xi > 0$ and a height $y_* > 0$ so small that, after shrinking ε, the estimate below gives $y \le y_*$ for $0 \le z \le \xi$, while $1 + bz - y \ge 3/4$ on that same rectangle. The choice of γ already makes $|P(\varepsilon U)|/V < 1/8$ there, so $F = \varepsilon^2(VQ + P(\varepsilon U)) \ge cy$ is valid on one common rectangle. The inequality $dy/dz \le Cz + C\delta/(\nu\varepsilon^2 + \gamma z^2)$ then gives $y \le C(\varepsilon^2 + z^2)$ and a finite hit of this fixed positive section. Hausdorff closeness places that hit inside its outgoing core.

On this positive arc, $z = (\log y)^{\cdot} - \delta/y$ and $\delta > 0$. Consequently

$$\int z\,dt \le C + 2|\log\varepsilon|, \qquad \int by\,dt \le C, \qquad \int \varepsilon^2 P'(z)\,dt \le C\varepsilon^2 \int \frac{dz}{\nu\varepsilon^2 + \gamma z^2} \le C\varepsilon.$$

The fixed regular fast arc contributes a bounded divergence integral. Choose the fixed entry height in Lemma A.2 small enough that the remaining descending relaxation lies in a fixed negative-z strip with nonpositive divergence. Adding these contributions gives

$$\log \mathfrak{m} \le -c_0\varepsilon^{-2} + C(1 + |\log\varepsilon|) < 0.$$

Choose the larger geometric annulus inside the common multiplier neighborhood, with a smaller annulus containing the counted cycles and their intervening annuli. The attracting estimate holds on that larger annulus throughout the coefficient neighborhood with $D > 0$. Choose the generic perturbations of Lemma A.4 inside this neighborhood. Together with the four-equilibrium bound in the whole positive half-plane, this gives at most two cycles. Every choice used only d_0, D_0, M and a fixed compact b interval, so the assertion is an open-neighborhood estimate at the stated limiting direction. □

B.1. **The ambient genericity window.** The raw-parameter genericity argument following Lemma A.4 makes nondegenerate fields dense in every open raw window. To apply it in the two-endpoint chart of Section 6, retain an ambient coefficient window before radial normalization. For fixed $\varepsilon > 0$ the two-endpoint chart has raw map

$$(A, C, c, D) \mapsto (\varepsilon^2 A^2 c, -\varepsilon^2 A, \varepsilon^2 C, \varepsilon^3 D), \qquad \det D\Psi = -A^2\varepsilon^9 \ne 0.$$

The nonzero Jacobian gives an open four-dimensional ambient window. Reserve a strict forcing bound larger than the inner bound, so that the multiplier estimate holds on the larger annulus throughout this window. Generic fields can then be chosen on the $D > 0$ side, also when the original field has $D = 0$. These perturbations are made in the ambient coefficient chart before sphere normalization.

The multiplier estimate on the larger annulus and this open ambient window supply all hypotheses of Lemma A.4, with Lemma A.5 giving the four-equilibrium bound. Thus there are at most two cycles. Each attracting hyperbolic cycle has actual multiplicity one.

Appendix C. Uniform Dulac estimates

The return comparisons require Dulac derivative estimates that remain uniform as the characteristic ratio passes through one. We construct common section coordinates on compact hyperbolic saddle families and bound the logarithmic curvature and its Euler derivative. All source derivatives keep the field parameters fixed.

Lemma C.1. *Fix an integer $J \ge 4$. Let p range over a compact subset of a finite-dimensional parameter manifold covered by finitely many charts. Assume that the vector field, its saddle branch, and the embeddings of two oriented transverse sections are jointly C^{J+12} in phase, section, and chart parameters, with common bounds. Suppose that the saddle eigenvalues are $\lambda_u(p) > 0$ and $-\lambda_s(p) < 0$, with $\lambda_u, \lambda_s \ge \lambda_* > 0$, and that $r(p) = \lambda_s(p)/\lambda_u(p)$ lies in $[r_-, r_+] \Subset (0, \infty)$. After shortening the sections uniformly, the local coordinates and section changes may be chosen jointly C^J, and the Dulac map D_p satisfies, for $0 < s \le s_0$,*

$$D_p' > 0, \qquad D_p' \le K s^{r(p)-1}, \qquad \frac{sD_p''}{D_p'} = r(p) - 1 + E_D(s,p), \qquad |E_D| + |sE_D'| \le K s^{\nu} \tag{C.1}$$

with common $K, s_0, \nu > 0$. Primes and E'_D denote source derivatives at fixed p. The estimates include $r = 1$; when r varies, no joint C^J extension of D_p to $s = 0$ is asserted.

Proof. Put $N = J + 12$. The hypotheses supply common C^N phase–parameter bounds after one common phase shrinking. Cover the compact parameter set by finitely many charts on which the two eigenline bundles have C^{N-1} frames with common bounds. Translate the saddle to the origin and use these frames on each chart to obtain

$$\dot\xi = \lambda_u(p)\xi + F_u(\xi,\eta,p), \qquad \dot\eta = -\lambda_s(p)\eta + F_s(\xi,\eta,p), \tag{C.2}$$

where $F(0,p) = D_{(\xi,\eta)}F(0,p) = 0$. Compact hyperbolicity and the finite cover give a common $\lambda_* > 0$ such that $\lambda_u, \lambda_s \ge \lambda_*$, common C^{N-1} bounds for the coefficients, and a common bound for the frames and their inverses. The loss of one derivative here accounts for differentiating the field when forming its spectral projections.

To obtain common separatrix jets in phase and parameters, use the Lyapunov–Perron equation for stable trajectories of (C.2). For an initial stable coordinate a, it is

$$\begin{aligned} \xi(t) &= -\int_t^\infty e^{\lambda_u(p)(t-q)} F_u(\xi(q),\eta(q),p)\,dq, \\ \eta(t) &= e^{-\lambda_s(p)t}a + \int_0^t e^{-\lambda_s(p)(t-q)} F_s(\xi(q),\eta(q),p)\,dq. \end{aligned} \tag{C.3}$$

In a fixed exponential trajectory norm with weight below λ_*, a common cutoff makes this operator a contraction with constant $q_0 < 1$, uniformly on each parameter chart. Differentiating its fixed-point equation gives

$$(I - D_w\mathcal{T}_p)D^m w = \mathcal{R}_m, \qquad \|(I - D_w\mathcal{T}_p)^{-1}\| \le (1-q_0)^{-1},$$

where $\mathcal{R}_m$ uses only lower trajectory jets and coefficients through order m. Kernel parameter derivatives add polynomial factors in time, absorbed by the spectral gap. Induction through $N-1$ gives common joint stable-graph jets; time reversal gives the unstable graph. Take finite minima of the chart radii. Local frames suffice, without a global orientation of either eigenline bundle.

Straighten the two invariant graphs successively. Invariance and Hadamard division then give

$$\dot x = xA(x,y,p), \qquad \dot y = -yC(x,y,p), \qquad A(0,0,p) = \lambda_u(p), \quad C(0,0,p) = \lambda_s(p).$$

After one common shrinking, $A > 0$; division by this positive time factor puts the equations in the orbital form

$$\dot x = x, \qquad \dot y = -yR(x,y,p), \qquad R(0,0,p) = r(p) = \lambda_s(p)/\lambda_u(p).$$

Absorb the pure-axis terms by one-dimensional coordinate changes. First put $y_1 = q(x)y$, where

$$q(x) = \exp\int_0^x \frac{R(t,0) - r}{t}\,dt.$$

Then $\dot y_1 = -y_1 R_1(x, y_1)$ with $R_1(x,0) = r$. The further change

$$y_2 = \phi(y_1), \qquad \phi(v) = v\exp\int_0^v \frac{r/R_1(0,t) - 1}{t}\,dt$$

makes the coefficient equal to r on both axes. Each quotient has a removable singularity and common finite derivatives. The remaining coefficient difference vanishes on both axes, so two successive Hadamard factorizations make it divisible by xy. Dropping subscripts, the resulting local orbital form is

$$\dot x = x, \qquad \dot y = -y[r + xyB(x,y,p)],$$

with common finite derivatives of B; the mixed term is retained, including at resonance. The derivative orders at each step are

operation	input order	output order
spectral projections and eigenframes	$J+12$	$J+11$
two graph straightenings and push-forwards	$J+11$	$J+9$
two initial Hadamard divisions by the axes	$J+9$	$J+7$
two one-variable quotient changes and push-forwards	$J+7$	$J+3$
two final Hadamard divisions producing xyB	$J+3$	$J+1$
regular section hit and coordinate transfer	$J+1$	J

Every division is applied only after the required vanishing identity has been established. The coordinate Jacobians and the transverse event derivatives are bounded away from zero on the finite cover, so inversion costs no additional derivative. Thus $N = J+12$ leaves the coordinates, their inverses, B, and the section maps jointly C^J, with common bounds.

We now estimate the Dulac map in these coordinates; see also [16]. Fix p and suppress it from the notation. For incoming (s, y_0) and outgoing $x = x_0$, factor out the saddle power by writing $y = (s/x)^r w$, $w(s; s) = y_0$, and $\mathcal{E} = s\partial_s$. Then

$$w_x = -(s/x)^r w^2 B(x, (s/x)^r w, p).$$

Choose the section sizes so that $y_0/2 \le w \le 2y_0$. For $0 < \nu < \min(1, r_-)$, reduced once if necessary,

$$\int_s^{x_0} (s/x)^r (1 + |\log(x/s)|^j)\, dx \le K_j s^\nu \quad (0 \le j \le 3).$$

The first three Euler differentiations give triangular variational equations whose coefficients and inhomogeneous terms are bounded by multiples of $(s/x)^r$. The initial point also moves with s, so its differentiated values satisfy

$$(\mathcal{E}^j w)(s; s) = s\frac{d}{ds}(\mathcal{E}^{j-1} w)(s; s) - s(\partial_x \mathcal{E}^{j-1} w)(s; s).$$

They are $O(s)$, successively for $j = 1, 2, 3$. Gronwall's inequality therefore gives $|w - y_0| + \sum_{j=1}^3 |\mathcal{E}^j w| \le K s^\nu$. At $x = x_0$ let $W = w(x_0; s)$ and $M = rW + \mathcal{E}W \asymp 1$. Since $D' = x_0^{-r} s^{r-1} M$,

$$E_D = \frac{\mathcal{E}M}{M}, \qquad \mathcal{E}^2 M = r\mathcal{E}^2 W + \mathcal{E}^3 W,$$

which proves (C.1) in each chart. On chart overlaps, the source and target coordinates differ by positive regular germs with common C^J bounds. The chain rule changes the displayed logarithmic jet, and its Euler derivative, by $O(s) + O(D(s))$. Decreasing $\nu < \min(1, r_-)$ once, these terms are absorbed in Ks^ν. Taking the maximum of the finitely many constants proves the uniform assertion. No exact analytic linearization or nonresonance hypothesis was used. □

Appendix D. Local passage tools at a simple contact

We treat a contact whose linear coefficient is negative, with estimates uniform as that coefficient tends to zero. This uniformity allows the same tools to be used at the quadratic contact in Section 5. Throughout this section the field is

$$F = y(1 + bz - y) + \varepsilon^2(Cz^2 - Az + A^2 c), \qquad G = zy + \varepsilon^3 A^3 d. \tag{D.1}$$

The constants in estimates are independent of the positive small parameters. The letter C in (D.1) always denotes the quadratic coefficient, not a generic constant. All derivatives in the estimates below are taken in phase or section variables, with the field parameters fixed.

To bound the zeros of the complete displacement, we compare a local contact passage with its complete complementary transition. We first estimate the negative and positive transitions in regular section coordinates, then construct the local saddle passages. These common estimates will be used in both contact-scale regimes below.

The return maps below are defined using the sections of Lemma A.2. In this section u denotes the physical ordinate on its outgoing section $z = z_c > 0$, where $F > 0$; it is not the coordinate on the upper section. We put the negative section at $z = -q_g$, $q_g > 0$. Every cycle being counted meets the positive section exactly once and has, in the contact neighborhood, $y > 0$, $F > 0$ and $y(-q_g) \le K_0\varepsilon^3$. We decrease that neighborhood so that $1/2 < 1 + bz - y < 2$ and $F_y > 1/2$ there.

D.1. **The negative transition.** The negative transition contracts strongly enough to absorb the shrinking transverse scale in both regimes. Write

$$c = t^2\kappa, \quad d = t^3\eta, \quad \varepsilon = te, \qquad 0 < e \le 1, \quad |\kappa|, |\eta| \le K, \quad n = At^2. \tag{D.2}$$

We shall take t small; e need not be bounded away from zero. Set $p = \varepsilon^2 P$ and $R = G/F$.

The central contact passage uses the finite-order saddle-node product of [8, Theorem 3 and Section 2.8], at pp. 6 and 11 of the authors' public manuscript. We apply it in the central contact chart of Appendix E. Its planar critical germ has one zero and one negative eigenvalue; the center reduction has a nonzero quadratic term. These data match the multiplicity-two translated critical field there; parameters form a finite-dimensional smooth unfolding. The differentiability order is prescribed before choosing its smaller parameter window. It supplies a finite-smooth orbital product. The fixed-gap product used away from the central saddle-node is proved in Section 3. Neither construction uses analytic linearization across degeneration or a positive-endpoint cyclicity bound.

Lemma D.1. *Assume $0 < A \le A_+ < \infty$ and either*

1. *C belongs to a fixed compact subinterval of $(0, \infty)$; or*
2. *$A \ge A_0 > 0$, C is bounded, and $P(z) \ge p_0 > 0$ on the fixed negative interval preceding the contact neighborhood.*

In the second case choose the fixed negative cut q_g sufficiently small in terms of these bounds. With the scales in (D.2)*, fix L sufficiently large and then take t sufficiently small. Every positive, right-going transition from the prescribed physical source, with $y(-q_g) \le K_0\varepsilon^3$, satisfies on its passage from $z = -q_g$ to $z = -Ln$*

$$0 < y < p, \qquad p \le F \le 3p, \qquad P(-x) \asymp x(x + A) \quad (Ln \le x \le q_g). \tag{D.3}$$

If $J_-(u) = y(-Ln; u)$ includes the preceding fast part of the orbit, then

$$0 < (J_-)_u \le K_1\varepsilon e^{-\gamma T_A/\varepsilon^2}, \qquad 0 \le \frac{(J_-)_{uu}}{(J_-)_u} \le K_1\varepsilon^{-2}, \qquad T_A = \log\frac{q_g + A}{A(1 + Lt^2)}. \tag{D.4}$$

Here $T_A \ge t_0 + t_1 \log(A_+/A)$ for some $t_0, t_1 > 0$. The transverse ordinate scale is $n^2e^2 = A^2\varepsilon^2t^2$. In particular,

$$\frac{(J_-)_u}{A^2\varepsilon^2t^2} \longrightarrow 0 \tag{D.5}$$

uniformly in A, t, e.

Proof. If C belongs to a fixed compact subinterval of $(0, \infty)$, $P(-x) \ge Cx^2 + Ax/2$ for $x \ge LAt^2$ after taking $L \ge 2K$. If instead $A \ge A_0 > 0$, C is bounded, and P has a fixed positive lower bound on the preceding negative interval, choose q_g with $|C|q_g \le A_0/8$ and $K/L \le 1/8$. Then $3Ax/4 \le P(-x) \le 5Ax/4$, which gives the same comparability in (D.3). At the graph $y = p(z)$,

$$G - p_zF \le \varepsilon^2 xP(-x)\left(-1 + \frac{3e^2}{L} + 6|C|\varepsilon^2 + \frac{2eK}{L^2}\right) < 0.$$

The incoming height is below $p(-q_g)$ for small ε. This proves the barrier by first contact.

For later use we record the exact derivatives. With $Q = 1 + bz$ and $\Delta = \varepsilon^3 A^3 t^3 \eta$,

$$R_y = \frac{z(y^2 + p) - \Delta(Q - 2y)}{F^2}, \tag{D.6}$$

$$R_{yy} = \frac{2\{z[y^3 + (3y - Q)p] + \Delta[p + Q^2 - 3Qy + 3y^2]\}}{F^3}. \tag{D.7}$$

On the small negative tail $|\Delta|/(xp) \le 2eK/L^2$. The numerator $z[y^3 + (3y - Q)p]$ is positive and comparable to xp. Increasing L, and decreasing the contact neighborhood, therefore gives

$$R_y \le -\frac{\gamma}{\varepsilon^2(x + A)}, \qquad 0 < R_{yy} \le \frac{K_2}{\varepsilon^4 x(x + A)^2}. \tag{D.8}$$

The complete negative transition also includes an entrance whose abscissa moves with u. Follow the descending part of the fast orbit to $y_e = K_e\varepsilon^3$. Its abscissa $\Lambda(u)$ lies in a fixed negative interval, with $0 < k_1 \le \Lambda_u \le k_2$ and $|\Lambda_{uu}| \le k_2$. Indeed, on the descending part,

$$\frac{dz}{dy} = \frac{1 + bz - y + \varepsilon^2 P(z)/y}{z + \Delta/y}, \qquad \int_{y_e}^{y_h} \left(1 + \frac{\varepsilon^2}{y}\right) dy \le K_3;$$

the denominator stays away from zero and the first two variational equations give these bounds. On the subsequent fixed negative interval, $R = -\Theta(\varepsilon)$ at the entrance, $R_y \le -k_3\varepsilon^{-2}$ and $|R_{yy}| \le K_3\varepsilon^{-4}$. The derivative and its logarithmic derivative, at a fixed terminal abscissa, are exactly

$$y_u(z) = -R(\Lambda, y_e)\Lambda_u \exp\int_\Lambda^z R_y, \tag{D.9}$$

$$\partial_u \log y_u(z) = \frac{\Lambda_{uu}}{\Lambda_u} + \left(\frac{R_z}{R} - R_y\right)(\Lambda, y_e)\Lambda_u + \int_\Lambda^z R_{yy} y_u. \tag{D.10}$$

Here $R_z/R = O(1)$ at the entrance. The last integral on the fixed interval has absolute value at most $K_4\varepsilon^{-4}\int_0^{O(1)} \varepsilon e^{-k_3 s/\varepsilon^2} ds = O(\varepsilon^{-1})$. The term $-R_y\Lambda_u$ is at least $k_4\varepsilon^{-2}$. Thus the map to $-q_g$ is increasing and convex, with logarithmic derivative bounded by $K_4\varepsilon^{-2}$. This argument does not require R_{yy} to be positive on the remote interval.

On the tail, (D.8) preserves convexity and bounds $(J_-)_u$. Its contribution to the second variation reduces, with $N = \gamma/\varepsilon^2$, to

$$\frac{K_5\varepsilon^{-3}}{(q_g + A)^N} \int_{LAt^2}^{q_g} \frac{(x + A)^{N-2}}{x}\, dx. \tag{D.11}$$

For $A \le q_g/4$, split at $x = A$. The upper integral, using $x^{-1} \le 2/(x + A)$, is $O(\varepsilon^{-1})$; the lower one is bounded by

$$K_5\varepsilon^{-3}|\log(Lt^2)|A^{-2}\left(\frac{2A}{q_g + A}\right)^N = o(1).$$

For large N the A-dependent factor is increasing up to $q_g/4$, where its exponential ratio is $2/5$. For $q_g/4 \le A \le A_+$ split instead at $x = q_g/2$. The upper integral is again $O(\varepsilon^{-1})$, and the lower one is bounded by $K_6\varepsilon^{-3}(1 + |\log\varepsilon|)\theta^N = o(1)$, where $\theta = (q_g/2 + A_+)/(q_g + A_+) < 1$. We used $t \ge \varepsilon$. In the signed-C case one may equivalently use $P(-x) \asymp Ax$, split at $q_g/2$, and obtain the same estimate. This proves (D.4).

Choose t so that $T_A(A_+, t) > t_0 > 0$. Differentiation with respect to $\log A$ gives $\partial_{\log A} T_A = -q_g/(q_g + A)$, proving its asserted lower bound. Finally the left side of (D.5) is at most

$$K_7\varepsilon^{-3}A^{-2}\left(\frac{A(1 + Lt^2)}{q_g + A}\right)^{\gamma/\varepsilon^2}.$$

For $N > 2 + 2A_+/q_g$, the function $A^{-2}[A/(q_g + A)]^N$ is increasing. Its maximum is at A_+, where the displayed exponential ratio is uniformly below one. The exponential therefore absorbs the shrinking ordinate scale, with no restriction on the relative rates of A, t, e. □

D.2. **Positive transitions and local passage maps.** On the positive transition, a relative transverse coordinate avoids a second-derivative loss over a long flight. The following variational estimate controls that coordinate by its total change.

Lemma D.2. *Suppose $v_x = \alpha(x,v)v$ on a forward interval $x_0 \le x \le x_1$, with $x_1 > x_0$ and $v(x_0,a) = a > 0$, and suppose the selected solutions are monotone in x and remain below v_0. If*

$$|v\alpha_v| + |v^2\alpha_{vv}| \le Kv|\alpha|,$$

then the endpoint map Φ satisfies $a\Phi'/\Phi = 1+O(v_0)$ and $a\Phi''/\Phi' = O(v_0)$. If $\alpha \le 0$, the stronger estimates

$$\Phi' > 0, \qquad \Phi' \le K_1, \qquad |\Phi''/\Phi'| \le K_1 \tag{D.12}$$

hold, independently of its flight time or of $\Phi'(0)$.

Proof. Put $E = av_a/v$ and $B = a\partial_a \log E$. Exact differentiation gives

$$(\log E)_x = v\alpha_v, \qquad B_x = E(v\alpha_v + v^2\alpha_{vv}), \qquad E(x_0) = 1, B(x_0) = 0.$$

Since $\int v|\alpha|dx = |v(x_1) - a| \le v_0$, $|\log E| \le Kv_0$ and $|B| \le Ke^{Kv_0}v_0$. The identity $a\Phi''/\Phi' = E - 1 + B$ proves the first assertion. If $\alpha \le 0$, replace v_0 by a and use $\Phi' = E\Phi/a \le E$. □

The next lemma converts this relative distortion bound into physical section derivatives. Its input is a reference outgoing orbit of the same field, with quadratic height bounds and positive speed. We will construct that orbit separately for the bounded and small saddle ratios.

Lemma D.3. *Put $z = nU$, $y = n^2V$, where $n > 0$ and either $n = A\varepsilon^2$ or $n = Ac$. Consider the scaled equations*

$$U' = V(1 + bnU - n^2V) + q(U), \qquad V' = UV + \ell_1, \tag{D.13}$$

where $q(U) = k - U + C\varepsilon^2U^2$, with bounded k, or $q(U) = e^2(1-U) + C\varepsilon^2U^2$. Let $\mathcal{F}(U,V)$ denote the right-hand side of the first equation. Suppose a reference outgoing orbit of the same system, in particular with the same value of ℓ_1, is written as $V = h(U)$ and reaches a fixed $U_0 > 0$ with a uniform positive speed and has the bounds

$$k_0(1 + U^2) \le h(U) \le K_0(1 + U^2), \qquad \mathcal{F}(U,h) \ge k_0 h, \tag{D.14}$$

up to a fixed small physical section $z = z_c$. For $|\ell_1|$ sufficiently small, a fixed small relative neighborhood of this reference reaches the same section. If its initial transverse coordinate a is oriented so that $V/h - 1 = j_0 a + O(a^2)$, with j_0 a positive unit and bounded first three derivatives, its physical transition $u = C_+(a)$ satisfies

$$0 < C_+' \le K_1 n^{3/2}, \qquad \left|\frac{C_+''}{C_+'}\right| \le K_1. \tag{D.15}$$

The conclusions hold for bounded signed C. The entire relative neighborhood lies in the fixed outgoing section near the limiting fast orbit, after decreasing the parameters. The reference graph and the constants in (D.14) *are required only on the compact U–interval ending at the fixed section $z = z_c$. In each application below that graph, its two barriers, and the positive-speed collar are constructed first on this interval; the parameter neighborhood is then shrunk so that the same collar is valid for all parameters under consideration. No assertion is made for U outside this interval.*

Proof. Write $V = h(1+v)$, $F_0 = \mathcal{F}(U,h)$, $F_v = \mathcal{F}(U,h(1+v))$ and $B_v = 1 + bnU - n^2h(2+v)$. Subtracting the two orbit equations gives

$$v_U = \alpha(U,v)v, \qquad \alpha = -\frac{(1+v)UhB_v + (\ell_1/h)F_0 + \ell_1(1+v)B_v}{F_vF_0}. \tag{D.16}$$

On the physical neighborhood, B_v is a positive unit. By (D.14), $F_v/F_0 = 1 + O(v)$ and h/F_0 is bounded. The two terms involving ℓ_1 are uniformly smaller than the first numerator term, since $U \geq U_0 > 0$ and $h \geq k_0(1 + U^2)$. Thus $\alpha < 0$ and direct differentiation gives

$$|v\alpha_v| + |v^2\alpha_{vv}| \leq K_2|v||\alpha|.$$

Apply Lemma D.2 to v or $-v$. The relative deviation contracts, so it cannot leave the chosen neighborhood; the second estimate in (D.15) follows, including the initial smooth change of section.

To recover the size of C'_+, use (D.6) along the reference neighborhood. The physical constant and forcing terms have sizes $O(n^2)$ and $O(n^3)$, and the term proportional to $-z^2$ has a favorable sign. Since $F \geq k_1 y$ and $y \geq k_1(n^2 + z^2)$,

$$R_y \leq K_3 z + K_3 \frac{\varepsilon^2 z^3 + n^2 z + n^3}{(n^2 + z^2)^2}.$$

Its integral to z_c is at most $K_4 + K_4\varepsilon^2 \log(1/n)$. The physical transverse derivative at the initial scaled section is $O(n^2)$. Hence $C'_+ \leq K_5 n^{2-K_4\varepsilon^2} \leq K_5 n^{3/2}$, where the last inequality uses one common choice ε_0 with $K_4\varepsilon_0^2 \leq 1/2$ and then the common shrinking $n < 1$. Finally $k_2 z^2 \leq y \leq K_6(n^2 + z^2)$ on fixed positive intervals. In the joint limit $n, \varepsilon \to 0$, any limiting graph therefore solves $y_z = z/(1 + bz - y)$ and has $y(0) = 0$. Uniqueness for this regular equation identifies it with the selected fast orbit. The bounds give compactness on every $[z_0, z_c]$ and then convergence as $z_0 \downarrow 0$, uniformly over the relative neighborhood. This proves the last assertion. □

At the central saddle-node, the exact product separates the scalar flight time from normal contraction. It also reduces the fixed-point count to the sign of one derivative involving the complement.

Lemma D.4. *Let X_p be a C^∞ local family, with p in a compact parameter germ, and suppose the phase germ of X_{p_0} is real analytic and has an ordinary multiplicity-two saddle-node with a negative nonzero eigenvalue and a positive quadratic center coefficient. For every integer $k < \infty$, after decreasing the phase–parameter neighborhood, there are parameter-dependent C^k orbital coordinates*

$$\dot{s} = W(s, p), \qquad \dot{w} = -w, \qquad W = u(s, p)((s - \sigma(p))^2 + \mu(p)), \quad u > 0.$$

The coordinates, their inverses, and the regular changes to fixed transverse sections have bounded derivatives through every order fixed before that shrinking. No C^∞ or analytic equivalence on one common phase–parameter neighborhood is asserted. Let $D(s) = w_0 \exp(-\int_s^{s_+} W^{-1})$ be a finite passage from $w = w_0 > 0$ to $s = s_+$. Then $WD_s = D$. On a sufficiently small incoming interval,

$$D(s) \leq w_0 \exp\left(-\frac{k_0}{|s - \sigma| + \sqrt{|\mu|}}\right). \tag{D.17}$$

If an increasing return transition H is defined on an interval of $a > 0$, and on one subinterval containing every solution of $D(H(a)) = a$ one has $W(H) > 0$ and

$$1 - W_s(H(a)) + aH''(a)/H'(a) > 0, \tag{D.18}$$

then the equation has total isolated zero multiplicity at most two on that actual finite-passage subinterval.

Proof. The finite-order orbital product form is the multiplicity-two specialization of [8, Section 2.1, equation (2.2), Theorem 3, and Section 2.8]. More precisely, the source treats a C^∞ unfolding of a real-analytic critical saddle-node germ. Its multiplicity is denoted by $\mu + 1$, so the present case has $\mu = 1$ and, after a possible reversal of time, its normal form is

$$\dot{z} = b_0(p) + \frac{z^2}{1 + \alpha(p)z}, \qquad \dot{w} = -w,$$

because its polynomial $P(p,z)$ has degree at most $\mu-1=0$ in the multiplicity-two case. This is an exact product, not a formal jet. Moreover

$$b_0+\frac{z^2}{1+\alpha z}=\frac{(z-\sigma)^2+\widehat{\mu}}{1+\alpha z},\qquad \sigma=-\frac{\alpha b_0}{2},\qquad \widehat{\mu}=b_0-\frac{\alpha^2 b_0^2}{4}.$$

Shrinking the phase neighborhood makes $(1+\alpha z)^{-1}$ a positive unit, which gives the asserted W after renaming z and $\widehat{\mu}$. The source states finite smoothness on a phase–parameter neighborhood and that decreasing this neighborhood permits a higher prescribed finite order; Section 2.8 treats a multidimensional unfolding. We therefore include all compact passive parameters in p before choosing k and shrinking. Only the resulting C^k coordinates and regular C^k section changes are used here; the source's analyticity assertion away from the critical stable manifold is not used.

In these canonical coordinates the second equation integrates exactly. Along a passage on which $W>0$,

$$\frac{dw}{ds}=-\frac{w}{W(s,p)},\qquad D(s)=w_0\exp\left(-\int_s^{s_+}\frac{dx}{W(x,p)}\right),\qquad WD_s=D.$$

Write $s_+=\sigma+r$ and $m=\max(|s-\sigma|,\sqrt{|\mu|})\le r/2$. Every finite passage under consideration contains the interval $[\sigma+m,\sigma+r]$ and has

$$\int_s^{s_+}\frac{dx}{W(x)}\ge\int_m^r\frac{dx}{u_1(x^2+|\mu|)}\ge\frac{1}{4u_1 m}.$$

If $\mu\le 0$, its starting point is necessarily to the right of the rightmost root; no root is crossed. This proves (D.17), including the passage from $s>\sigma$ at $\mu=0$. For $L(a)=\log D(H(a))-\log a$ and $Z(a)=aH'(a)-W(H(a))$, exact differentiation gives

$$L'=\frac{Z}{aW(H)},\qquad Z'=H'\left(1-W_s(H)+aH''/H'\right)>0.$$

Thus L' has at most one zero. A fixed point with $L'\neq 0$ is simple; if $L=L'=0$, then $L''=Z'/[aW(H)]>0$, so it is exactly quadratic and the only tangent zero. Since the critical point is a strict minimum, the sum of isolated fixed-point multiplicities is at most two on this actual finite-passage interval. □

When the saddle ratio tends to zero, we instead need product coordinates whose phase derivatives remain bounded in that limit. The following lemma constructs them and gives the corresponding weighted curvature criterion for the complement.

Lemma D.5. *For the system*

$$X'=e^2\{Y(1+bnX-n^2e^2Y)+1-X+CcX^2\},\qquad Y'=XY+e\eta, \tag{D.19}$$

near $X=1,Y=0$, with $n=Ac$, $0<A\le A_+$, $c,e,|\eta|$ small and bounded C, there are uniform C^j orbital coordinates, for any fixed j, in which

$$\dot{\xi}=-r\xi,\quad \dot{\omega}=\omega,\qquad r\asymp e^2,\qquad D(s)=\xi_0(s/\omega_0)^r. \tag{D.20}$$

Here D is the exact passage between fixed positive sections $\xi=\xi_0$ and $\omega=\omega_0$. If an increasing complete complementary transition H has

$$r|H''/H'|<\frac{1-r_0}{2\xi_0},\qquad 0<r\le r_0<1, \tag{D.21}$$

then $D(H(a))=a$ has total isolated zero multiplicity at most two on each actual finite-passage interval where these maps are defined.

Proof. It suffices to establish the assertion for $j\ge 4$. Fix an integer $K\ge 2j+11$ before shrinking the parameters, and choose nested intervals $I_0\Subset I_1\Subset I_2\Subset(0,\infty)$ about $X=1$. The interval I_0 contains the saddle and the local sections to be used below. Write

$$\mathcal{F}(X,Y)=Y(1+bnX-n^2e^2Y)+1-X+CcX^2,\qquad w=Y+\frac{e\eta}{X}.$$

In the original time the translated equations are exactly

$$X' = e^2\mathcal{F}(X, w - e\eta/X), \qquad w' = Xw - \frac{e^3\eta}{X^2}\mathcal{F}(X, w - e\eta/X).$$

Thus, after reversing time, they have the form

$$\dot{X} = e^2\mathcal{A}(X, w), \qquad \dot{w} = -Xw + q\mathcal{R}(X, w), \qquad q = e^3\eta, \tag{D.22}$$

where $\mathcal{A} = -\mathcal{F}(X, w - e\eta/X)$ and $\mathcal{R} = X^{-2}\mathcal{F}(X, w - e\eta/X)$ have common finite-order phase bounds on a fixed tube over I_2. These bounds also hold at $\eta = 0$; none of the changes divides by η, e, or n.

To construct an invariant graph with common finite derivatives, embed I_2 as an arc of a circle and extend the coefficients with a fixed smooth cutoff equal to one on I_1. Extend the normal coefficient X to a function $\widetilde{B} \geq b_0 > 0$ on the circle tube. The extended equations are

$$\dot{\theta} = e^2\widetilde{A}(\theta, w), \qquad \dot{w} = -\widetilde{B}(\theta, w)w + q\widetilde{R}(\theta, w).$$

The extension retains both small factors e^2 and q, and all its coefficient norms are common over the prescribed compact parameter sets. At $e = 0$ the zero circle has zero tangential rate and a normal contraction rate bounded below by b_0. For the fixed order $K+1$, the normal-hyperbolicity inequalities persist after one parameter shrinking. Fenichel's theorem [14, Theorem 3, p. 215] therefore gives a C^{K+1} invariant circle, whose projection is a graph $w = \phi(\theta)$. It has common finite phase bounds: in the graph transform the normal variations decay at least as $e^{-b_0 t/2}$, while each fixed number of base variations is bounded by a polynomial in t times $e^{C_K e^2 t}$. Taking $C_K e^2 < b_0/4$ gives common bounds for the successive jet equations. The invariance equation then retains the factor q, which is needed when the selected graph is translated back to the physical variables.

That equation is

$$e^2\widetilde{A}(\theta, \phi)\phi' + \widetilde{B}(\theta, \phi)\phi = q\widetilde{R}(\theta, \phi). \tag{D.23}$$

The differentiated-maximum induction in the proof of Lemma 5.5 now retains q on the right-hand side of (D.23): at order $m \leq K$ the transport term vanishes at a maximum of $|\phi^{(m)}|$, the normal coefficient is at least $b_0/2$, and every remaining term is $O_m(|q|)$ once the lower jets are bounded. The preliminary phase bounds control the first-order quadratic term. Thus $\|\phi\|_{C^K} \leq C_K|q|$, uniformly also at $q = 0$. Restriction to I_0 and reversal of the translation therefore give

$$Y_s(X) = -\frac{e\eta}{X} + O_{C^K}(e^3|\eta|). \tag{D.24}$$

In particular $\phi = 0$ when $\eta = 0$, so this estimate includes that parameter face with the same constants.

Put $u = Y - Y_s(X)$ and $\chi(X, u) = 1 + bnX - n^2e^2(2Y_s(X) + u)$. Subtracting the graph invariance equation gives, in the original time,

$$X' = e^2\mathcal{F}(X, Y_s(X)) + e^2\chi(X, u)u, \qquad u' = \{X - e^2Y_s'(X)\chi(X, u)\}u.$$

The normal coefficient is a positive unit. The graph estimate supplies common C^{2j+10} bounds for these coefficients, so we may apply Lemma 3.1 with $\tau = \rho = e^2$ at order $J = j + 2$. This gives exact product coordinates

$$\theta' = e^2 f(\theta), \qquad \omega' = a(\theta)\omega, \quad f = 1 - \theta + Cc\theta^2 + O_{C^1}(e|\eta|), \quad a \geq a_* > 0.$$

Their inverse satisfies $X_\omega = O_{C^j}(e^2)$ and $Y = Y_s(\theta) + \omega u_1(\theta, \omega)$, with u_1 and X_θ positive units. Since $f_\theta = -1 + O(c + e|\eta|)$, its root is simple. The scalar linearization at the end of that lemma, with $g = f/a$ and $\beta = -g'(\theta_s) > 0$, gives (D.20) with $r = e^2\beta \asymp e^2$. The two extra product orders leave the asserted C^j bounds after this scalar coordinate change.

The estimate $X_\omega = O_{C^j}(e^2)$ also controls changes to physical sections despite the weak longitudinal speed. Fix a canonical base θ_b a distance comparable to $\sigma > 0$ from the simple root, and then a normal collar of width $\delta \ll \sigma$. In the inverse product coordinates $X = \theta + e^2\omega L(\theta, \omega)$, the longitudinal displacement across this collar is $O(e^2\delta)$. A vertical section just to the left of

the incoming canonical section can therefore be chosen so that the intervening forward flight has duration $O(\delta/\sigma)$. For a nearby physical section $X = \theta_b + e^2\ell$, the hitting equation, starting on $\theta = \theta_b$, becomes

$$\int_0^T f(\Theta(t))\,dt + \omega(T)L(\Theta(T),\omega(T)) - \ell = 0$$

after division by e^2. Its time derivative is separated from zero on the chosen collar because $|f(\theta_b)| \asymp \sigma$. The implicit-function theorem therefore gives common derivatives through the required order for this section map and its inverse. The intervals, σ, and δ are fixed before shrinking $e, c, |\eta|$; no relation between their rates of approach to zero is used.

Finally put $\mathcal{P} = D \circ H$. At a point where $\mathcal{P}' = 1$,

$$\mathcal{P}'' = \frac{D''}{D'^2} + \frac{H''}{H'} = \frac{r-1}{rD(H)} + \frac{H''}{H'} < 0,$$

since $D \le \xi_0$ and (D.21) holds. Every zero of $\mathcal{P}' - 1$ is a strict downward crossing. The single-maximum argument of Lemma 4.1 therefore gives at most two simple return zeros or one exactly quadratic zero on the actual interval, with the multiplicity transfer of Section 2. □

Appendix E. Simple-contact zero count

We now compose the local passages of Section D with the complete complementary transition. Bounded saddle ratios use Lemma C.1; small ratios use the exact power passage just constructed. Before estimating zeros, we specify the return domain on which these compositions preserve physical fixed points and their multiplicities.

Proposition E.1. *Suppose that $0 < A \le A_+ < \infty$ and that either*

1. *C belongs to a compact subinterval of $(0,\infty)$; or*
2. *$A \ge A_0 > 0$, C is bounded, and the polynomial satisfies $P(z) \ge p_0 > 0$ on the fixed negative interval preceding the small contact neighborhood used below.*

There are positive constants $c_0, d_0, \varepsilon_0, \rho_{\mathrm{H}}$ and a neighborhood U of the whole graphic such that, for $|c| < c_0$, $|d| < d_0$ and $0 < \varepsilon < \varepsilon_0$, the total actual finite minimal-return multiplicity of the isolated periodic orbits $L \subset U$ satisfying $d_{\mathrm{H}}(L, \Gamma_b) < \rho_{\mathrm{H}}$ is at most two. The choices are locally uniform in $b \in (0,2)$ and uniform on every compact set in the second case having the stated positive lower bound.

Domain of the outgoing inverse. Let D denote the local passage and N the transition from the physical source to its incoming coordinate. The conjugate return $D \circ N \circ C_+$ can be used only where the outgoing continuation C_+ and its inverse are both defined. Fix the original positive low section core I_0, the local outgoing section Ω, and a larger ordinary outgoing collar containing $[0, a_{\mathrm{out}}]$. Write $I_{\mathrm{out}} = (0, a_{\mathrm{out}})$ for its counting interval. The ordinary positive continuation $C_+ : I_{\mathrm{out}} \to J \subset I_0$ is defined on the entire collar, with $C_+' > 0$, before the return is defined. Reference barriers and the regular fast tube are constructed for that whole collar and the same field. A smaller incoming collar puts every finite local output in I_{out}; there is no orbit-dependent collar choice.

The signed transition along the fast arc, descent and negative predecessor is increasing in the physical source ordinate. Let I_{pass} be the set of incoming ordinates admitting a positive finite contact passage with $F > 0$ to the target core. Between two passing graphs, $F_y > 0$ bounds the intermediate speeds below by the positive minimum on the lower graph, for this fixed field. Scalar order then continues every intermediate graph to the target; transverse finite hits give openness. Thus I_{pass}, and its source preimage $I_{\mathrm{ret}} \subset I_0$, are open intervals, possibly empty. The regular fast clearance, negative predecessor and rightgoing contact event show that this is a minimal first return R on I_{ret}.

For the central or compact passage, or the late weak-saddle class, write $I_{\mathrm{ret,local}}$ for the actual local-return interval. In the weak case it is the lower, late-class interval, excluding the finite

threshold assigned to the early class. The local entry and outgoing-collar estimates give

$$R(I_{\mathrm{ret,local}}) \subset J.$$

This containment is asserted only for these local passages. The early weak class retains its separate physical displacement and the signed gluing argument below; no small weak-saddle collar is asserted for it.

Consequently C_+^{-1} is used only on $I_{\mathrm{ret,local}} \cap J$:

$$I_{\mathrm{adm}} = C_+^{-1}(I_{\mathrm{ret,local}} \cap J), \qquad H = N \circ C_+, \quad \mathcal{P}_a = D \circ H,$$
$$R(C_+(a)) = C_+(\mathcal{P}_a(a)) \qquad (a \in I_{\mathrm{adm}}).$$

Every point of this interval follows the same sequence of positive finite passages. Flow uniqueness therefore gives the conjugacy on this domain. The conjugacy preserves actual minimal-return multiplicity by Section 2; its derivative need only be nonzero for each fixed field. Axis hits, zero amplitudes on equilibria or stable separatrices, missing targets and infinite flights are outside the actual return domain. The same domain requirements apply to bounded signed C under the second hypothesis of the proposition.

E.1. **Bounded ratios to the contact scale.** We first bound the rescaled constant term and forcing: $|k|, |\ell| \le K$, where $k = c/\varepsilon^2$ and $\ell = d/\varepsilon^3$. Putting $n = A\varepsilon^2$, $z = nU$, $y = n^2V$ gives

$$U' = \mathcal{F} = V(1 + bnU - n^2V) + k - U + C\varepsilon^2U^2, \qquad V' = UV + \ell. \tag{E.1}$$

The tail estimate with $t = \varepsilon$ gives $V(-L) \le L + K + 1$. On $[-L, 0]$, above a sufficiently large fixed level T, one has $\mathcal{F} \ge V/4$ and $V' \le K$. Thus $(V^2)_U \le 8K$ on each excursion above T, and

$$V(U) \le B_* := \sqrt{T^2 + 8KL} \qquad (-L \le U \le 0). \tag{E.2}$$

Thus all incoming contact passages enter a bounded scaled strip. For signed C, take $|C|\varepsilon^2L^2 < 1$ and increase T by a fixed amount; the argument is unchanged.

Parameters separated from $\ell = 0$. When the rescaled forcing is separated from zero, negative forcing prevents passage and positive forcing makes the complete return contracting. Fix $\ell_0 > 0$. If $\ell \le -\ell_0$, start at $U = -2L$, where $V \le 2L + K + 1 \le 3L$. On $[-2L, -L]$ a hypothetical passing orbit has $V' \le -LV - \ell_0$ and $0 < \mathcal{F} \le 12L$, after increasing the harmless numerical constant if necessary. Consequently

$$V(-L) \le (3L + \ell_0/L)e^{-L/12} - \ell_0/L < 0$$

for a sufficiently large fixed L, a contradiction.

For $\ell \ge \ell_0$, the entrance instead lies below the positive nullcline $m(U) = -\ell/U$. To see this uniformly, on the first half of $[-2L, -L]$ the level $2\ell/(3L)$ is an upper barrier and

$$V(-3L/2) \le 2\ell/(3L) + 3Le^{-L/16} < 3\ell/(4L).$$

If the orbit remained above m on the second half, V would be nonincreasing, contradicting $m(-L) = \ell/L$. At a meeting with m, $d(V - m)/d\sigma = -m'\mathcal{F} < 0$, so it cannot cross back upward. It follows that $UV + \ell > 0$ and

$$\partial_V \frac{UV + \ell}{\mathcal{F}} = \frac{U\mathcal{F} - (UV + \ell)\mathcal{F}_V}{\mathcal{F}^2} < 0 \qquad (-L \le U < 0). \tag{E.3}$$

If there is an equilibrium at a negative abscissa, the first one, U_n, prevents passage: the region $\{-L \le U \le U_n,\ 0 \le V \le m(U)\}$ is forward invariant. Its upper boundary points inward, and on the right boundary $\mathcal{F}(U_n, V) \le \mathcal{F}(U_n, m(U_n)) = 0$. This includes the coalescence of two equilibria.

Every remaining crossing has $V(0) \ge v_0 > 0$ uniformly. Indeed, on $[-1, 0]$, below $\min(1, \ell_0/2)$, one has $V' \ge \ell_0/2$ and $0 < \mathcal{F} \le K + 4$. Choosing v_0 smaller than both half that level and $\ell_0/[4(K + 4)]$, a terminal value below v_0 would force $V(-1) < 0$. The level v_0 cannot be crossed downward there. For $U \ge 0$, a small parabola $V_b = v_0/2 + \gamma U^2$ points inward:

$$V' - V_b'\mathcal{F} \ge \ell_0 + U\{(1 - 4\gamma)v_0/2 - 2\gamma K\} + \gamma(1 - 4\gamma - 2|C|\varepsilon^2)U^3 > 0.$$

Take U_1 large enough that $\mathcal{F} \geq V/4$ for $U \geq U_1$. On $[0, U_1]$ the bounded-height argument used in (E.2) gives $V \leq B_1$; since $V' \geq \ell_0$, its flight time is at most B_1/ℓ_0. The vertical-section derivative on this compact part is

$$\frac{\mathcal{F}(0, V(0))}{\mathcal{F}(U_1, V(U_1))} \exp \int \operatorname{div}(\mathcal{F}, UV + \ell) \leq K_1.$$

No inverse of the possibly small speed at $U = 0$ is used. On the rest of the positive part, (D.6) and the lower parabola give a logarithmic derivative loss at most $K_2 + K_2\varepsilon^2 \log(1/(A\varepsilon^2))$. The negative transition has derivative at most $K_3\varepsilon \exp[-(t_0 + t_1 \log(A_+/A))/\varepsilon^2]$, and (E.3) adds no loss. The derivative of the complete return is therefore bounded above by

$$K_4\varepsilon \exp\left\{-\frac{t_0 + t_1 \log(A_+/A)}{\varepsilon^2} + K_4\varepsilon^2 \log \frac{1}{A\varepsilon^2}\right\} < 1. \tag{E.4}$$

Choose ε so that $K_4\varepsilon^4 < t_1/2$; then the coefficient of $\log(A_+/A)$ is negative and the remaining logarithms are absorbed by $t_0/(2\varepsilon^2)$. The displacement is therefore strictly decreasing on its single return interval and has at most one zero.

Small forcing and negative k. For small forcing, a negative constant term is excluded by reducing the incoming height below the level needed for positive speed. Fix $k_0 > 0$, let $k \leq -k_0$, and take $|\ell| \leq k_0^2/64$. Put $H = k_0/8$. On $[-2L, -L]$, as long as $V \geq H$,

$$V' = UV + \ell \leq -LV + k_0^2/64 \leq -(L/2)V$$

for fixed large L. A passing orbit has $0 < \mathcal{F} \leq BL$ and enters this strip below $3L$. Until its first hit of $V = H$ it therefore satisfies

$$\frac{dV}{dU} \leq -\frac{V}{2B}, \qquad V(-L) \leq 3Le^{-L/(2B)} < H.$$

If it hits H earlier, the vector field on that boundary points down and prevents a later upcrossing. Thus in all cases $V(-L) < H$. For $U \leq -k_0/4$, $V'|_{V=H} \leq -k_0 H/4 + k_0^2/64 < 0$. At $U = -k_0/4$ this implies $\mathcal{F} \leq 2H - k_0 + k_0/4 + O(\varepsilon^2) < 0$. Thus this parameter range has no passing orbit. The saddle-node at $(k, \ell) = (0, 0)$. At these parameters the origin has eigenvalues $-1, 0$ and center equation $V' = V^2 + O(V^3)$; $V = 0$ is its strong invariant line. Lemma D.4 applies, with the center coordinate oriented by $s_V(0) > 0$. To use its small-passage estimate, we must first put every physical entry in a common small incoming range. For a small fixed $d_0 > 0$, descent on $[-2L, -L]$ and the constant upper barrier on $[-L, -d_0]$ give

$$V(-d_0) \leq \max\{(2L + K + 1)e^{-k_5 L},\ 2|\ell|/d_0\}. \tag{E.5}$$

For $|k| < d_0/4$, the speed on the latter interval is at least $d_0/2$. Choose L large, then $|k|, |\ell|$ small. In product coordinates the entry has $w \in [3d_0/4, 5d_0/4]$ and $|s - \sigma| \leq K_5(V + |k| + |\ell|)$. It reaches $w_0 = d_0/2$ in the bounded time $\log(w/w_0)$, with a positive transverse derivative and bounded derivatives of the section change. At zero parameters, positive physical entries have $s > 0$; their potentially infinite time from the right is included in (D.17).

For the positive reference, use the linear and parabolic barriers in the saddle-node part of the proof of Lemma 6.3, with $p(U) = k - U + C\varepsilon^2 U^2$, $\ell_* = \ell$ and $n = A\varepsilon^2$. Start at $U = \delta$ on $h = \delta + \delta^2 + O(\delta^3 + |k| + |\ell| + \varepsilon^2)$. The lower line $(1 + \delta/4)U$ has a strict crossing margin for $|k| + |\ell| \ll \delta^2$; the small and large parabolas then give $h \asymp 1 + U^2$ and $\mathcal{F}/h \geq k_6 > 0$. These inequalities do not use a lower bound for A: their physical smallness is $y \leq M(n^2 + z^2)$, with $n \leq A_+\varepsilon^2$. They also allow signed bounded C, since its adverse contribution is $O(\varepsilon^2 h)$. Fix z_c before shrinking n. Thus Lemma D.3 applies uniformly in A.

Let $N(u)$ be the complete transition from the positive section to the incoming center coordinate. The convex map of Lemma D.1 remains convex up to $U = -d_0$: in (D.7) the positive term $xp(1 + bz)$ dominates the forcing, since $|\ell| \ll d_0^2$, and all higher physical-height terms tend uniformly to zero on the bounded core. Its scaled first derivative is $o(1)$ by (D.5). The final

bounded section change therefore gives $N' > 0$ and $N''/N' \geq -o(1)$. Combining this incoming estimate with the outgoing transition gives, for $H = N \circ C_+$,

$$aH''/H' = aC_+''/C_+' + aC_+'(N''/N') \geq -K_7 a. \tag{E.6}$$

First choose the product neighborhood with $W_s \leq 1/4$, then a_0 with $K_7 a_0 \leq 1/4$. Decrease the incoming range and the parameters in (E.5) so that (D.17) puts every fixed point in $0 < a < a_0$. The actual passing condition restricts H to one component to the right of any scalar root, hence to an interval. Intersecting with $(0, a_0)$ keeps an interval containing every fixed point. Equations (E.6) and (D.18) give at most two cycles.

Small forcing and positive k. For $k \in [k_0, K]$ the saddle near $(U, V) = (k, 0)$ persists, with

$$U_s = k + O(\varepsilon^2 + |\ell|), \quad V_s = -\ell/U_s, \quad \lambda_- = -1 + O(\varepsilon^2 + |\ell|), \quad \lambda_+ = k + O(\varepsilon^2 + |\ell|).$$

Its characteristic ratio $r = -\lambda_-/\lambda_+$ lies in a compact subinterval of $(0, \infty)$, including values on either side of one. The descent (E.5), followed to $U = U_s - \delta$, makes the incoming unstable coordinate s as small as prescribed. That continuation has $\mathcal{F} \geq \delta/2$; its derivatives are uniformly bounded. Choose the saddle coordinates with $U_x > k_8 > 0$, $U_y < -k_8$. In the prepared form used in Lemma C.1, a negative entry s exits on the left with stable coordinate at most $y_0(|s|/x_0)^{r-/2}$. For small $|s|$, its physical longitudinal speed is negative there. Such a trajectory, and the infinite-time entry $s = 0$, cannot be a counted contact passage. Hence $s > 0$.

At $U = U_s + \delta$ the reference satisfies $h = (k+1)\delta + O(\delta^2 + |\ell| + \varepsilon^2)$. Use the bounded-ratio construction in the proof of Lemma 6.3, replacing k_c by k_0 and ℓ_* by ℓ. The lower line $(1 + k_0/2)(U - U_s)$ has a strict positive crossing margin; the same parabolas give $h \asymp 1 + U^2$ and $\mathcal{F}/h \geq k_9 > 0$. Again only $n = A\varepsilon^2 \leq A_+\varepsilon^2$ enters the physical smallness, and bounded signed C contributes $O(\varepsilon^2 h)$. The positive transverse orientation and compact local section change give the incidence required by Lemma D.3. Hence (D.15) holds with $n = A\varepsilon^2$.

For the negative transition to the saddle's incoming section, Lemma D.1 and its final bounded section change give $N' > 0$, $N' = o(1)$ and $|N''/N'| \leq K_9\varepsilon^{-2}$. Consequently

$$0 < H' \leq m(\varepsilon) \longrightarrow 0, \qquad |H''/H'| \leq K_{10} + K_{10} n^{3/2}\varepsilon^{-2} \leq K_{11}, \qquad H = N \circ C_+. \tag{E.7}$$

The identity $n^{3/2}\varepsilon^{-2} = A^{3/2}\varepsilon$ is what makes the bound uniform in A.

The smallness of H' allows the two-zero comparison to include $r = 1$. At a point where $(D \circ H)' = 1$, set $s = H(a)$, $T = \log(1/s)$ and $M = \log(1/m) - \log K_{12}$. Lemma C.1 gives $(1 - r)T \geq M$. For large M this is impossible if $r \geq 1$. Otherwise $\beta = 1 - r \geq M/T$ and $sD''/D' \leq -\beta/2$, since $Te^{-\nu T}$ is bounded. At that point

$$(D \circ H)'' = \frac{D''}{D'^2} + \frac{H''}{H'} \leq -\frac{\beta}{2sD'} + K_{11} < 0, \qquad \frac{\beta}{sD'} \geq \frac{Me^{r-T}}{K_{12}T}.$$

Taking m small proves the last strict inequality. All zeros of $(D \circ H)' - 1$ are downward crossings, so there is at most one. A displacement zero at this stationary point is exactly quadratic and the sole zero; all other zeros are simple. Hence the total actual fixed-point multiplicity is at most two. Fix the outgoing collar first; choose the incoming collar small enough that D maps into it and that the negative-entry exclusion holds; then choose $L, |\ell|, \varepsilon$. Monotonicity makes the resulting return domain one interval. This order places every counted cycle in the domain on which the estimates were proved.

These cases cover every fixed bounded (k, ℓ) set with common thresholds. Choose a small central neighborhood, then $k_* > 0$ inside it, and finally $\ell_* > 0$ below its threshold, $k_*^2/64$, and the positive-saddle threshold. The five parameter ranges

$$\ell \geq \ell_*, \quad \ell \leq -\ell_*, \quad |\ell| < \ell_*, k \leq -k_*, \quad |\ell| < \ell_*, |k| < k_*, \quad |\ell| < \ell_*, k \geq k_*$$

have bounds $1, 0, 0, 2, 2$, respectively. Equalities are assigned to a range whose estimate holds with a strict margin.

E.2. **Ratios larger than the contact scale.** Put

$$\rho = \max\{\varepsilon, \sqrt{|c|}, |d|^{1/3}\}, \qquad e = \varepsilon/\rho, \quad \kappa = c/\rho^2, \quad \eta = d/\rho^3.$$

The compact direction set is

$$0 \le e \le 1, \quad |\kappa|, |\eta| \le 1, \qquad \max\{e, \sqrt{|\kappa|}, |\eta|^{1/3}\} = 1. \tag{E.8}$$

For e bounded below, $k = \kappa/e^2$ and $\ell = \eta/e^3$ are bounded, so the preceding subsection applies. It remains to treat small $e > 0$, when the constant term or forcing determines the larger scale ρ. Negative forcing or a negative constant term. On the larger scale, the same incoming-height argument excludes negative forcing and a negative constant term. Use the exact scaling

$$X = \frac{z}{A\rho^2}, \quad Y = \frac{y}{A^2\varepsilon^2\rho^2}, \quad X' = e^2\mathcal{F}, \quad Y' = XY + e\eta,$$

where

$$\mathcal{F} = YQ + \kappa - X + C\rho^2X^2, \qquad Q = 1 + bA\rho^2X - A^2\rho^4e^2Y. \tag{E.9}$$

Lemma D.1, with $t = \rho$, gives $Y(-L) \le L + K + 1 =: B_0$. If $\eta \le -\eta_0 < 0$, on the unit interval $[-L, -L+1]$ one has $Y' \le -e\eta_0$, $Y \le B_0$ and $0 < \mathcal{F} \le B_1$. Thus $Y_X \le -\eta_0/(eB_1)$, which is incompatible with $Y > 0$ when e is small.

If $\kappa \le -\kappa_0 < 0$, take the section $X_g = \kappa/2$. Its polynomial value is at most $-\kappa_0/4$ after decreasing ρ. Choose $\theta > 0$ with $2\theta < \kappa_0/8$. On the same first unit interval, as long as $Y \ge \theta$, $Y' \le -(L-1)Y/2$; hence

$$Y(-L+1) \le B_0 \exp[-(L-1)/(2e^2B_1)] < \theta$$

unless it already met that level. For $X \le -\kappa_0/2$, the boundary $Y = \theta$ still points down when $e|\eta| < \kappa_0\theta/4$. It therefore reaches X_g with $\mathcal{F} \le 2\theta - \kappa_0/4 < 0$, another contradiction. Both exclusions use the incoming-height bound from the fixed physical section; that bound is essential to the scaled argument.

Positive forcing separated from zero. Assume $\eta \ge \eta_0 > 0$. To retain contraction between the negative tail and the core, we first place the physical incoming orbit below the positive nullcline. On the far negative tail set $h_s(z) = -\Delta/z$, $w = y - h_s$ and

$$E(z, s) = \exp\int_s^z \frac{\zeta}{F(\zeta, y(\zeta))}\, d\zeta.$$

The exact difference equation and its integral are

$$w_z = (z/F)w - h_s', \qquad w(z_g) = E(z_g, -q_g)w(-q_g) - \int_{-q_g}^{z_g} E(z_g, s)h_s'(s)\, ds, \quad z_g = -LA\rho^2. \tag{E.10}$$

The first term has absolute value at most $K\varepsilon^3e^{-\gamma T_A/\varepsilon^2}$. On the final interval of length $A\varepsilon^2$, the kernel is bounded below by a positive constant and $h_s' \ge k\Delta/(A^2\rho^4)$. The integral is consequently at least $kA^2\varepsilon^5\eta/\rho$. The ratio of the first term to this lower bound is at most

$$K\frac{\rho}{\varepsilon^2}A^{-2}\left(\frac{A(1+L\rho^2)}{q_g + A}\right)^{\gamma/\varepsilon^2} \longrightarrow 0.$$

The maximization in A from Lemma D.1 makes this limit uniform for arbitrarily small relative scales. Thus $w(z_g) < 0$. At $w = 0$ its derivative is $-h_s' < 0$, so the passing orbit remains below the nullcline up to zero. In particular $G > 0$ there, and $R_y = (zF - GF_y)/F^2 < 0$ on the entire negative contact interval.

For the positive part, the forcing determines the scale $s = A\varepsilon\rho$. Put $z = sU$, $y = s^2V$. The equations are

$$U' = V(1 + bsU - s^2V) + \kappa - eU + C\varepsilon^2U^2, \qquad V' = UV + \eta. \tag{E.11}$$

The low-height argument on $[-1,0]$ used above gives $V(0) \geq 2v_0 > 0$. With $\gamma > 0$ small, $V_b = v_0 + \gamma U^2$ is a strict lower barrier: its crossing expression is bounded below by $\eta_0 + U(v_0/2 + \gamma U^2/2) > 0$. After excluding the already treated parameters $\kappa \leq -\kappa_* < 0$, choose $\kappa_* \leq v_0/8$ and $e \leq \sqrt{v_0\gamma}/4$. Then $U' \geq V/4$ throughout the positive part. For signed C, absorb $|C|\varepsilon^2 U^2 \leq (|C|\varepsilon^2/\gamma)V$ and replace $V/4$ by $V/8$. The estimate $y_z \leq Kz + Ks^3/(s^2v_0 + \gamma z^2)$ keeps a sufficiently small contact height in the physical neighborhood up to z_c.

In the scalar variational numerator of (E.11), the terms $-eU^2$ and $-\eta(1 + bsU - s^2V)$ are nonpositive. The other terms, divided by the square of the speed, have integrals bounded by

$$K\int_0^{z_c/s} \frac{U\,dU}{(v_0+\gamma U^2)^2} + K\varepsilon^2 \int_0^{z_c/s} \frac{U^3\,dU}{(v_0+\gamma U^2)^2} + Ks^2 \int_0^{z_c/s} \left(U + \frac{1}{v_0+\gamma U^2}\right) dU.$$

This is at most $K + K\varepsilon^2 \log(1 + z_c/s)$. Combining it with the negative estimate gives

$$\log \mathcal{P}' \leq K - \frac{\gamma}{\varepsilon^2}(t_0 + t_1 \log(A_+/A)) + K\varepsilon^2\{\log(A_+/A) + 2|\log\varepsilon| + K\} < 0.$$

Here $\rho \geq \varepsilon$ was used in the last logarithm. All scale factors cancel in vertical-section derivatives. Thus the complete return has at most one fixed point on this parameter range.

A positive constant term and small forcing. The last case has $\kappa = 1$ and $|\eta|$ small. Write $c = \rho^2 > 0$, $n = Ac$, $e = \varepsilon/\sqrt{c}$. The equations are (D.19). Orbits may leave a small neighborhood of $Y = 0$ before reaching the saddle, so a local saddle map alone does not cover the return domain. We separate these early departures from the late passages near the saddle. Fix a small $\sigma > 0$, then a smaller $\delta > 0$. The separating event is the first upward intersection with $Y = \delta$: an intersection at $X_b \leq 1 - \sigma$ defines the first class. The second, late class enters a saddle neighborhood near $X = 1 - 2\sigma$ below $Y = \delta$; its fixed outgoing normal section has height between 3δ and 5δ.

We first show that these classes exhaust the incoming orbits and form ordered source intervals. For $\eta \leq 0$ the homogeneous negative estimate gives

$$Y(-L) \leq \frac{Ke}{A^2\sqrt{c}} \exp\left(-\frac{\gamma T_A}{ce^2}\right) \leq e^{-B/e^2} \tag{E.12}$$

for any prescribed $B > 0$, by taking the upper bound for c small first. The factor A^{-2} is absorbed by maximizing as before, and the remaining $c^{-1/2}$ by $\exp[-k/(ce^2)]$. On $[-L, 0]$ the ordinate decreases; on $[0, 1-\sigma]$ it satisfies $Y_X \leq KY/(\sigma e^2)$. Taking $B > K/\sigma$ proves that this sign of forcing cannot belong to the first class. For $\eta \geq 0$, the height on $[-L, 0]$ is uniformly bounded. For example an upward crossing of a large fixed level T requires $X \geq -e\eta/T$; while $Y \geq T$, one has $\mathcal{F} \geq Y/2$ and $(Y^2)_X \leq 4\eta/e$. The remaining interval has length at most $e\eta/T$. This bounds the height and gives, by an integrating factor,

$$Y(0) \leq B_0 e^{-L^2/(K_1e^2)} + \frac{K_1\eta}{e}\int_{-L}^{0} e^{-X^2/(K_1e^2)}\,dX \leq B_0 e^{-L^2/(K_1e^2)} + K_2\eta < \delta/4.$$

An upward crossing of $Y = \delta$ at negative X could not be followed by a downward crossing before zero, since $X\delta + e\eta$ increases with X. Its first upward intersection therefore has $X_b > 0$. Every counted cycle eventually has such an intersection because its outgoing physical height is separated from the shrinking level $n^2e^2\delta$. Write q for the increasing physical source ordinate. Scalar ordering gives $Y_q > 0$ at every fixed pre-hit abscissa, and at the upward event $Y_X > 0$. Thus

$$(X_b)_q = -Y_q/Y_X < 0.$$

The second class is therefore the lower source interval and the first class the upper interval, separated, when both occur, by one finite transverse threshold.

For the early class, $\eta \geq 0$. We estimate the complete multiplier by following the growing height through the core. Use Y as independent variable up to $Y = e^{-2}$. A first zero of $\mathcal{F}$ is impossible, because its derivative there is $\mathcal{F}_Y > 0$. Moreover,

$$\frac{d(X^2)}{dY} \leq K_3e^2 + \frac{K_3e^2}{Y} + \frac{K_3ce^2X^2}{Y}. \tag{E.13}$$

Gronwall's inequality keeps X bounded. Before a fixed large level Y_0, its squared increment is $O(e^2)$, leaving $X \le 1-\sigma/2$. Choose Y_0 above four times the resulting global upper bound for X; above it $\mathcal{F} \ge Y/4$. Thus $\mathcal{F} \ge k_1(1+Y)$ throughout. The terminal X is bounded below as well. If $XY < e\eta$ above Y_0, then $X_Y \ge eY/(8\eta)$, which forces $XY = e\eta$ before Y_0+1, for sufficiently small η. Thereafter $(X^2)_Y \ge e^2/4$. For $\eta = 0$ this last estimate applies directly. It follows that at $Y = e^{-2}$,

$$0 < x_- \le X_t \le x_+ < \infty, \qquad y = n^2, \quad z = nX_t.$$

On the whole scaled positive part the exact density is

$$\partial_Y \frac{XY + e\eta}{e^2\mathcal{F}} = \frac{X[1 - X + CcX^2 + n^2e^2Y^2] - e\eta(1 + bnX - 2n^2e^2Y)}{e^2\mathcal{F}^2}.$$

Its integral is at most K_4/e^2. From the terminal point a small lower parabola and a large upper parabola in $U = z/n, V = y/n^2$ give $y \asymp n^2 + z^2$ and $F \ge k_2 y$ to the fixed positive section. At a lower meeting the term $UV(1-2\gamma Q)$ is positive; the terms of orders e^2U and ε^2U^3 are absorbed after fixing γ. After shrinking the physical neighborhood, at the upper parabola $V = M(1+U^2)$ one has $F/n^2 \ge V/2$, or equivalently $F \ge y/2$, and $dV/dU < 2MU$. The remaining density integral is therefore at most $K_5 + K_5\varepsilon^2 \log(1/n)$, by the integration in Lemma D.3. Since the negative part has $R_y < 0$, the complete derivative satisfies

$$\log \mathcal{P}' \le \log(K_6\varepsilon) - \frac{\gamma T_A}{ce^2} + \frac{K_6}{e^2} + K_6 + K_6\varepsilon^2 \log(1/n) < 0. \tag{E.14}$$

Indeed $\log(1/n) \le T_A + K_7 + \log(1/c)$. First take c small to absorb K_6/e^2, then ε small to absorb $\varepsilon^2 T_A$; the remainder $ce^2 \log(1/c)$ tends uniformly to zero. There is at most one cycle in the first interval.

For the late class, we apply Lemma D.5 after verifying the physical amplitude and the derivatives of both complementary transitions. The incoming amplitude must be positive. For $\eta \ge 0$, the same selected weak graph is

$$Y_s(X) = -e\eta/X + O_{C^K}(e^3\eta).$$

It is below the axis for $\eta > 0$ and equals the axis for $\eta = 0$. The product inverse preserves the sign of $Y - Y_s$; hence $\omega \le 0$ is nonphysical in $Y > 0$. For $\eta < 0$, zero amplitude approaches the saddle in infinite time, while a negative amplitude, if initially physical, reaches the axis while growing in modulus. The axis-hit argument is used only for this negative sign of the forcing. The incoming tilted section is reached with bounded derivatives by the explicit $O(e^2\delta)$ section adjustment in that lemma. The transition from $-L$ to that section stays to the left of $1-\sigma$ with bounded height and $\mathcal{F} \ge k_\sigma > 0$. Writing this scalar transition as Q_e and its equation as $Y_X = f_e(X, Y)$, direct differentiation on the fixed strip gives

$$|f_{e,Y}| + |f_{e,YY}| \le K_8 e^{-2}, \qquad 0 < Q_e' \le e^{K_8/e^2}, \qquad \left|\frac{Q_e''}{Q_e'}\right| \le K_8 e^{-2} e^{K_8/e^2}.$$

The rescaled incoming map v supplied by (D.4) has

$$v' \le L_0(A, c, e) e^{-\gamma T_A/(ce^2)}, \qquad |v''/v'| \le K_8\varepsilon^{-2}.$$

Here L_0 has at most polynomial growth in the inverse small scales. After taking c small, the negative action absorbs the factor e^{K_8/e^2} and every polynomial loss, uniformly as $A \downarrow 0$. Consequently $(Q_e \circ v)' = o(1)$ and

$$\frac{(Q_e \circ v)''}{(Q_e \circ v)'} = \frac{v''}{v'} + \left(\frac{Q_e''}{Q_e'} \circ v\right) v' = O(\varepsilon^{-2}).$$

Including the ordinate rescaling and the regular tilted-section change therefore gives

$$N' > 0, \qquad |N''/N'| \le K_9\varepsilon^{-2}. \tag{E.15}$$

To bound the outgoing transition independently of the weak eigenvalue, move the normal outlet to a horizontal section $Y = \delta_1 \asymp \delta$, slightly above it. Its X-coordinate is a decreasing

smooth function of the outgoing weak coordinate a, with derivative a negative unit. The inverse-coordinate bounds in Lemma D.5 give uniform first three derivatives for this conversion; shrink $e|\eta| \ll \delta$ after fixing δ. Continue with Y as independent variable to e^{-2}. Here X stays in a positive compact interval, $XY + e\eta \geq k_3 XY$, and for $j \leq 3$,

$$\left| \partial_X^j \frac{e^2 \mathcal{F}}{XY + e\eta} \right| \leq K_{10} e^2 (1 + 1/Y).$$

Its integral is bounded by $K_{11} + K_{11} e^2 |\log \delta|$. After the last shrinking of e this gives a nonzero first derivative and bounded second and third derivative ratios to the terminal horizontal section. The parabolic bounds just established, a fixed compact transition to a vertical U-section, and Lemma D.3 give $0 < C_+' \leq K_{12} n^{3/2}$ and $|C_+''/C_+'| \leq K_{12}$. The coordinate $a = 0$ is the actual outgoing unstable separatrix. Its fixed small collar can be chosen by first fixing σ, then $\delta \ll \sigma$, then the parameters: the horizontal-section conversion has bounded derivative ratios independently of the last choice of δ, since its two heights have a bounded ratio. This avoids a dependence of the allowable collar on an inverse weak speed.

For the complete complementary transition $H = N \circ C_+$,

$$r|H''/H'| \leq K_{13} e^2 + K_{13} \frac{e^2 n^{3/2}}{\varepsilon^2} = K_{13} e^2 + K_{13} A^{3/2} \sqrt{c} \longrightarrow 0. \tag{E.16}$$

The weight $r \asymp e^2$ cancels the loss from the incoming second derivative; the available unweighted estimate need not tend to zero. Lemma D.5 gives at most two cycles in the second interval. The equality $X_b = 1 - \sigma$ is assigned to the first interval; its transverse event and all estimates extend there, so it adds no cycle.

Lemma E.2 (Gluing the two simple-contact classes)**.** *The two classes just constructed have total actual minimal-return multiplicity at most two, including a finite threshold only once.*

Proof. For $\eta \leq 0$ the first class is absent, so only the second-class bound is needed. For $\eta > 0$, let $\mathcal{P}(q)$ be the complete return on the increasing physical ordinate and put $\mathcal{F}(q) = \mathcal{P}(q) - q$. The two class descriptions are restrictions of this same finite flow return. Its physical target is transverse, so its hitting-time formula gives a C^1 return through the finite threshold. The actual passing domain is one interval by scalar order and $F_y > 0$, as in Lemma A.2. On the upper interval, (E.14) gives $\mathcal{F}' < 0$, including the threshold assigned to that class.

On the lower, late interval use the increasing physical continuation $q = C_+(a)$ only on $I_{\mathrm{adm}} = C_+^{-1}(I_{\mathrm{ret,local}} \cap J)$, where $J = C_+(I_{\mathrm{out}})$. The complete outgoing collar was fixed first and contains all late-class local outputs; its inverse is not used outside J. The amplitude return is $\mathcal{P}_a(a) = D(H(a))$, with $H = N \circ C_+$; the chosen outgoing collar contains its outputs. The proof of Lemma D.5 shows that every zero of $\mathcal{P}_a' - 1$ has $\mathcal{P}_a'' < 0$. Hence the amplitude displacement has at most one stationary point, a strict maximum. If it has two zeros, they cross first upward and then downward; a tangent zero is a strict maximum and is the sole zero in that interval. Since

$$\mathcal{F}(C_+(a)) = C_+(\mathcal{P}_a(a)) - C_+(a),$$

the increasing coordinate change preserves these signs. Writing $J = (j_-, j_+)$, the containment $\mathcal{P}(I_{\mathrm{ret,local}}) \subset J$ also gives $\mathcal{F}(q) > 0$ for $q \leq j_-$ and $\mathcal{F}(q) < 0$ for $q \geq j_+$ wherever the late return is defined. Thus these signs extend across the parts of the late interval outside J.

Every lower zero is simple unless it is the sole, exactly quadratic strict-maximum zero. If there are two simple lower zeros, or one quadratic lower zero, the physical displacement is negative after the last lower zero and up to the threshold. A zero at that threshold would have negative derivative and therefore positive displacement immediately to its left, a contradiction. The strictly decreasing upper displacement has no zero either. Otherwise there is at most one simple lower zero and at most one simple upper zero. Thus the sum of actual multiplicities is at most two. The same strict derivative statements exclude intervals of zeros. A finite threshold zero is simple because $\mathcal{F}' < 0$ there and is counted once in this total multiplicity count; zero-amplitude separatrices and infinite-time passage boundaries are not periodic orbits. □

All these arguments remain valid in the signed-C case of the proposition. On fixed scaled intervals replace $1 - X + CcX^2 \geq 1 - X$ by half that bound, after fixing the intervals and decreasing c. Near the saddle its derivative is still $-1+O(c+e|\eta|)$. On the positive parabolas the only new adverse term is $-|C|\varepsilon^2U^2 = O(\varepsilon^2V)$. The negative estimates were already proved with $P(-x) \asymp Ax$, and $A \geq A_0$ only improves their scale cancellation. These explicit replacements cover every sign-sensitive inequality above.

Proof of Proposition E.1. Choose a small fixed $\eta_* > 0$ for the positive-constant, small-forcing argument, then $e_* > 0$ small enough for it and both separated-forcing arguments. In (E.8), $e \geq e_*$ gives a fixed bounded (k, ℓ) set and hence the bound two from Subsection E.1. For $e < e_*$, $\eta \geq \eta_*$ gives the bound one and $\eta \leq -\eta_*$ gives zero. In the remaining range $|\eta| < \eta_*$, compact normalization forces $|\kappa| = 1$: $\kappa = -1$ is excluded and $\kappa = 1$ gives the bound two by Lemma E.2. The parameter thresholds are fixed before taking the common small upper bound on ρ. Only finitely many choices are needed. Within each range, fixed section sizes, barrier levels, and collar widths are selected first. The common upper bounds for L^{-1} and the small parameters are then chosen in the order used in the corresponding proof, so estimates such as $n^{2-K\varepsilon^2} \leq n^{3/2}$ hold simultaneously on the whole compact parameter set.

For each fixed field the subdivision is made by its coefficients, so bounds from different parameter ranges are not added. Within the last range the two first-intersection intervals are the only subdivision of the physical source. To verify the interval assertions used throughout, two passing scalar graphs bound every intermediate graph by uniqueness. Since $F_y > 0$, the lower graph supplies positivity of F and y, and the upper graph prevents an earlier upper exit. Intermediate graphs therefore reach the same section. Conditions on an incoming amplitude or on the ordered first intersection are inverse images of intervals under monotone maps. Rolle's theorem has always been applied on these intervals, not across missing portions of a return map. By Lemma A.2, one fixed physical source contains the single intersection of every sufficiently close whole-graphic cycle. Thus no additional sum over source sections is required, and the proposition follows. □

Appendix F. Contact-scale forcing and return bounds

We prove the exclusions and complete-return contraction used in the directional cover of Section 7. Let

$$P_R(z) = Cz^2 + Raz + R^2c, \qquad D = R^3d, \qquad 0 < C_- \leq C \leq C_+, \tag{F.1}$$

where a, c, d lie in fixed compact sets and $R, \varepsilon > 0$ are independently small. The fixed negative interval has a positive polynomial margin. Thus Lemma A.2 supplies the same complete positive-speed contact interval for all these parameters. All return maps in this section are restricted to that common interval. The constants L, the contact height, and the section markers are fixed after the compact coefficient sets are chosen and before R, ε are shrunk. Consequently every barrier and variational estimate below has the same source and target sections for the whole parameter family.

On this common interval, barriers exclude negative forcing or a negative polynomial value. For positive forcing, a rescaled core supplies a lower height bound, which controls expansion on the positive arc. Contraction on the negative arc then dominates in the complete return, uniformly in the relative sizes of R and ε.

Lemma F.1. *There is no counted cycle in either of the following cases:*

1. $d = -1$;
2. $P_R(-\alpha R) \leq -pR^2$, *where* $p > 0$ *and* $0 < \alpha_0 \leq \alpha \leq \alpha_1$ *are fixed.*

In particular the second case includes $c = -1$ *with bounded* a, *and, when* $C = 1$, $a = 1$, $c \leq 1/8$.

Proof. Both exclusions begin with an upper height bound on the negative tail. Choose L sufficiently large that, for $x \geq LR$, $P_R(-x) \asymp x^2$ and $|P_R'(-x)| \leq C_1x$. The graph $B(z) = \varepsilon^2P_R(z)$

is an upper barrier there:

$$G - B'F \le \varepsilon^2 x P_R(-x)[-1 + C_2\varepsilon^2 + C_3\varepsilon L^{-3}] < 0.$$

At the fixed negative entry, $y \le K\varepsilon^3$; consequently $y(-LR) \le K_1\varepsilon^2 R^2$ on every putative counted passage. If $d = -1$, then on $[-LR, -(L-1)R]$ the height decreases and

$$G \le -\varepsilon^3 R^3, \qquad 0 < F \le K_2\varepsilon^2 R^2, \qquad \frac{dy}{dz} \le -\frac{\varepsilon R}{K_2}.$$

Integration over a length R would make y negative for small ε. This proves the first assertion.

For a negative polynomial value, we first improve the height bound by one power of ε. Enlarge L past $\alpha_1 + 2$ and take $H > 2\sup|d|/\alpha_0 + 1$. While $y \ge H\varepsilon^3 R^2$ on the first coefficient-scale strip,

$$G \le -\tfrac{1}{2}(L-1)Ry, \qquad \frac{d\log y}{dz} \le -\frac{L-1}{2K_2\varepsilon^2 R}.$$

Therefore the orbit enters $y \le H\varepsilon^3 R^2$ before $z = -(L-1)R$, since otherwise its height is reduced by $\exp[-(L-1)/(2K_2\varepsilon^2)]$, contradicting the assumed lower height. This horizontal upper bound is invariant for $z \le -\alpha_0 R$. At $z = -\alpha R$ it gives $F \le 2H\varepsilon^3 R^2 - p\varepsilon^2 R^2 < 0$, contrary to the counted positive-speed passage. If $c = -1$, choose a fixed small α with $C_+\alpha^2 + \sup|a|\alpha \le 1/2$. If $C = a = 1$ and $c \le 1/8$, use $\alpha = 1/2$. □

Proposition F.2. *For $d = 1$ in* (F.1)*, there is at most one cycle sufficiently close to the graphic, uniformly for bounded a, c and independently small R, ε.*

Proof. Write $\delta = \varepsilon^3 R^3$ and $\mathcal{R} = G/F$. On a positive-speed contact graph, the exact variational coefficient is

$$\mathcal{R}_y = \frac{z(y^2 + \varepsilon^2 P_R(z)) - \delta(1 + bz - 2y)}{F^2}. \tag{F.2}$$

We will bound its integral over the full contact arc: the negative part supplies a logarithmic contraction, while the positive part has only a smaller logarithmic loss.

On the fixed negative interval down to $-LR$, the preceding upper barrier gives $0 < y \le \varepsilon^2 P_R$ and $\varepsilon^2 P_R \le F \le 3\varepsilon^2 P_R$. Thus

$$\mathcal{R}_y \le -\frac{c_1}{\varepsilon^2|z|}, \qquad -q_g \le z \le -LR. \tag{F.3}$$

To keep the variational coefficient negative from $-LR$ to zero, compare the orbit with the positive nullcline. Let $h = -\delta/z$ and $w = y - h$. Exactly,

$$w_z = (z/F)w - h', \qquad h' = \delta/z^2 > 0.$$

With $E(z,s) = \exp\int_s^z \zeta/F(\zeta, y(\zeta))\,d\zeta$,

$$w(-LR) = E(-LR, -q_g)w(-q_g) - \int_{-q_g}^{-LR} E(-LR, s)h'(s)\,ds.$$

The first term is at most $C\varepsilon^3(LR/q_g)^{c_2/\varepsilon^2}$. On the last interval of length $\varepsilon^2 R$, the lower bound for F gives $E(-LR, s) \ge c_3$ and $h'(s) \ge c_4\delta/R^2$. The negative integral is therefore at least $c_5\varepsilon^5 R^2$. It dominates the first term uniformly: for fixed R_+ with $LR_+/q_g < 1/2$ and $N = c_2/\varepsilon^2 > 2$,

$$\sup_{0<R\le R_+} \varepsilon^{-2}R^{-2}(LR/q_g)^N = \varepsilon^{-2}R_+^{-2}(LR_+/q_g)^N \longrightarrow 0.$$

Hence $w(-LR) < 0$, and $w_z|_{w=0} = -h' < 0$ keeps $y < h$ up to zero. While h remains in the small contact rectangle, $F_y > 0$ and $F(z,h) > F(z,y) > 0$. The identity

$$\mathcal{R}_y = \frac{z[(y-h)^2 + F(z,h)]}{F(z,y)^2} \tag{F.4}$$

then gives $\mathcal{R}_y < 0$. When h exceeds the fixed height η, use (F.2) instead: $|z| \le \delta/\eta$ and its numerator is at most $\delta(\varepsilon^2|P_R|/\eta - c_6) < 0$. At $z = 0$ it equals $-\delta F_y < 0$. We obtain

$$\int_{-q_g}^{0} \mathcal{R}_y \, dz \le -\frac{c_1}{\varepsilon^2} \log \frac{q_g}{LR}. \tag{F.5}$$

The positive forcing determines the core scale $s = \varepsilon R$. For the positive arc put $z = sU$, $y = s^2V$, and $d\tau = s\,dt$. The equation becomes

$$U' = VQ + C\varepsilon^2U^2 + a\varepsilon U + c, \qquad V' = UV + 1, \qquad Q = 1 + bsU - s^2V.$$

We first obtain a positive core height, then propagate it with a lower parabola. Every counted passage crosses $-1 \le U \le 0$ with $U' > 0$. When $V \le 1/4$, $V' \ge 3/4$ and $U' \le M_0$ for a fixed M_0. Thus it cannot remain below $v_0 = \min(1/4, 3/(8M_0))$ across that interval, and it cannot recross the level v_0 downward. Hence $V(0) \ge v_0$. For a sufficiently small fixed $\gamma > 0$, the lower parabola $V = v_0 + \gamma U^2$ satisfies

$$V' - 2\gamma UU' = 1 + UV(1 - 2\gamma Q) - 2\gamma C\varepsilon^2U^3 - 2\gamma a\varepsilon U^2 - 2\gamma cU \ge 1 + v_0U/4 + \gamma U^3/4.$$

Indeed choose $\gamma \le \min(1/6, v_0/[16(\sup|c| + 1)])$, use $2\gamma|a|\varepsilon U^2 \le \gamma U^3/8 + 8\gamma a^2\varepsilon^2U$, and then shrink ε. Consequently $y \ge v_0s^2 + \gamma z^2$.

Fix $c_0 \le v_0/8$. If $c \le -c_0$, a fixed small $\alpha > 0$ gives $P_R(-\alpha R) \le -c_0R^2/2$, and Lemma F.1 excludes the cycle. For $c \ge -c_0$, Young's inequality and the parabola give $U' \ge V/4$, hence $F \ge y/4$. In (F.2) discard the negative forcing term to obtain

$$\mathcal{R}_y \le 16z + C\varepsilon^2 \frac{z^3 + Rz^2 + R^2z}{(s^2 + z^2)^2}.$$

The three integrals up to a fixed positive ξ are bounded respectively by $\log(1 + \xi/s)$, $\pi/(4s)$, and $1/(2s^2)$. As $s = \varepsilon R$, this proves

$$\int_0^{\xi} \mathcal{R}_y \, dz \le C_7 + C_8\varepsilon^2 \log(1 + \xi/(\varepsilon R)). \tag{F.6}$$

It remains to include the fast arc and the moving entry in the complete multiplier. Take the fixed positive marker as the source of the return $\mathcal{P}$. Its fast arc to the descending entry has bounded derivatives. At $y = K\varepsilon^3$, differentiating the moving initial abscissa gives $J = -\mathcal{R}\Lambda_q = O(\varepsilon)$, with $J > 0$ in this orientation. On the remaining fixed negative interval $\mathcal{R}_y < 0$ for small ε, since the $z\varepsilon^2P_R$ term dominates the bounded forcing term. Hence $0 < J(-q_g) \le C\varepsilon$. Combining this with (F.5)–(F.6) yields, with $T_R = \log(q_g/(LR)) \ge \log 2$,

$$\log \mathcal{P}' \le \log(C\varepsilon) + C_7 - c_1\varepsilon^{-2}T_R + C_8\varepsilon^2 \log(1 + \xi/(\varepsilon R)) < 0.$$

For the last inequality use $\log(1 + \xi/(\varepsilon R)) \le T_R + \log(1 + L\xi/(q_g\varepsilon))$ and then shrink ε. This choice is uniform even for arbitrarily small R relative to ε. The complete domain is an interval by Lemma A.2, after the fixed regular source change. Thus $0 < \mathcal{P}' < 1$ gives at most one fixed point and proves the proposition. □

Appendix G. The full signed complement for the realizing family

We prove Lemma 8.1 on the full source rectangle. The main estimate compares contraction on a fixed negative slow interval with the amplification between that interval and the positive saddle. The former has a fixed action, whereas the latter has action $O(c)$; choosing c first leaves enough contraction to absorb every finite-power loss as ε decreases. All passages use positive physical time. Time is reversed only to construct normal coordinates, with finite phase order, cuts, and normal gap fixed before the small parameters.

G.1. **The same stable graph and the exact local power.** On a fixed interval about $X = 1$, put $Y = \lambda_+ k$. The auxiliary equations

$$X' = e^2\mathcal{F}(X, \lambda_+ k), \qquad k' = Xk - 1$$

extend regularly to $\lambda_+ = 0$. Translation $k = 1/X + v$ gives

$$v' = Xv + e^2\mathcal{F}(X, \lambda_+(1/X + v))/X^2.$$

Reverse time, extend the base with a fixed cutoff to a circle, and keep its normal coefficient at most $-\sigma_0 < 0$. At $e = 0$ the critical circle has zero tangential speed and a fixed normal gap. For a prescribed finite order $K \geq 2j + 11$, $j \geq 4$, the source order is fixed at $K + 2$. Fenichel's persistence theorem [14, Theorem 3, p. 215] supplies a small finite-order graph on this compact boundaryless extension. The limiting circle is a compact invariant manifold with hyperbolic normal splitting. Its tangential rates vanish and remain $O_K(e^2)$ under perturbation, so the fixed normal gap gives the finite-order domination required by the theorem. The circle, cutoff, and finite order are chosen before e is made small.

Its graph equation is $e^2A(X, v)v' = B(X)v + e^2R(X, v)$, with $|B| \geq \sigma_0$. The differentiated-maximum induction of Lemma 5.5, with forcing size e^2, gives $\|v\|_{C^K} \leq C_K e^2$; the transport coefficients are $O_K(e^2)$ and the top normal coefficient stays separated from zero. Reconstructing the physical graph retains the forcing factor:

$$h(X) = \lambda_+/X + O_{C^K}(e^2\lambda_+), \qquad h|_{\lambda_+=0} = 0. \tag{G.1}$$

Thus the reconstructed graph also covers $\lambda_+ = 0$ without division by the forcing parameter.

Let $v_b(X) = \mathcal{F}(X, h(X))$. Then $v_b = 1 - X + cX^2 + O_{C^1}(\lambda_+)$ and $v_b' = -1 + O(c + \lambda_+)$ near one. It has a unique simple root $X_s = 1 + O(c + \lambda_+)$. Graph invariance gives $X_s h(X_s) = \lambda_+$, so this is an actual equilibrium. Its tangent eigenvalue $e^2 v_b'(X_s)$ is negative for $e > 0$, and its normal eigenvalue is a positive unit. The graph approaching it from the left is its actual weak stable branch.

With $u = Y - h(X)$ the exact translated equations are

$$X' = e^2 v_b(X) + e^2 C(X, u)u, \qquad u' = B(X, u)u,$$
$$C = 1 + bcX - c^2e^2(2h + u), \qquad B = X - e^2h'C.$$

They meet Lemma 3.1 on the same selected graph, with $\tau = \rho = e^2$ and B a positive unit. The product construction preserves

$$\theta' = e^2 v_b(\theta), \quad \omega' = b_N(\theta)\omega, \quad X = \theta + e^2\omega L(\theta, \omega), \quad Y = h(X) + \omega A(\theta, \omega), \quad A > 0. \tag{G.2}$$

In particular $X_\omega = O_{C^j}(e^2)$.

The common phase and amplitude estimates of Lemma 3.1, with $\tau = \rho = e^2$, give these bounds directly on the selected graph. In particular its phase integral retains the outside factor e^2, so the section shear is $X_\omega = O_{C^j}(e^2)$ without an inverse weak eigenvalue.

After the positive rescaling $ds = b_N(\theta)d\tau$, apply the scalar simple-root linearizer for $g = v_b/b_N$, using the removable integrand $g'(\theta_s)/g(\theta) - 1/(\theta - \theta_s)$. It gives the exact equations (8.3), with $\beta_s = -g'(\theta_s)$ a positive unit. Take $\theta_b = 1 - 2\sigma$. The incoming value $\xi_0(p) = \xi(\theta_b)$ has common bounds $0 < \xi_- \leq \xi_0(p) \leq \xi_+$ by the scalar linearizer above. Fix a small $\omega_0 > 0$ and then $a_2 < \xi_-/4$. Include the unstable outlet $\xi = 0$ and choose the cuts so that the entire outgoing interval has $\theta \geq \theta_b + \sigma$.

G.2. **Every outgoing source reaches the same fast core.** Choose a horizontal $Y = \delta_1$ above the entire outgoing cut and the incoming cut for $0 \leq \omega \leq \omega_0/4$, and still in the local tube. Restrict λ_0 below a fixed multiple of ω_0. On the outgoing cut, $\mathcal{F} \geq k\omega_0$ and $Y' \geq kY$. Thus every $a \in [0, a_2]$, including zero, reaches this horizontal cut in bounded positive τ time. Its abscissa $x_1(a, p)$ has $-C \leq x_{1,a} \leq -k < 0$. Indeed the source tangent $(1, 0)$ has determinant $-\omega_0$ with the canonical field, while the (ξ, ω) to (X, Y) Jacobian is negative. The target determinant is $-(X\delta_1 - \lambda_+)x_{1,a}$. The finite-flight determinant identity preserves its strict sign and unit bounds.

Use Y as phase from δ_1 to e^{-2}:

$$X_Y = \frac{e^2\mathcal{F}}{XY - \lambda_+}.$$

The speed $\mathcal{F}$ cannot have a first zero: at such a zero $X_Y = 0$ and $d\mathcal{F}/dY = 1 + bcX - 2c^2e^2Y > 1/2$. Consequently X increases and has a fixed positive lower bound. Choose $\lambda_0 < x_{\min}\delta_1/2$. Young's inequality $cX \le 1 + c^2X^2$ gives

$$(X^2)_Y \le Ce^2(1 + Y^{-1}) + Ce^2(c^2 + c/Y)X^2.$$

The coefficient integrals up to e^{-2} are bounded. Gronwall gives a common upper bound for X. The integral of the first-variation coefficient is likewise bounded by the integral of $Ce^2(1+Y^{-1})$. At the terminal physical point $y = c^2$ the abscissa is cx_t, where x_t remains in a fixed positive compact interval and $x_{t,a}$ is a negative unit. The physical flight time is $O_c(1 + |\log\varepsilon|)$.

For the continuation put $z = cU$, $y = c^2V$. The equations are

$$U' = V(1 + bcU - c^2V) + e^2(1 - U + cU^2), \qquad V' = UV - e^2\lambda_+.$$

From $V = 1$, $U = x_t$, ordinary compact continuous dependence gives passage to a fixed $U_0 > x_{\max} + 1$: at $c = e = 0$ its graph is $V = 1 + (U^2 - x_t^2)/2$ with both speeds positive. Choose a small γ and large M placing these initial points between $\gamma(1 + U^2)$ and $M(1 + U^2)$. Fix z_c small enough that $M(c_0^2 + z_c^2) < 1/4$. Only then make e small. On $U_0 \le U \le z_c/c$, the ratio of the horizontal speed to V is between $1/2$ and 3, and $UV - e^2\lambda_+ \ge UV/2$. The lower barrier has slope $2\gamma U < U/6$, while the actual slope is at least $U/6$. The upper barrier has slope $2MU > 2U$, while the actual slope is at most $2U$. Neither barrier can be crossed at a first exit. This proves the parabolic passage and its $O(1/c)$ physical time bound.

On any fixed $d \le z \le z_c$, the scalar equation converges to $y_z = z/(1 + bz - y)$. At $z = d$ every initial height is at most $M(c^2 + d^2)$. First take d small and then c, e, λ_+ small. The fast scalar Lipschitz bound identifies the limiting curve with the zero-height fast solution. The entire outgoing collar therefore reaches a smaller interior core of the fixed fast section in the joint limit $c, e, \lambda_+ \to 0$.

G.3. **Moving descent and its finite-power variation.** Apply the ordinary fast holonomy of Lemma A.2 from that core to a fixed small descending height d_0. Choose it so that the fast foot strip is strictly left of $-q_0 < 0$. On this strip $z \le -q_* < 0$, $\dot{y} = zy - \nu \le -q_* y$ until $y = K\varepsilon^3$. The time is $O(1 + |\log\varepsilon|)$ and

$$|\Delta z| \le Cd_0/q_* + C\varepsilon^2(1 + |\log\varepsilon|).$$

Fix d_0 first and then shrink ε. A first-exit argument keeps every source in the strip and proves the transverse finite hit $z = \Lambda_p(a)$.

On the horizontal positive source, $z_a = cx_{1,a} < 0$ and $G = c^2\varepsilon^2(X\delta_1 - \lambda_+)$ is a positive unit times $c^2\varepsilon^2$. The whole upstream physical time is at most $T_c(1 + \log(1/\varepsilon))$, and the divergence is bounded by M_K on a fixed compact phase set. The determinant identity at the moving hit is

$$-G_{\text{hit}}\Lambda_p'(a) = \det(X_{\text{source}}, \gamma_a') \exp \int_0^{T(a)} \operatorname{div} X \, dt. \tag{G.3}$$

Since $-G_{\text{hit}} \ge q_* K\varepsilon^3$, it gives $0 < \Lambda_p'(a) \le C_c\varepsilon^{-N_c}$. The determinant accounts for the derivative of the hitting time.

G.4. **The common negative action.** On a fixed negative interval from all descending entries to $-q_g$, $P_c = z^2 - z + c$ has a uniform positive margin. The auxiliary substitution $y = \nu k$ has limiting graph $k = 1/z$ and a fixed negative normal gap. The same compact-circle and differentiated-maximum argument used above gives a single graph $h_\nu = \nu/z + O_{C^K}(\varepsilon^2\nu)$ when $\nu \le \varepsilon^4$. For $W = y - h_\nu$ the exact equations are

(G.4)

$$W_z = \frac{z - h_\nu'(1 + bz - 2h_\nu - W)}{F} W, \qquad R_y = \frac{z(y^2 + \varepsilon^2 P_c) + \nu(1 + bz - 2y)}{F^2}, \quad R = G/F.$$

Choose $K_0 > K + C_h$ before shrinking ε. The tube $0 < W < K_0\varepsilon^3$ is forward invariant, has $F \asymp \varepsilon^2$, and includes all descending sources. It allows negative y; uniqueness of the scalar flow preserves $W > 0$. At the fixed cut $-q_0$, the height $B_p(a)$ satisfies

$$B_p'(a) = -R(\Lambda, K\varepsilon^3)\Lambda_p'(a) \exp \int_{\Lambda}^{-q_0} R_y \, dz > 0.$$

Here $R(\Lambda, K\varepsilon^3) < 0$, its magnitude is $O(\varepsilon)$, and the integral is nonpositive after shrinking.

On $I_N = [-q_0, -q_g]$, fix $p_* \le P_c \le p^*$ before the final c choice. The exact identities give $W_z \le -\kappa_W W/\varepsilon^2$ and $R_y \le -\kappa_J/\varepsilon^2$, with $\kappa_W = q_g/(4p^*)$ and $\kappa_J = q_g p_*/[8(p^*)^2]$. Set $m_N = \frac{1}{2}\min(\kappa_W, \kappa_J)(q_0 - q_g) > 0$. Consequently

$$0 < W(-q_g) \le K_0\varepsilon^3 e^{-2m_N/\varepsilon^2}, \qquad 0 < y_a(-q_g) \le C_c\varepsilon^{-N_c} e^{-2m_N/\varepsilon^2}. \tag{G.5}$$

Now define ν_+ by (8.5). For each fixed c, decreasing ε puts the entire ν interval in the auxiliary and saddle windows and makes $\nu_+ \le \varepsilon^4$. In particular $|y(-q_g)| \le A_0\varepsilon^3 e^{-m_N/\varepsilon^2}$, where A_0 was fixed beforehand.

G.5. **A positive-time adjustment of signed entry.** The incoming cut is at $\theta_b = 1 - 2\sigma$. In physical coordinates the same stable graph is $h_s(z) = c\varepsilon^2 h(z/c)$. Choose $\ell > 2\delta \sup|L| + 1$ and the physical vertical cut $z_v = c\theta_b - \varepsilon^2\ell > 0$. If $u = (y - h_s(z_v))/(c\varepsilon^2)$ is its signed normal displacement, (G.2) gives $\theta_0 < \theta_b - e^2/2$. On the intervening strip $v_b \ge k\sigma$, so the finite positive τ time to θ_b is at most a fixed T_v. Choose δ_v small enough that normal growth over T_v keeps every $|u| \le \delta_v$ within $\delta/4$. Choose this common δ_v before any trajectory. The hitting equation divided by e^2 has time derivative $v_b(\theta_b) \ge k\sigma$, while $\theta_u = O(e^2)$. The implicit-function and determinant identities give a map

$$E_p(0) = 0, \qquad 0 < k_E \le E_p'(u) \le C_E \quad (|u| \le \delta_v).$$

This positive-time adjustment is defined for positive, zero, and negative signed inputs. At zero it reaches the incoming cut on the stable branch before the saddle.

G.6. **The entire signed connector and its variation.** Let $t_\sigma = 1 - 2\sigma$, $I_\sigma = -t_\sigma - \log(1 - t_\sigma) > 0$, and $J_c = \log(1 + q_g/c) + \log(1/(2\sigma))$. On $[-q_g, z_v]$,

$$P_c \ge c - z \ge 2\sigma c, \quad \int_0^{ct_\sigma} \frac{z}{P_c}\, dz \le cI_\sigma, \quad \int_{-q_g}^{z_v} \frac{dz}{P_c} \le J_c.$$

Choose $q_g, c_0 < 1/8$ and $\varepsilon^2 \sup P_c \le 1$. In the temporary tube $|y| \le \varepsilon^2 P_c/4$ one has $|1 + bz - y| \le 2$ and $\varepsilon^2 P_c/2 \le F \le 3\varepsilon^2 P_c/2$. We use these bounds to close the height estimate. Variation of constants in the exact equation $y_z = (z/F)y - \nu/F$, with F evaluated along the orbit, gives

$$|y(z)| \le (A_0\varepsilon^3 + 8q_g K_0 J_c \varepsilon) e^{-(m_N - 2I_\sigma c)/\varepsilon^2}. \tag{G.6}$$

Choose c so that $6I_\sigma c < m_N/4$, then make ε small enough that this is below $\varepsilon^2\sigma c/8$. It is strictly inside the temporary tube, so first exit is impossible. All signed sources reach z_v in time at most $2\varepsilon^{-2}J_c$, finite for each field. Their final signed input belongs to $|u| < \delta_v/2$. Thus $H_p(a) = E_p(u_v(a))$ exists on the full claimed rectangle and has $|H_p(a)| < \omega_0/4$.

The positive part of R_y on the negative connector is at most $8\nu/(\varepsilon^4 P_c^2)$, and on the positive part at most $6z/(\varepsilon^2 P_c) + 8\nu/(\varepsilon^4 P_c^2)$. Using $\int_{-q_g}^0 P_c^{-2}\, dz \le 1/c$ and $\int_0^{z_v} P_c^{-2}\, dz \le t_\sigma/(4\sigma^2 c)$ yields

$$\int_{-q_g}^{z_v} R_y \, dz \le 6I_\sigma c/\varepsilon^2 + B_\sigma \nu/(c\varepsilon^4) \le 6I_\sigma c/\varepsilon^2 + 1. \tag{G.7}$$

The last inequality holds for the entire ν interval. Combining the moving-entry derivative with the physical scaling gives

$$H_p'(a) = \frac{E_p'(u_v)}{c\varepsilon^2} B_p'(a) \exp \int_{-q_0}^{z_v} R_y \, dz > 0.$$

Together with (G.5), this is bounded by $C_c\varepsilon^{-(N_c+2)}e^{-(m_N-6I_\sigma c)/\varepsilon^2}$. For this fixed c, take ε small enough that $\log C_c + (N_c+2)\log(1/\varepsilon) + 1 \le m_N/(4\varepsilon^2)$. It gives (8.6). Thus the fixed negative action dominates the positive $O(c)/\varepsilon^2$ amplification, forcing cost, and finite-power upstream loss on the full complement.

G.7. **Continuity, endpoint signs, and whole images.** On a fixed circle extension the small graph is unique. Subtracting two invariance equations at a maximum of their difference leaves the fixed normal gap; the transport difference has an $O(e^2)$ coefficient and is absorbed. Compactness of the common finite-order graph bounds therefore gives continuity for fixed c, ε as ν varies. The product kernels are dominated and continuous. All complementary section hits are finite and transverse; their parameter continuity and source differentiability follow. Continuity in ν is all that the splitting argument requires.

At $\nu = 0$ the unstable positive branch stays positive through the full complement and the stable branch is the axis, giving $H_0(0) > 0$. At $\nu = \nu_+$, $h_\nu(-q_g) \le -\nu/(2q_g)$ and (G.5) give $y(-q_g) < 0$. The axis field has $G = -\nu$, so this output cannot become positive again. The signed connector carries it to the same incoming cut, whose stable height is positive. Hence $H_{\nu_+}(0) < 0$.

The local images are $O(c)$ in abscissa and $O(c\varepsilon^2)$ in height. The height-phase exit has $|z| \le Cc$, $y \le c^2$; the parabolic passage has the fixed bounds already proved. The ordinary outer tube covers each compact fast subarc, and the scalar descending comparison of Lemma A.2 gives $\sup_a |\Lambda_p(a) - f_b| \to 0$ in the joint limit. Its final tail displacement is $O(\rho) + O(\varepsilon^2(1 + |\log\varepsilon|))$. The negative graph and (G.6) bound the whole slow image, while $F > 0$ covers all its abscissae. The estimates are uniform on the rectangle and give both Hausdorff inclusions for each return orbit in Section 8, completing the proof of Lemma 8.1.

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